

# A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

Kai Arseneau

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# Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick-slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

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# Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

## Symbol Summary

Symbols used throughout the derivation are summarized in [Table 1](#).

| Symbol                                         | Meaning                                                               |
|------------------------------------------------|-----------------------------------------------------------------------|
| <b>State and kinematics</b>                    |                                                                       |
| $\mathbf{x}(t)$                                | state vector                                                          |
| $t$                                            | time                                                                  |
| $\omega_p, \omega_s$                           | primary and secondary angular speed                                   |
| $s, \dot{s}, \ddot{s}$                         | shift position, velocity, and acceleration                            |
| $R, \dot{R}$                                   | CVT ratio and ratio rate                                              |
| $v_b$                                          | belt transport speed                                                  |
| $v_b^*, T_b$                                   | slip-branch target belt speed and belt-speed relaxation time constant |
| $v_\Delta$                                     | slip metric: primary-imposed minus secondary-imposed belt-line speed  |
| <b>Geometry</b>                                |                                                                       |
| $r_{p,\text{outer}}(s), r_{s,\text{outer}}(s)$ | primary and secondary outer radius                                    |
| $r_{p,\text{eff}}(s), r_{s,\text{eff}}(s)$     | primary and secondary effective torque radius                         |
| $r_{p,\text{outer},\text{min}}$                | minimum primary outer radius                                          |
| $r_{s,\text{outer},\text{max}}$                | maximum secondary outer radius                                        |
| $s_{\text{dz}}, s_{\text{max}}$                | deadzone threshold and maximum shift                                  |
| $\beta$                                        | sheave face half-angle (radial reference)                             |
| $C$                                            | pulley center distance                                                |
| $L_b$                                          | total belt length                                                     |
| $\phi$                                         | belt wrap angle used in distributed contact model                     |
| $r, \dot{r}$                                   | local belt path radius and its time derivative                        |
| $r_{\text{cm}}$                                | belt element center-of-mass radius                                    |
| <b>Constraint and mapping functions</b>        |                                                                       |
| $f(r_p, r_s, C)$                               | belt-length compatibility root function ( $f = 0$ )                   |
| $\text{root}_{r_s}(\cdot)$                     | numerical root solve for secondary radius                             |
| $r_f(s), \frac{dr_f}{ds}$                      | primary flyweight radius map and gradient                             |
| $\theta_s(s), \frac{d\theta_s}{ds}$            | secondary helix angle map and gradient                                |

| Symbol                                           | Meaning                                                                  |
|--------------------------------------------------|--------------------------------------------------------------------------|
| <b>Forces and torques</b>                        |                                                                          |
| $F_{p,ax}, F_{s,ax}$                             | primary and secondary axial mechanism force                              |
| $F_{c,p}, F_{c,s}$                               | primary and secondary belt centrifugal axial contribution                |
| $F_{ax}$                                         | generic axial clamp force at one pulley contact                          |
| $\tau_{eng}, \tau_{load}$                        | engine input torque and reflected load torque                            |
| $\tau_p, \tau_s$                                 | coupling torque seen at primary and secondary                            |
| $\tau, \tau_c$                                   | generic interface torque and applied coupling/friction torque            |
| $\tau_{ns}$                                      | no-slip requested coupling torque                                        |
| $\tau_{-}^{stick}, \tau_{+}^{stick}$             | aggregate lower and upper coupling-torque bounds for stick admissibility |
| $\tau_{-}^{slip}, \tau_{+}^{slip}$               | aggregate lower and upper coupling-torque bounds on slip branches        |
| $\tau_{c,p,\pm}^{stick}, \tau_{c,s,\pm}^{stick}$ | primary/secondary pulley-wise stick-branch lower/upper bounds            |
| $\tau_{c,p,\pm}^{slip}, \tau_{c,s,\pm}^{slip}$   | primary/secondary pulley-wise slip-branch lower/upper bounds             |
| <b>Mass/inertia and material parameters</b>      |                                                                          |
| $I_p, I_s$                                       | primary and secondary rotational inertia                                 |
| $m_f$                                            | effective flyweight mass                                                 |
| $m_{p,m}, m_{s,m}$                               | equivalent primary and secondary translating mechanism mass              |
| $\rho_b$                                         | belt material density                                                    |
| $A_b$                                            | belt cross-sectional area                                                |
| $\mu$                                            | belt-pulley traction coefficient                                         |
| $k_p, x_{p,0}$                                   | primary spring stiffness and preload displacement                        |
| $k_{s,\theta}, \theta_{s,0}$                     | secondary torsional stiffness and preload angle                          |
| $k_{s,x}, x_{s,0}$                               | secondary axial stiffness and preload displacement                       |

Table 1: Core symbols used throughout the derivation.

## Constants and Parameters

All parameters in [Table 2](#) are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

| Symbol                   | Name                           | Unit | Description                                      |
|--------------------------|--------------------------------|------|--------------------------------------------------|
| <b>Geometry and belt</b> |                                |      |                                                  |
| $L_b$                    | Belt length                    | m    | Fixed total belt length                          |
| $\beta$                  | Sheave half-angle              | rad  | Cone half-angle used for radial/axial conversion |
| $h_b$                    | Belt thickness                 | m    | Radial belt thickness                            |
| $b_{out}$                | Belt outer width               | m    | Belt width at outer face                         |
| $b_{in}$                 | Belt inner width               | m    | Belt width at inner face                         |
| $r_{p,outer,min}$        | Primary minimum outer radius   | m    | Radius at fully open primary                     |
| $r_{s,outer,max}$        | Secondary maximum outer radius | m    | Radius at fully closed secondary                 |
| $s_{dz}$                 | Shift deadzone threshold       | m    | Onset of effective radial motion                 |

| Symbol                             | Name                                     | Unit              | Description                                        |
|------------------------------------|------------------------------------------|-------------------|----------------------------------------------------|
| $s_{\max}$                         | Maximum shift                            | m                 | Physical upper travel bound                        |
| <b>Primary/secondary actuation</b> |                                          |                   |                                                    |
| $m_f$                              | Flyweight mass                           | kg                | Effective flyweight mass                           |
| $r_{f,0}$                          | Initial flyweight radius                 | m                 | Flyweight radius offset at $s = 0$                 |
| $k_p$                              | Primary spring stiffness                 | N/m               | Primary compression spring rate                    |
| $x_{p,0}$                          | Primary spring preload                   | m                 | Initial compression of primary spring              |
| $k_{s,\theta}$                     | Secondary torsional stiffness            | N m/rad           | Torsional spring rate in helix assembly            |
| $\theta_{s,0}$                     | Secondary torsional preload              | rad               | Initial torsional deflection preload               |
| $r_h$                              | Helix reference radius                   | m                 | Effective cylindrical radius in helix kinematics   |
| $k_{s,x}$                          | Secondary axial stiffness                | N/m               | Secondary axial compression spring rate            |
| $x_{s,0}$                          | Secondary axial preload                  | m                 | Initial compression of secondary axial spring      |
| <b>Inertia and drivetrain</b>      |                                          |                   |                                                    |
| $I_{\text{eng}}$                   | Engine inertia                           | kg m <sup>2</sup> | Engine rotational inertia                          |
| $I_{\text{prim}}$                  | Primary pulley inertia                   | kg m <sup>2</sup> | Primary CVT pulley inertia                         |
| $I_p$                              | Effective primary inertia                | kg m <sup>2</sup> | Implemented primary-side inertia in current setup  |
| $I_{\text{sec}}$                   | Secondary pulley inertia                 | kg m <sup>2</sup> | Secondary CVT pulley inertia                       |
| $I_{\text{gb}}$                    | Gearbox inertia                          | kg m <sup>2</sup> | Gearbox/intermediate shaft inertia                 |
| $I_{\text{wheel}}$                 | Wheel inertia                            | kg m <sup>2</sup> | Wheel/hub inertia about wheel axis                 |
| $I_s$                              | Effective secondary inertia              | kg m <sup>2</sup> | Lumped secondary-side inertia about secondary axis |
| $G$                                | Final drive ratio                        | 1                 | Fixed ratio defined by $\omega_s = G \omega_w$     |
| $m$                                | Vehicle mass                             | kg                | Total vehicle mass                                 |
| $r_w$                              | Wheel rolling radius                     | m                 | Effective tire rolling radius                      |
| <b>Contact and environment</b>     |                                          |                   |                                                    |
| $\rho_b$                           | Belt density                             | kg/m <sup>3</sup> | Belt material density                              |
| $\mu_s$                            | Belt-pulley static friction coefficient  | 1                 | Static traction coefficient used at impending slip |
| $\mu_k$                            | Belt-pulley kinetic friction coefficient | 1                 | Kinetic traction coefficient used on slip branches |
| $C_{\text{rr}}$                    | Rolling resistance coefficient           | 1                 | Tire-road rolling resistance coefficient           |
| $C_d$                              | Drag coefficient                         | 1                 | Aerodynamic drag coefficient                       |
| $A_f$                              | Frontal area                             | m <sup>2</sup>    | Vehicle aerodynamic reference area                 |
| $\rho$                             | Air density                              | kg/m <sup>3</sup> | Ambient air density used in drag model             |
| $g$                                | Gravitational acceleration               | m/s <sup>2</sup>  | Gravity constant                                   |

Table 2: Canonical non-varying parameters used by the CVT model equations.

# Chapter 1

## Introduction & Background

### 1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt-pulley contacts, and the vehicle drivetrain to which it is attached.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, while drivetrain inertia, road grade, and external vehicle load react back through the secondary side. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from vehicle response alone. A shift curve, engine-speed trace, or loss of acceleration may reflect the combined effect of primary engagement, secondary reaction, belt traction, shift motion, and changing drivetrain load. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during vehicle testing. This is especially consequential in racing-oriented applications such as Baja SAE, where launch behaviour, shift response, and torque transfer strongly affect performance, while changes to springs, flyweights, ramps, helices, or other hardware may require fabrication and repeated vehicle testing before their effect can be assessed.

Addressing this problem requires a physically interpretable dynamic framework that connects the internal CVT mechanisms to the vehicle-level response they produce. The model should be able to begin from varied initial conditions and respond to changing engine torque, road grade, external load, and other time-varying inputs, so that launch, clutching, shifting, backshift, traction loss, and vehicle acceleration emerge from the same coupled system response.

The remainder of this chapter develops the context for that framework. [Section 1.2](#) introduces the physical layout and operating principles of the belt-sheave CVT, including ratio change, traction,

mechanical actuation, and the surrounding drivetrain. [Section 1.3](#) contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, shift dynamics, and drivetrain coupling. Given this context, [Section 1.4](#) then states the aim, scope, assumptions, and contributions of the present formulation.

## 1.2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and why the CVT must be understood as part of a larger drivetrain system.

### 1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in [Figure 1.1](#). A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. [Figure 1.2](#) contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.

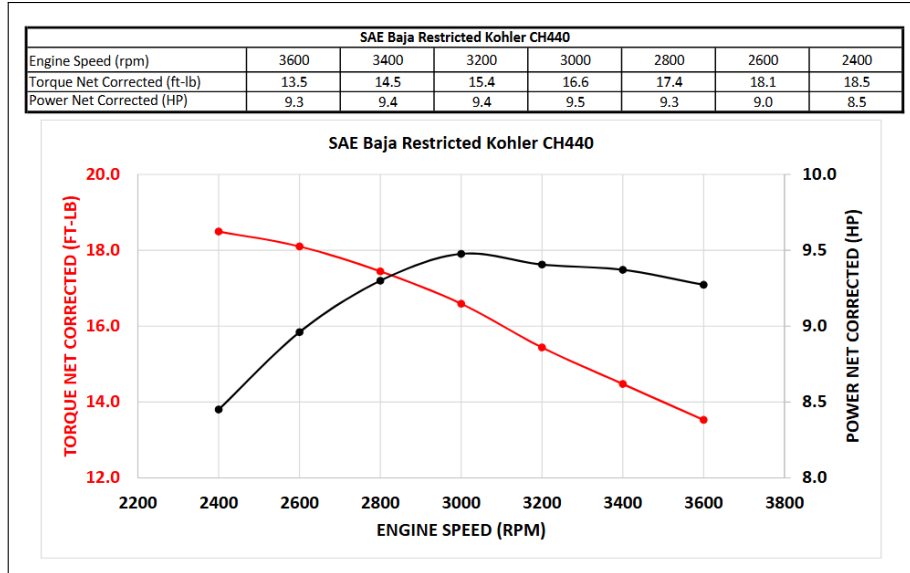


Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. ([Baja SAE, 2022](#))

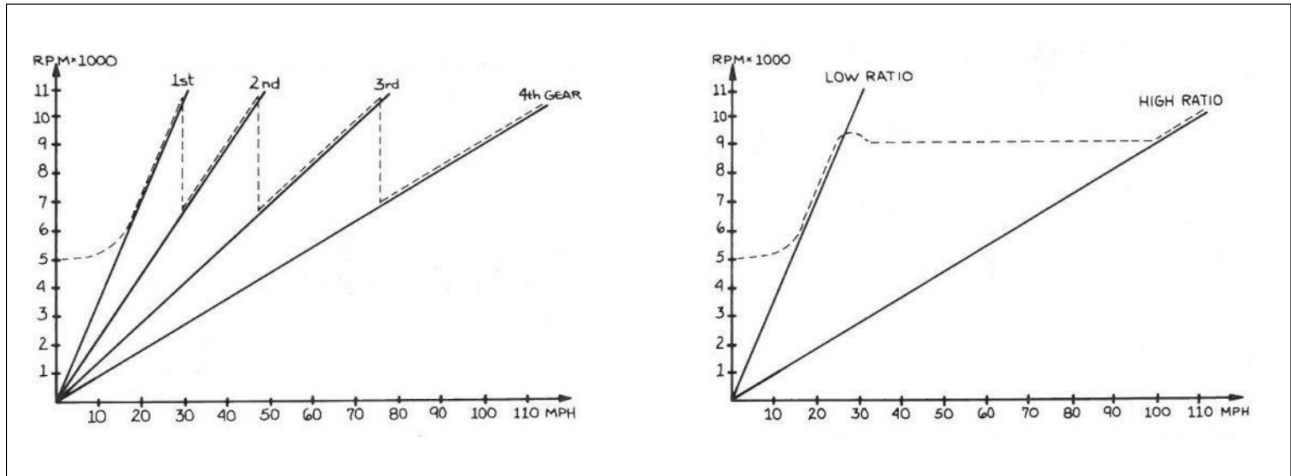


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. A stepped gearbox requires repeated engine-speed sweeps and drops between gears, whereas a CVT can vary its ratio continuously to keep the engine closer to a desired operating speed as vehicle speed increases. ([Aaen, 2007](#))

### 1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in [Figure 1.3](#). In this arrangement, two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.



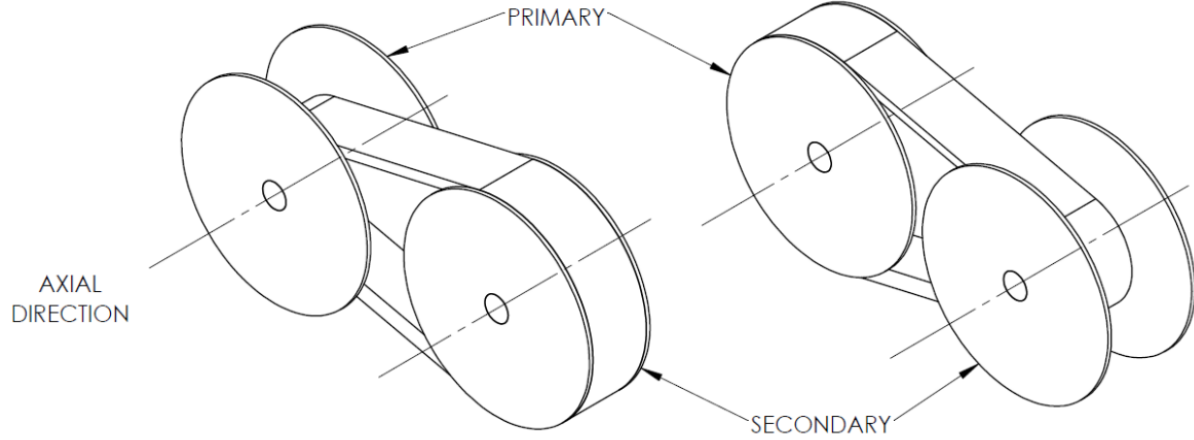


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in [Figure 1.4](#).

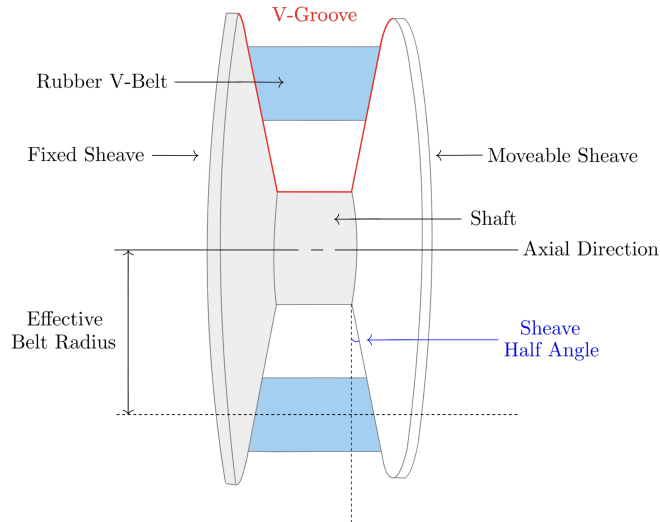


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower

groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in [Figure 1.5](#).

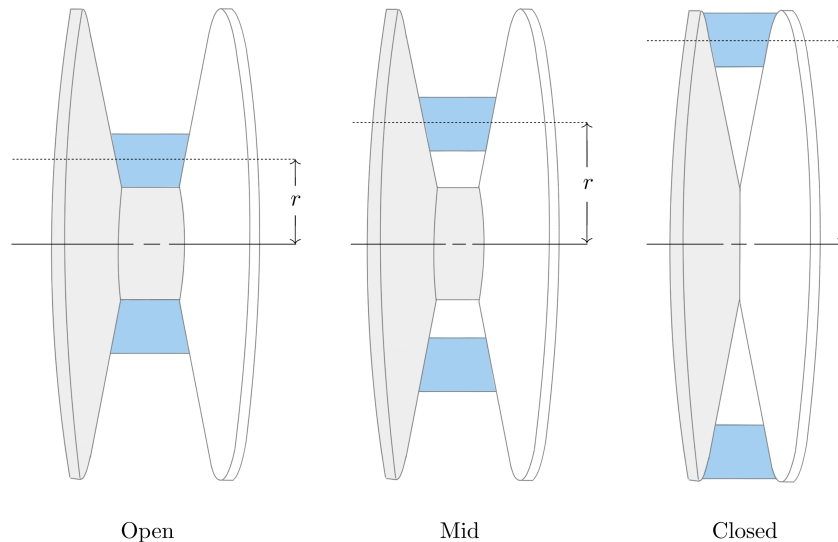


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

### 1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its

launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in [Figure 1.6](#).

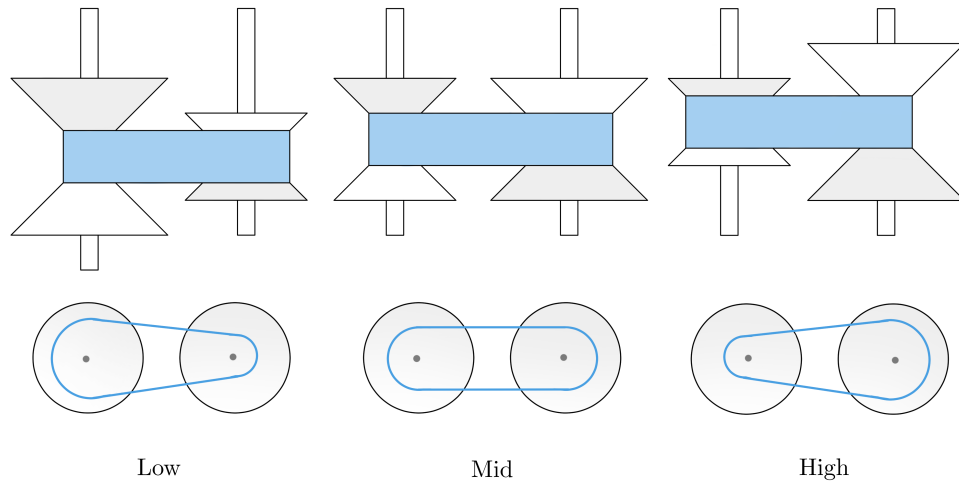


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt-sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

#### 1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. [Figure 1.7](#) shows how this happens at a small portion of the belt-pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt-pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without

excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

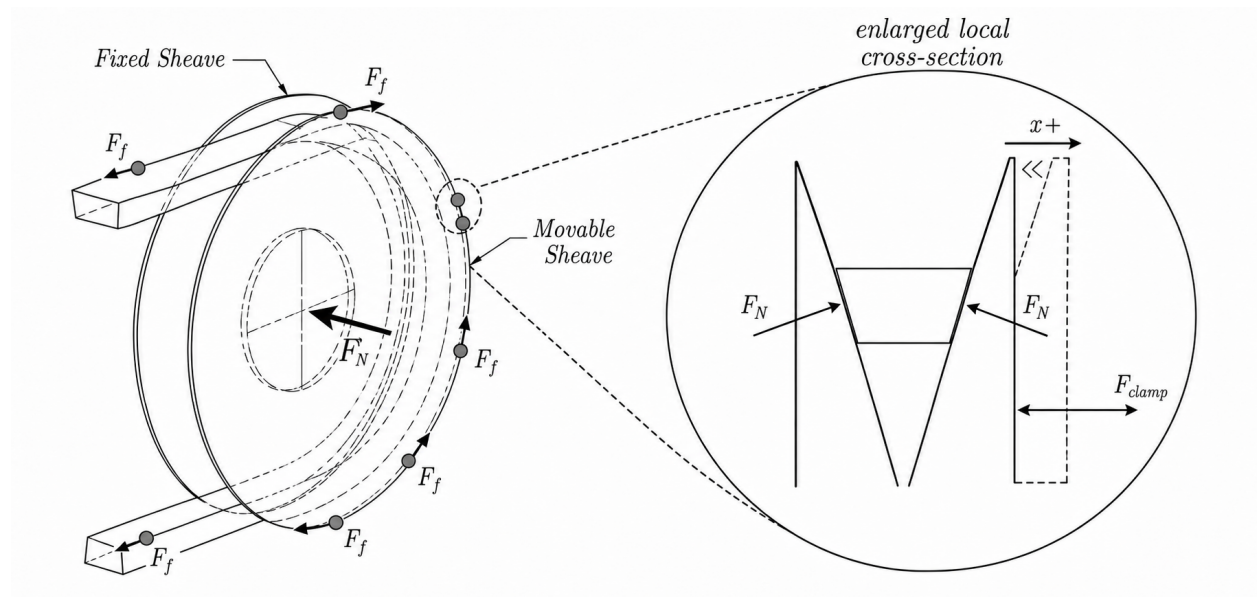


Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

### 1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in [Figure 1.6](#). The primary actuation mechanism is shown conceptually on the left side of [Figure 1.8](#).

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of [Figure 1.8](#).

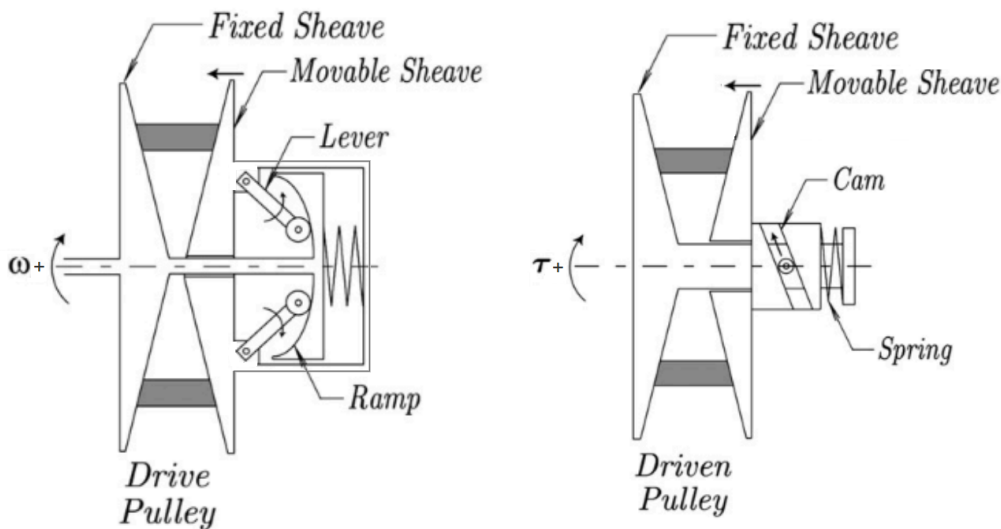


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. ([Juli'o and Plante, 2011](#))

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. The primary is mainly driven by speed, the secondary is mainly driven by transmitted torque, and the resulting CVT behavior depends on their combined interaction with the belt, pulley geometry, drivetrain inertia, external load, and available clamping force.

### 1.2.6 Full Drivetrain Architecture

The CVT does not operate in isolation. The physical drivetrain architecture considered in this work is shown in [Figure 1.9](#). The figure shows the CVT within the complete vehicle context, including the downstream gearbox, disconnect, shafts, bevel units, axles, and all four wheels.

Engine torque enters the primary pulley, passes through the belt to the secondary pulley, and then enters the downstream driveline. This driveline contains a fixed-ratio gearbox and the remaining shaft, bevel, and axle components that transmit torque to the wheels. At the tire-ground interface, tractive force accelerates the vehicle, while vehicle inertia, grade resistance, rolling resistance, and aerodynamic drag produce the load that feeds back into the drivetrain through the secondary side.

Although the full physical architecture is shown in [Figure 1.9](#), the dynamic model does not resolve the individual motion of every U-joint, bevel unit, axle, or wheel. Instead, their combined effect is represented through the downstream fixed reduction, equivalent drivetrain inertia, and vehicle longitudinal load. This preserves the coupling that matters to the CVT while avoiding unnecessary detail outside the scope of the present model.

Many drivetrain architectures are possible, but the engine-CVT-downstream driveline-wheel arrangement shown here is the system modeled in this work. This system-level view is important because the CVT ratio is not simply chosen independently of the rest of the vehicle. Engine speed affects primary actuation. Transmitted torque affects secondary clamping. Vehicle load affects the torque carried by the secondary. Belt traction determines whether that torque can be carried without slip. All of these quantities evolve together during a launch.

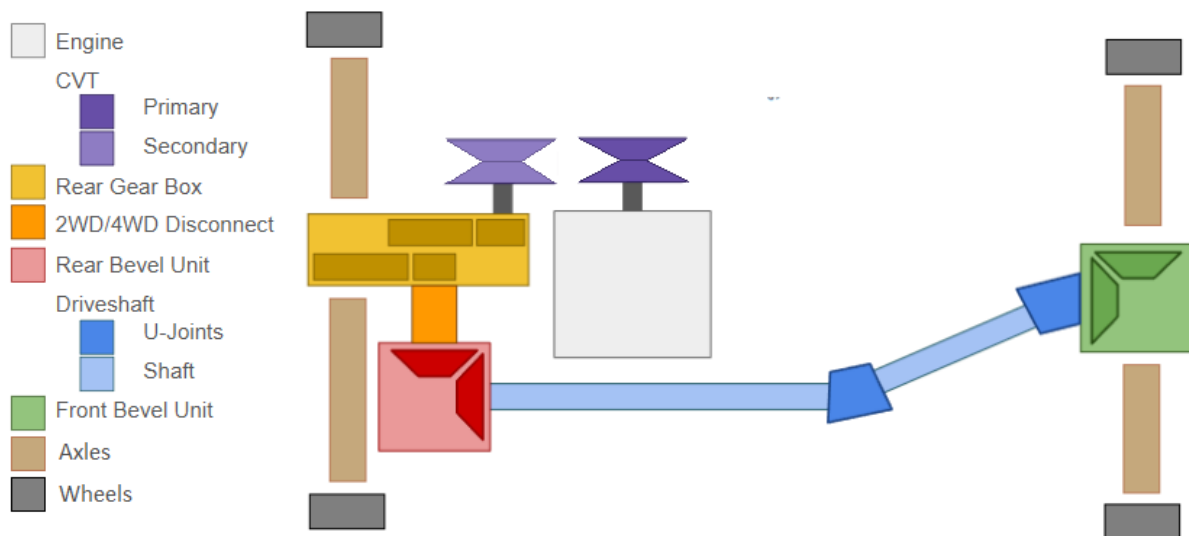


Figure 1.9: Physical drivetrain architecture considered in this work. The figure shows the CVT within the complete four-wheel vehicle layout, including the downstream gearbox, disconnect, shaft, bevel, and axle components. In the dynamic model, these downstream elements are represented by an equivalent fixed reduction, reflected inertia, and vehicle load acting through the secondary side of the CVT.

The rest of this document turns this physical picture into a dynamic model. The geometry describes how pulley radii change during shift. The force models describe how the primary and secondary sheaves are driven. The traction model describes when torque transfer can stick and when slip occurs. The drivetrain model closes the loop between engine torque, CVT behavior, vehicle load, and vehicle acceleration. Together, these pieces explain how the CVT behaves as a coupled mechanical system during launch.

## 1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict launch behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt-pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, to full transient launch behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt-sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system whose engine speed, belt state, shift position, and vehicle motion evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

### 1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a drivetrain problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model tries to explain how the system moves from one condition to another.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but whether clamping force is prescribed from outside the model or emerges from the same system being simulated.

A third choice is how much of the belt-pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove. More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so



that local belt motion and contact behaviour are solved directly (Juli'o and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it also changes the purpose of the model: a model built for efficiency or local belt force prediction may not be the most transparent way to study the full drivetrain response.

A fourth choice is how the rotating system is coupled. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that input speed, output speed, and transmitted torque evolve together (Srivastava and Haque, 2006; Ballew, 2015; Duan et al., 2016). Extending this further, the input may be coupled to an engine model and the output may be coupled to a dynamometer or vehicle load. This determines whether the CVT is being studied as an isolated variator or as one part of a larger powertrain.

These choices build on one another. A model can have detailed belt contact but prescribed pulley speeds. Another can include vehicle dynamics but use a simplified belt law. Another can model mechanical actuation carefully but treat shifting as quasi-static. None of these choices is inherently wrong; they reflect the question being asked. The important point is that the assumptions define the model's domain of usefulness.

Launch and clutching sit near the intersection of these choices. In that regime, ratio change, actuation, belt contact, rotational dynamics, and external loading can all matter at once. This does not make launch the only goal of CVT modelling, but it does make launch a useful stress test of model closure: a formulation that prescribes too many of these quantities may still be valuable, but it is not resolving the same coupled problem. The following review uses this framework to move from practical and quasi-static design models, to mechanical actuation and shift mechanics, to belt contact and efficiency models, to transient rubber-belt formulations, and finally to related dynamic models in metal-belt and chain CVTs.

### 1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett (2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (1979) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the full system response under changing operating conditions.



### 1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models attempt to convert practical component knowledge into quantitative design procedures. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. They commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of [Arango and Mu noz Alzate \(2020\)](#), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. [Cammalleri \(2005\)](#) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. [La Battaglia et al. \(2022\)](#) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a full transient simulation of the drivetrain.

Within Baja-style applications, [Skinner \(2020\)](#) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally describe the complete time history of clutching, slip, shift motion, and vehicle acceleration.

### 1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

[Cammalleri \(2005\)](#) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are

treated as part of the variator design, rather than as external force inputs. This helps establish the link between actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT.

[Kim et al. \(2002\)](#) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their work is useful because it connects the mechanically generated driver and driven pulley forces to the resulting CVT behaviour. Although these earlier mechanical-actuator models include torque-reactive secondary behaviour, they simplify how torque is shared and reacted through the driven sheave assembly.

[Sorge and Cammalleri \(2011\)](#) address this secondary-side issue in more detail by studying the helical shift mechanics of rubber V-belt variators. Their work shows that the relative rotation between the driven sheaves is important to the secondary response, and that the helix should not be treated only as a simple torque-to-axial-force conversion. This makes the secondary a coupled torque-reactive mechanism whose axial force depends on helix geometry, transmitted torque, spring loading, and the internal motion of the driven pulley.

Together, these works show that mechanical actuation is more than a source of prescribed clamp force. The primary and secondary mechanisms are part of the dynamics of the variator: speed, torque, spring preload, ramp or helix geometry, and pulley motion all influence the axial forces applied to the belt. This establishes the actuator-side foundation needed before considering the belt contact and transient shift response.

### 1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. [Gerbert \(1996\)](#) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that “slip” is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. [Kim and Kim \(1989\)](#) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. [Sorge \(1996\)](#) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by [Kong and Parker \(2005\)](#), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. [Kong and Parker \(2006\)](#) extend this

framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick’s work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete drivetrain to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that efficiency depends on multiple interacting mechanisms, not just on whether the belt remains below a gross slip threshold.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also shows why contact fidelity must be chosen carefully. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or local belt mechanics. A lower-dimensional contact model may be more appropriate when the goal is to preserve physical intuition while coupling the belt to actuator dynamics, rotational dynamics, and drivetrain response.

### 1.3.6 Transient Rubber-Belt Models and the Launch Problem

Transient rubber-belt models address time-dependent behaviour more directly, but “transient” can mean several different levels of closure. A model may study how the belt migrates radially during shift, how local belt elements move through the pulley contact, how the variator responds to prescribed axial forces, or how the CVT interacts with an engine and vehicle load. Launch lies near the most coupled end of this spectrum. During launch, the belt may be slipping, the engine may be accelerating, clamping forces may be changing, the secondary torque reaction depends on transmitted torque, and the vehicle load evolves as the vehicle begins to move. A launch model therefore has to manage torque transfer and ratio change while the drivetrain state is still developing.

A useful bridge between steady force-balance models and fully transient belt simulations is the shift-mechanics work of Sorge (2008). Sorge shows that ratio change in a rubber belt variator is not simply a sequence of steady operating points. During shift, the belt moves radially through the sheave grooves while also travelling circumferentially and transmitting torque, so belt mass conservation, radial motion, sliding, and circumferential transport interact. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for this shift-mechanics problem, making the theory more accessible for engineering calculations. These works isolate the mechanics of ratio change itself, without attempting to close the full engine–vehicle launch problem.

Juli’o and Plante (2011) developed an experimentally validated transient model of rubber-belt CVT mechanics using a discretized belt formulation. This work represents the belt as a distributed mechanical body rather than relying only on lumped contact assumptions. Its strength is that local belt motion and contact response are resolved directly. The tradeoff is that the global mechanism chain becomes less compact, and the relationship between model behaviour and a small set of

practical tuning variables becomes less direct.

Ballew (2015) continues in the discretized-belt direction and expands the model to include engine torque output and vehicle loads such as rolling resistance and aerodynamic drag. Ballew also argues against prescribing input pulley speed, since doing so removes an important rotational degree of freedom. This makes the work especially relevant to drivetrain-level rubber-belt CVT simulation. At the same time, the reported simulations use a PI controller with feed-forward gain to generate primary axial force, fix the secondary axial force to simplify the simulation, and introduce damping between belt nodes to manage numerical behaviour. Thus, the model includes important engine and vehicle coupling, but it does not close the launch problem through a mechanically self-actuated flyweight and helix model.

The transient rubber-belt literature therefore provides important precedents, while also showing why the problem remains difficult. Shift-mechanics models clarify how ratio change occurs, discretized belt models resolve local belt behaviour, and drivetrain-coupled models show how engine and vehicle dynamics can be included. The remaining challenge is not simply to make the belt model more detailed, but to choose a level of fidelity that preserves the coupling between actuation, torque transfer, ratio evolution, and vehicle response without losing physical interpretability.

### 1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive dynamic modelling tradition for metal push-belt and chain CVTs, especially in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, but they are useful adjacent references because they show how other CVT architectures have handled shift dynamics, slip, clamping force, deformation, and validation.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs, covering modelling and control strategies for friction-limited transmissions. Their review is useful not only because it summarizes the field, but because it shows that CVT modelling has developed unevenly across architectures. Different CVT types are modelled with different assumptions, and fully dynamic, experimentally validated models remain less common than steady-state or control-oriented formulations. The review also reinforces an important distinction: rubber V-belts, metal push belts, and chains should not be treated interchangeably because their contact mechanics, actuation methods, deformation behaviour, torque capacity, and control objectives differ.

For metal belt CVTs, Carbone et al. (2005) study the influence of pulley deformation on shifting behaviour. Their model explains the distinction between creep-mode and slip-mode shifting and shows that pulley deformation can be essential to reproducing slow-shift behaviour under Coulomb-friction assumptions. Yildiz et al. (2017) later validate the Carbone–Mangialardi–Mantriota model under steady-state conditions and shift manoeuvres with torque load, supporting the usefulness of this modelling framework for controlled metal-belt or chain CVTs.

Other metal-belt studies provide useful analogues for traction-limited behaviour and ratio-change dynamics. Srivastava and Haque (2006) model the dynamics of metal V-belt CVTs, while Bonsen et al. (2004) analyze slip behaviour and the relationship between slip and transmitted torque. These works are relevant because they treat slip and dynamic ratio change as part of the variator mechanics, even though the transmission architecture differs from a dry rubber V-belt system.

Chain CVTs provide another adjacent comparison. Duan et al. (2016) develop a physics-based chain CVT model beginning from the kinematics and dynamics of an infinitesimal chain segment. Their model includes speed-dependent friction, pulley axial dynamics coupled with chain radial

movement, and pulley deformation. This makes the work a useful example of a physically coupled CVT formulation in which rotational dynamics, axial dynamics, contact mechanics, and friction state are treated together.

The important distinction is that these models are usually developed for lubricated automotive metal-belt or chain systems with hydraulic or electronic clamping. Their methods are therefore valuable, but their assumptions cannot be imported directly into dry rubber V-belt CVTs with mechanical flyweights, springs, and torque-sensitive helices. They show what is possible in adjacent CVT architectures, while also reinforcing the need for a formulation matched to rubber-belt contact and mechanical actuation.

### 1.3.8 Synthesis of Prior Work

The literature establishes a complementary set of foundations rather than a single preferred model family. Practical and design references identify the hardware variables and force relationships that govern rubber V-belt CVTs. Mechanical-actuation studies show how primary and secondary clamping forces can arise from the pulley mechanisms themselves. Contact and efficiency models resolve the belt–pulley interface in increasing detail, while transient formulations address ratio change, belt motion, and, in some cases, drivetrain response. Adjacent metal-belt and chain formulations further demonstrate how axial, rotational, and traction dynamics can be coupled when the transmission architecture and actuation system are different.

The remaining difficulty is not any single missing equation, but the closure of the whole system under assumptions appropriate to a mechanically actuated dry rubber V-belt CVT. Launch and clutching sit at the intersection of engine acceleration, primary engagement, secondary torque reaction, belt traction, shift motion, and vehicle load. Treating one or more of these quantities as a prescribed input can be appropriate for a focused study, but it changes the problem being solved. The central modelling challenge is therefore to retain the dominant couplings required for transient vehicle response without obscuring the system behind unnecessary local detail.

Table 1.1 compares selected formulations that most directly bracket the present modelling objective. The preceding sections discuss the individual models in detail; the table brings together their principal modelling choices and contributions. It therefore provides a comparison of modelling scope across the selected formulations.

| <b>Selected formulation</b>                 | <b>Ratio / shift</b>                     | <b>Actuation closure</b>                         | <b>Belt / contact treatment</b>                            | <b>Rotational vehicle coupling</b>                 | <b>Principal contribution</b>                                                          |
|---------------------------------------------|------------------------------------------|--------------------------------------------------|------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------|
| <a href="#">Cammalleri (2005)</a>           | Quasi-static equilibrium                 | Mechanical primary and secondary                 | Analytical belt-force model                                | Steady imposed conditions                          | Mechanical variator design with centrifugal and torque-sensitive actuation             |
| <a href="#">Kim et al. (2002)</a>           | Steady plus empirical transient response | Mechanical primary and secondary                 | Simplified belt-force model                                | Externally supplied conditions                     | Empirical transient response of a mechanically actuated rubber-belt CVT                |
| <a href="#">Sorge (2008)</a>                | Derived transient shift mechanics        | Pulley loading prescribed                        | Radial belt motion and sliding during shift                | No engine-vehicle model                            | Derived mechanics showing that ratio change is inherently transient                    |
| <a href="#">Sorge and Cammalleri (2011)</a> | Secondary helix motion                   | Torque-reactive helix and spring                 | Belt loads supplied to secondary                           | No complete drivetrain model                       | Secondary helix resolved as a coupled torque-rotation-axial mechanism                  |
| <a href="#">Messick (2018)</a>              | Quasi-equilibrium operating states       | Axial loading prescribed                         | Detailed distributed contact, thrust, and efficiency model | Dynamometer operating points                       | Experimentally validated rubber-belt contact, axial-thrust, and efficiency formulation |
| <a href="#">Juli'o and Plante (2011)</a>    | Transient discretized-belt response      | Axial and kinematic inputs prescribed            | Local belt inertia, elasticity, and contact                | External speed and load conditions                 | Experimentally validated transient rubber-belt mechanics                               |
| <a href="#">Ballew (2015)</a>               | Transient discretized-belt response      | Primary controlled; secondary force fixed        | Nodal stick-sliding contact with damping                   | Engine torque and vehicle road loads included      | Engine-vehicle coupling with controlled or prescribed pulley clamping                  |
| <a href="#">Duan et al. (2016)</a>          | Dynamic axial and radial shift           | Applied clamping and axial dynamics              | Chain elasticity, slip, and pulley deformation             | Coupled pulley and chain dynamics                  | Coupled axial, rotational, contact, and friction dynamics in a chain CVT               |
| Present formulation                         | Dynamic shift coordinate                 | Mechanical flyweight primary and helix secondary | Lumped geometry and gross-slip traction model              | Engine, belt, pulleys, and vehicle coupled in time | Mechanically actuated, vehicle-coupled transient rubber-belt framework                 |

Table 1.1: Selected CVT formulations most relevant to the present modelling objective. The table summarizes the principal closure choices and contribution of each formulation.



## 1.4 Aim, Scope, and Contributions

The literature review in Section 1.3 motivates a specific modelling objective: to develop a physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT coupled to a vehicle drivetrain. The formulation is intended to close the dominant system-level interactions within a single time-domain model, allowing the transmission and vehicle response to evolve together under changing operating conditions. It is not intended to provide the highest-fidelity representation of local rubber-belt contact or a complete efficiency-loss model; instead, it focuses on the coupled mechanisms needed to understand how actuation, shift behaviour, torque transfer, and vehicle loading produce transient CVT behaviour.

### 1.4.1 Aim

The aim of this work is to derive a coupled nonlinear dynamic model of a mechanically actuated rubber V-belt CVT and its longitudinal vehicle drivetrain. The model extends from the engine-side pulley, through the belt and downstream driveline, to the vehicle response. Pulley speeds, belt transport, torque transfer, axial shift position, and vehicle motion evolve from the governing equations rather than being prescribed independently.

The formulation is intended to describe how the drivetrain responds from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Primary actuation, secondary torque reaction, belt traction, ratio change, rotational inertia, and vehicle loading are treated as interacting parts of the same system: a change in one mechanism alters the conditions under which the others act. This allows launch, clutching, shifting, backshift, gross slip, engine braking, and vehicle acceleration to be studied as outcomes of a common coupled response.

A further aim is to preserve the physical interpretation of that response. The derivation is organized so that the role of pulley geometry, primary speed-sensitive actuation, secondary torque-reactive actuation, belt traction, rotational inertia, and vehicle loading remains visible throughout the model. Its assumptions are stated explicitly, together with the mechanisms they exclude, so that simulation results can be interpreted in terms of both the physics retained and the limits of the formulation. The goal is therefore not only to predict system behaviour, but also to clarify why a particular response occurs and how a mechanical or operating change propagates through the drivetrain.

### 1.4.2 Scope

The system considered in this work is a mechanically actuated dry rubber V-belt CVT coupled to a simplified longitudinal vehicle drivetrain. The CVT consists of a speed-sensitive primary pulley, a torque-sensitive secondary pulley, a rubber V-belt, and a downstream fixed-ratio driveline that transmits power to the vehicle wheels. The vehicle is represented through one-dimensional longitudinal motion.

The model includes the following effects:

- variable pulley geometry, belt-length closure, and ratio kinematics;
- a dynamic shift coordinate relating sheave motion to pulley radii, wrap angles, and transmission ratio;
- mechanically generated primary and secondary axial forces;

- flyweight, ramp, spring, helix, and torque-reactive effects within the pulley actuation mechanisms;
- effective inertia associated with axial shift motion;
- explicit rotational dynamics of the primary pulley, belt, and secondary pulley;
- belt–pulley torque transfer subject to traction limits and gross-slip conditions;
- engine torque and engine-side inertia;
- downstream fixed reduction, reflected driveline inertia, and longitudinal vehicle road load; and
- transient response to specified initial conditions and time-varying input histories.

The belt is represented through a lumped geometric and traction model rather than a fully distributed contact formulation. The model determines whether the belt–pulley contacts can support the torque required by the coupled system, but it does not separately resolve detailed belt-slip and loss mechanisms such as creep, radial compliance, shear deformation, seating and unseating, local rubber hysteresis, or thermal effects. Local belt deformation, detailed radial penetration, belt wear, sheave compliance, detailed tribology, and three-dimensional vehicle dynamics are also outside the present scope. Tire–ground contact is assumed to provide sufficient traction for the longitudinal force demanded by the drivetrain; wheel slip and tire-force limits are not represented.

The formulation retains these assumptions explicitly so that their consequences remain clear when interpreting a simulation result. Its structure also permits individual closures to be refined in later work. For example, the traction relation could be replaced by a distributed contact model, belt elasticity or pulley deformation could be introduced, or thermal, efficiency-loss, and tire-traction models could be incorporated while preserving the surrounding drivetrain framework.

### 1.4.3 Contributions

The primary contribution of this work is the closure of a coupled mechanism chain for a mechanically actuated dry rubber V-belt CVT operating within a vehicle drivetrain. Individual elements of the formulation build on established ideas in CVT and belt-drive modelling. The contribution lies in combining these elements into a physically interpretable time-domain framework in which the response of the transmission and vehicle emerges from their interaction.

The main contributions are as follows:

1. A vehicle-coupled dynamic formulation that closes the speed-sensitive primary mechanism, torque-reactive secondary mechanism, belt-mediated torque transfer, ratio change, pulley dynamics, and longitudinal vehicle response within a single system of governing equations.
2. A dynamic ratio-change and belt-transport treatment in which axial sheave motion, pulley radii, belt geometry, and belt speed evolve as physical states of the system rather than being imposed through a prescribed ratio map or instantaneous equilibrium condition.
3. A traction-limited torque-transfer formulation that compares the torque required for no-slip motion with the torque supportable by the belt–pulley contacts. This permits gross-slip behaviour to be represented when the sticking response is not admissible.
4. A transient drivetrain framework that evolves from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Launch, clutching, shifting, backshift, engine braking, traction loss, and vehicle acceleration can therefore be studied as responses of the same coupled system.



5. A derivation structure that preserves the physical role of the major mechanisms and states the modelling assumptions explicitly. This supports interpretation of the predicted response and provides a modular basis for future extensions, including distributed belt contact, belt elasticity, pulley deformation, thermal effects, efficiency losses, and tire-traction limits.

Together, these contributions provide a framework for tracing how a change in component geometry, spring loading, actuator parameters, operating conditions, or vehicle load propagates through the CVT. The resulting effect on clamping, traction state, shift behaviour, torque transfer, engine speed, and vehicle acceleration can then be examined within the assumptions of the model.

## Chapter 2

# Model Definition and Simulation Problem

Chapter 1 established the need for a first-principles formulation in which shaft motion, belt motion, and ratio evolution emerge from the mechanics of the CVT. This chapter turns that objective into a simulation problem. It defines what is included in the model, how systems outside the CVT act at its two shaft boundaries, the assumptions and sign conventions used throughout the formulation, and the state variables advanced through time.

At any instant, the current state and the mechanical conditions at the two shaft boundaries are known. The formulation developed in Chapter 3 determines the corresponding state derivative, and numerical integration advances the solution. This chapter therefore defines the system to be simulated; Chapter 3 derives the mechanics that govern its evolution.

### 2.1 Modeled System and External Boundaries

The modeled system is the complete mechanically actuated rubber V-belt CVT. It contains the primary and secondary pulley assemblies, the complete belt loop, and both belt–pulley contacts. The primary flyweights, ramp, and spring, together with the secondary spring and helix, are included because they act directly on the pulley assemblies. The masses and rotational inertias of the belt, pulleys, movable components, and actuation hardware also lie within the model boundary.

The boundary is cut through the primary and secondary shafts, as shown in [Figure 2.1](#). Systems attached beyond those cuts are represented through their mechanical action at the corresponding shaft. This keeps the CVT formulation independent of a particular vehicle, engine, or test apparatus.

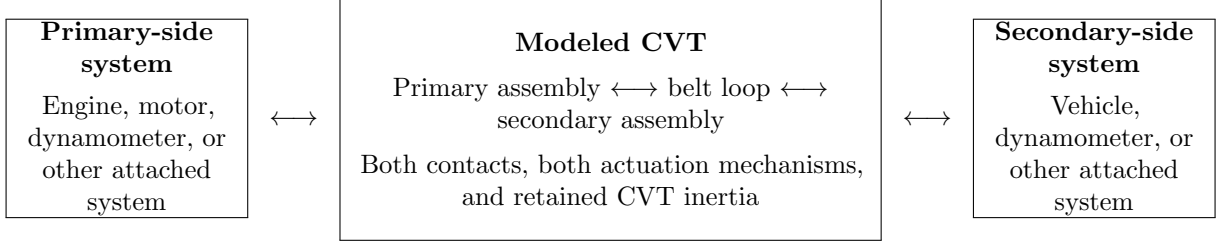


Figure 2.1: System boundary of the CVT formulation. The primary and secondary shaft cuts separate the modeled CVT from interchangeable attached systems. At either cut, the attached system is represented by the torque and equivalent inertia that it presents to the shaft. The shaft motion remains part of the CVT simulation.

### 2.1.1 Torque–Inertia Boundary Contract

At either shaft cut, everything beyond the modeled CVT is reduced to the mechanical load it presents at that shaft: an applied torque and an attached or reflected inertia. The two boundary descriptions are written as

$$\mathcal{B}_p = (\tau_{B,p}, I_{B,p}), \quad \mathcal{B}_s = (\tau_{B,s}, I_{B,s}), \quad (2.1.1)$$

and are summarized in [Table 2.1](#).

| Table 2.1: Mechanical information supplied at the external shaft boundaries. |                |                       |
|------------------------------------------------------------------------------|----------------|-----------------------|
| Boundary                                                                     | Applied torque | Inertia seen at shaft |
| Primary-side shaft                                                           | $\tau_{B,p}$   | $I_{B,p}$             |
| Secondary-side shaft                                                         | $\tau_{B,s}$   | $I_{B,s}$             |

The shaft speeds are retained and advanced by the CVT simulation. During an evaluation, the current primary and secondary speeds are made available to the corresponding attached systems. Each attached system may use that speed, time, and its own internal information to compute the torque and equivalent inertia that it presents at the shaft cut. Only this torque–inertia pair is returned to the CVT formulation. The pair enters the coupled shaft dynamics, which determine the shaft acceleration subsequently used by the time integrator to update the speed.

A prescribed torque curve or load law therefore prescribes the boundary relation, not the shaft motion. Neither  $\omega_p$  nor  $\omega_s$  is imposed by an attached system; both remain dynamic results of the CVT formulation.

A positive boundary torque acts in the positive rotational direction defined for that shaft, and the equivalent inertia is expressed about the same shaft coordinate. Pulley, sheave, flyweight, helix, and belt inertias are assigned to the CVT;  $I_{B,p}$  and  $I_{B,s}$  account only for inertia beyond the two shaft cuts.

Engine, vehicle, electric-dynamometer, and hydraulic-dynamometer implementations are given separately in the appendices as particular realizations of this common contract.

The two cuts are called the primary-side and secondary-side boundaries so the terminology remains valid under either direction of power flow. Any attached system that can supply the torque–inertia description in Equation (2.1.1) can be coupled to the same CVT equations.

## 2.2 Modeling Assumptions

The following assumptions define the physical resolution of the formulation and the operating scope within which its results are interpreted.

### 2.2.1 Material and Structural Idealizations

#### Assumption 1 (Constant Material Properties)

---

The material properties assigned to the modeled components remain constant during a simulation. Temperature-dependent material variation and thermal state evolution are outside the present formulation.

#### Assumption 2 (Rigid Structural Components)

---

The pulleys, shafts, housing, and actuation hardware are represented as rigid bodies. The belt is inextensible along its length while remaining free to bend around the pulleys. Elastic deformation of the hardware and longitudinal belt stretch are neglected.

#### Assumption 3 (Negligible Wear and Surface Evolution)

---

Component wear, belt degradation, and pulley-surface evolution are negligible over the simulated interval. Component geometry and surface condition are therefore treated as unchanged.

### 2.2.2 Contact and Friction Modeling

#### Assumption 4 (Idealized Belt–Pulley Friction Law)

---

Each belt–pulley contact is represented by a gross stick–slip law. A sticking contact develops the tangential traction required by the coupled motion up to its static limit. A slipping contact develops signed kinetic traction opposing the relative motion. Microscopic contact mechanics, rubber hysteresis, local creep, partial slip within a wrap, lubrication, and detailed Stribeck behavior are not resolved.

### **Assumption 5 (Constant Effective Friction Coefficients)**

The effective static and kinetic belt–pulley friction coefficients remain constant during a simulation. Their dependence on temperature, wear, belt conditioning, contamination, normal pressure, and slip speed is neglected.

### **Assumption 6 (Integrated Contact Representation)**

The distributed loading over each pulley wrap is represented through its integrated normal and tangential effects. Local pressure variation, edge loading, and spatial redistribution of contact stress are not resolved separately.

### **Assumption 7 (Circumferential Traction Approximation)**

Belt–pulley traction is resolved in the circumferential transport direction. Radial belt migration through the sheave groove is not assigned a separate friction demand. Local radial sliding, local creep, and traction redistribution associated with radial motion are outside the present contact representation.

### **Assumption 8 (Equal Secondary Sheave-Face Torque Partition)**

Torque transferred from the belt to the secondary pulley is shared equally by the fixed and movable sheave faces. Face-level redistribution caused by helix motion, belt deformation, pressure redistribution, or microslip is neglected.

## **2.2.3 Geometric and Kinematic Idealizations**

### **Assumption 9 (Aligned Pulley Axes and Fixed Center Distance)**

The primary and secondary pulley axes remain parallel, and their center distance remains fixed. Shaft misalignment, eccentricity, bearing motion, and geometric runout are neglected.

### **Assumption 10 (Fixed Belt Cross-Section)**

The belt retains the same cross-sectional shape and dimensions as it shifts. Its inner width  $b_{\text{in}}$  and radial height  $h_b$  remain constant. Compression between the sheave faces, lateral bulging, and centrifugal growth of the cross-section are neglected, while bending along the belt path remains permitted.

### **Assumption 11 (Constant Belt Length)**

---

The belt length remains constant. Longitudinal elasticity, transient stretch, permanent creep, and progressive elongation are neglected. Every admissible pulley configuration must therefore satisfy the same closed belt length.

### **Assumption 12 (Single Belt Transport Speed)**

---

The belt is represented by one transport speed around the complete loop. Internal shear, span vibration, local stretch, and different transport speeds on the two sheave faces are not resolved.

### **Assumption 13 (Uniform Contact Radius Around Each Wrap)**

---

At any instant, the belt follows one contact radius around each pulley wrap. Spiral or progressive radial variation through a wrap is neglected. The radius may evolve as the CVT shifts while remaining spatially uniform around the contact at that instant.

## **2.2.4 Dynamic and Inertial Modeling**

### **Assumption 14 (Lumped Inertia Representation)**

---

The inertia of each retained CVT component is represented by an equivalent lumped mass or rotational inertia. Inertia belonging to an attached system is supplied separately through  $I_{B,p}$  or  $I_{B,s}$ . Distributed shaft flexibility and torsional compliance are neglected.

### **Assumption 15 (Negligible High-Frequency Vibration)**

---

High-frequency belt, shaft, and structural vibration is neglected. The formulation retains the dominant rigid-body shaft rotation, belt transport, and sheave-shift dynamics.

### **Assumption 16 (Negligible Unmodeled Parasitic Friction)**

---

Bearing, seal, and auxiliary-hardware losses are neglected unless introduced explicitly through an actuator or boundary model. Belt traction remains part of the core CVT mechanics.

### **Assumption 17 (No Separate Empirical Loss or Thermal Model)**

The formulation resolves torque transfer and gross sliding at the belt contacts. Sliding work is not propagated into temperature, wear, or temperature-dependent properties, and additional loss mechanisms such as belt hysteresis and flexural loss are not represented by an empirical efficiency model.

## **2.2.5 Boundary and Operational Scope**

### **Assumption 18 (Negligible Gravitational Influence on CVT Internals)**

Gravitational forces on internal CVT components are negligible relative to the centrifugal, spring, inertial, and contact loads developed during operation.

### **Assumption 19 (Reference Full-Throttle Engine Boundary)**

The reference vehicle simulations use a prescribed full-throttle engine torque curve. Engine speed remains dynamically coupled to the CVT, while part-throttle and throttle-transient states are outside that particular boundary model. Other primary-side boundary models may be coupled through [Equation \(2.1.1\)](#).

## **2.3 Coordinate and Sign Conventions**

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions. Coordinates belonging only to a particular engine, vehicle, or dynamometer are defined with the corresponding boundary model.

### **2.3.1 Rotational Sign Convention**

All rotating components in the reference drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion. Their positive directions are chosen such that positive drivetrain rotation produces forward vehicle motion.

As illustrated in [Figure 2.2](#), positive wheel angular velocity  $\omega_w > 0$  corresponds to wheel rotation that drives the vehicle forward. The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley,  $\omega_s > 0$ , is the direction that produces this wheel motion through the final reduction and axle. Positive rotation of the primary pulley,  $\omega_p > 0$ , is the direction that drives the secondary pulley in this same sense through the CVT belt.

Torque follows the same sign convention. A torque is positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity. The coordinate signs remain fixed when a torque resists the current motion or when power flow reverses.

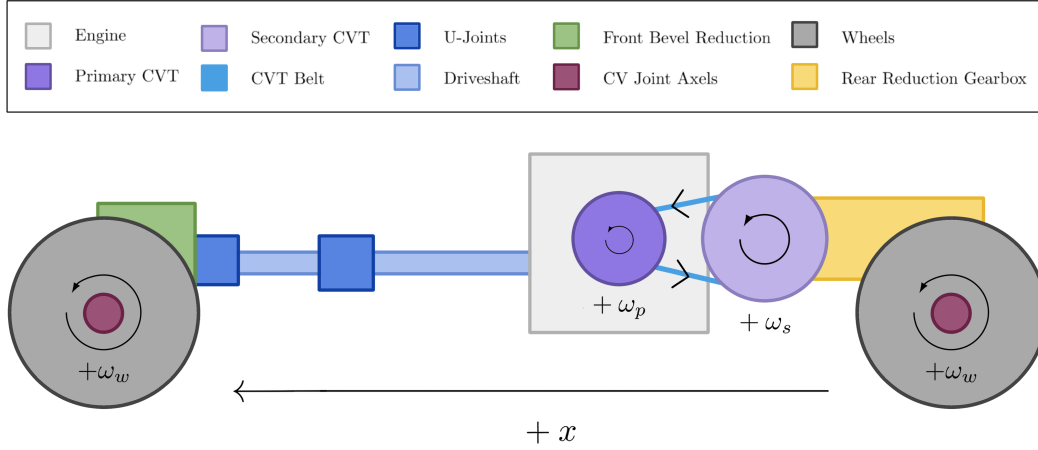


Figure 2.2: Rotational sign convention for the reference drivetrain. Positive angular velocities for the wheels ( $\omega_w$ ), secondary pulley ( $\omega_s$ ), and primary pulley ( $\omega_p$ ) are defined such that drivetrain rotation produces forward vehicle motion.

The belt loop is assigned a positive path direction consistent with transport produced by positive primary rotation. The belt transport speed  $v_b$  is positive in this direction and negative in the opposite direction. This sign choice defines belt motion independently of whether either belt–pulley contact is sticking or slipping.

### 2.3.2 Axial Sheave Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in [Figure 2.3](#) (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate. The axial variable  $s$  measures the translation of the movable sheave relative to this open configuration:

$$s = 0 \quad : \text{fully open configuration.}$$

A positive value of  $s$  corresponds to the movable sheave translating toward the fixed sheave along the pulley axis:

$$s > 0 \quad : \text{sheave-closing motion.}$$

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission. The maximum displacement  $s = S_{\max}$  corresponds to the fully closed configuration shown in [Figure 2.3](#) (right).

Although  $s$  is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open.



The precise geometric relationship between these motions will be derived later. The primary pulley is used here as the example; the same local convention—positive axial motion closes the sheave pair—applies when the secondary pulley is considered.

Axial velocity follows the same sign convention:  $\dot{s} > 0$  corresponds to the primary movable sheave translating toward its fixed sheave.

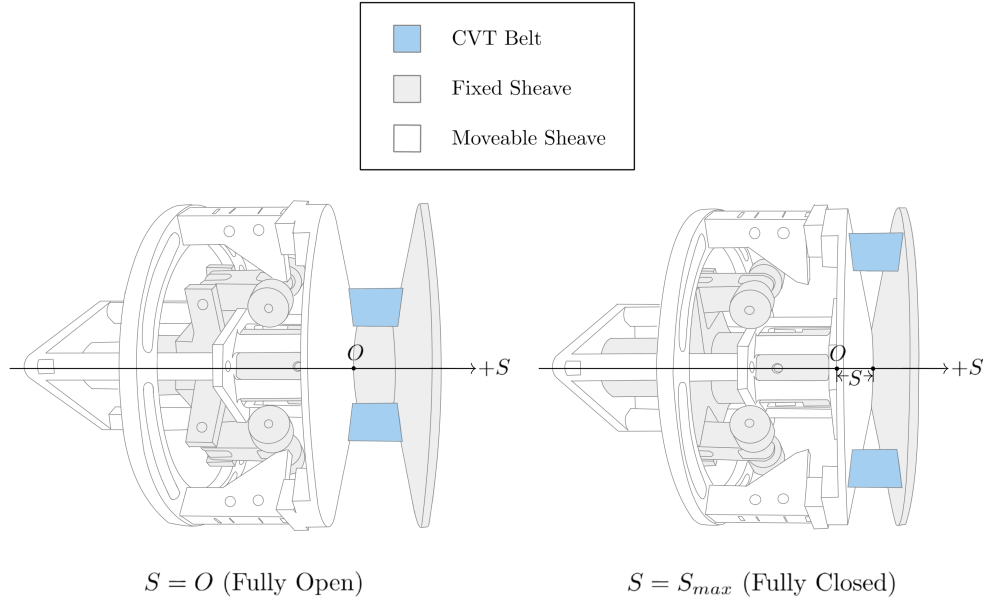


Figure 2.3: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement  $s$  measures how much the movable sheave has closed from this initial position. Increasing  $s$  increases primary effective radius and is coupled to secondary axial motion.

### 2.3.3 Local Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction  $\hat{e}_r$  and the tangential direction  $\hat{e}_\theta$ . These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus,  $F_r > 0$  corresponds to a force directed outward along  $\hat{e}_r$ .

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction  $\hat{e}_\theta$  is defined to be consistent with the positive rotational convention introduced in [Section 2.3.1](#). Tangential motion in this direction corresponds to motion produced by positive angular velocity.

[Figure 2.4](#) illustrates these directions using examples from the primary pulley. The centrifugal flyweights generate outward radial forces  $F_r$ , while the belt moves along the tangential direction with velocity  $v_t$ . These quantities serve as representative examples of forces and motion that act along the radial and tangential directions. The same local basis applies at the secondary pulley.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

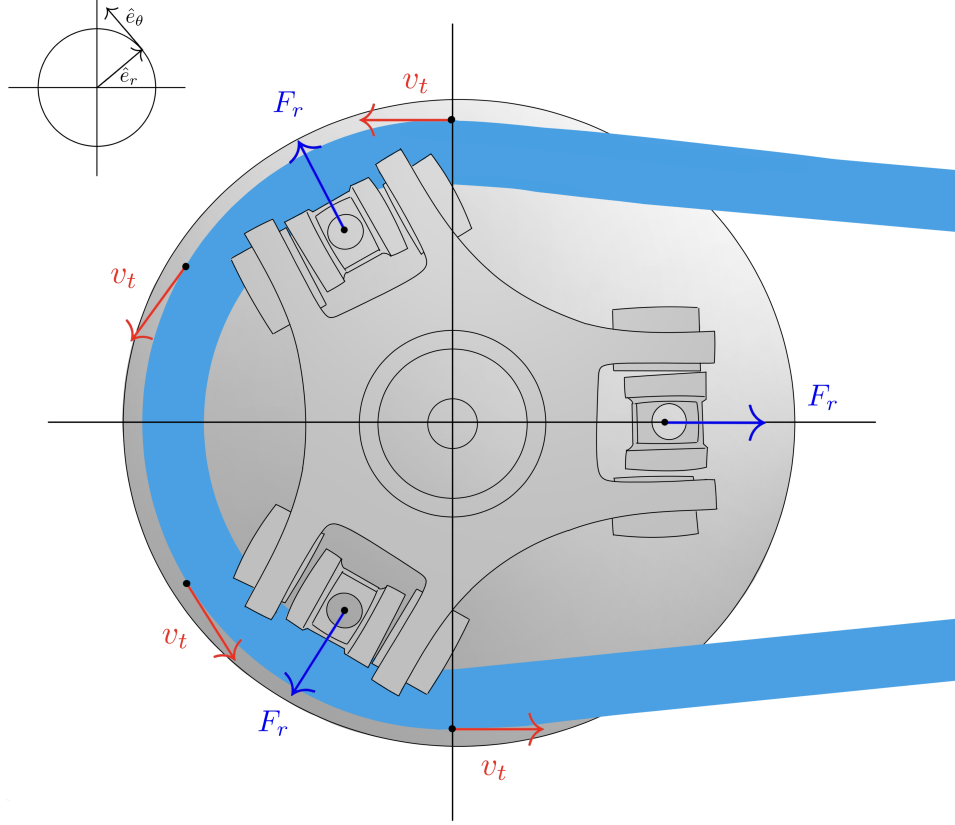


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The radial direction  $\hat{e}_r$  points outward from the center of the pulley, while the tangential direction  $\hat{e}_\theta$  is perpendicular to  $\hat{e}_r$  and aligned with positive rotation. Example quantities illustrated on the primary pulley include outward radial forces  $F_r$  generated by the flyweights and tangential belt velocity  $v_t$ . The same local directions apply at the secondary pulley. For reference, the axial coordinate  $s$  is perpendicular to the page in this view and points along the pulley shaft, into the page.

## 2.4 System State Definition

The simulation advances a state vector through time. At the beginning of each model evaluation, these state variables are known and describe the current motion and shift configuration of the CVT.

### 2.4.1 State Vector

|                                                                                                                                                                                                                                                                                  |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <div style="border-bottom: 1px solid black; margin-bottom: 10px;"> <b>State Vector</b> </div> <div style="text-align: center; padding-top: 20px;"> <math display="block">\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}</math> </div> | <b>(2.4.1)</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|

Here,  $\omega_p$  and  $\omega_s$  are the primary and secondary shaft speeds,  $s$  is the primary axial shift position,  $\dot{s}$  is its velocity, and  $v_b$  is the belt transport speed.

These are the quantities whose current values must be carried into the next instant. The rotational mechanics depend on shaft speed rather than absolute shaft angle, so absolute angular positions need not be stored. Similarly, the lumped belt representation depends on transport speed rather than which material point occupies a particular location, so no belt-phase coordinate is required. Shift position is retained because the current geometry and actuation depend on where the pulleys are in their travel, and  $\dot{s}$  is retained because the shift motion is governed by second-order dynamics.

### 2.4.2 Initial-Value Problem and Time Integration

At a chosen starting time  $t_0$ , the simulation is initialized with

$$\mathbf{x}(t_0) = \mathbf{x}_0. \quad (2.4.2)$$

The initial state may be any mechanically admissible condition within the modeled regimes. From that point onward, the formulation evaluates the time derivative of the state from the current state, the fixed CVT data, and the torque–inertia pairs returned at the two shaft boundaries. At the highest level, this evolution law is written as

$$\dot{\mathbf{x}}(t) = \mathcal{F}(t, \mathbf{x}(t); \boldsymbol{\theta}, \mathcal{B}_p, \mathcal{B}_s), \quad (2.4.3)$$

where  $\boldsymbol{\theta}$  collects the fixed physical and configuration data, and  $\mathcal{B}_p$  and  $\mathcal{B}_s$  are the boundary pairs evaluated using the current shaft speeds. Chapter 3 constructs  $\mathcal{F}$  from the CVT geometry, pulley dynamics, belt mechanics, and contact laws.

A numerical time integrator repeatedly evaluates [Equation \(2.4.3\)](#) and advances the state according to

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_{t_0}^t \dot{\mathbf{x}}(\xi) \, d\xi. \quad (2.4.4)$$

Contact transitions, engagement, and travel boundaries may change the particular form of  $\mathcal{F}$  during the simulation. Their event conditions and regime-specific equations are developed after the corresponding mechanics have been derived.

This construction is fully dynamic. The boundary models define the external torque and inertia seen by the CVT, while the coupled mechanics and time integration determine the evolving shaft speeds, belt speed, shift position, and transmission ratio.

## 2.5 Model Quantity Roles and Evaluation Flow

The constants and parameters collected in ?? and ?? serve different computational roles from the state and boundary data. Table 2.2 places these quantities in the high-level simulation workflow.

Table 2.2: High-level roles of quantities used in the simulation.

| Quantity role         | Source                                                       | Use during simulation                                                                                        |
|-----------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Fixed model data      | Reference data and selected CVT configuration                | Define the modeled hardware and remain fixed for the selected simulation.                                    |
| Initialized data      | Fixed model data                                             | Established once for the selected configuration and reused during time integration.                          |
| State variables       | Initial condition and integrator                             | Record the current CVT motion and shift configuration at time $t$ .                                          |
| Boundary data         | Attached-system models evaluated at the current shaft speeds | Give the applied torque and attached or reflected inertia seen at each shaft cut for the current evaluation. |
| Instantaneous results | Current model evaluation                                     | Provide the state derivative and event information requested by the integrator.                              |

The simulation proceeds at two levels. Initialization checks the selected CVT data, establishes quantities fixed by that configuration, assigns the initial state, and identifies the initial mechanical regime. Time integration then repeats the following high-level evaluation:

1. The integrator supplies a trial time and the corresponding current state  $\mathbf{x}(t)$ , including  $\omega_p$  and  $\omega_s$ .
2. Each attached system uses the corresponding current shaft speed to evaluate the torque and equivalent inertia it presents at the shaft cut.
3. The coupled CVT mechanics use the current state, fixed model data, and both boundary pairs to evaluate  $\dot{\mathbf{x}}(t)$  and the event conditions for the current mechanical regime.
4. The integrator advances every state variable, including both shaft speeds. If an event is encountered, the applicable regime is updated and integration continues from that event state.

This process can be summarized as

$$\begin{aligned}
\omega_p(t) \longrightarrow \mathcal{B}_p, \quad \omega_s(t) \longrightarrow \mathcal{B}_s & \quad \text{boundary evaluation,} \\
\{t, \mathbf{x}(t), \boldsymbol{\theta}, \mathcal{B}_p, \mathcal{B}_s\} \longrightarrow \dot{\mathbf{x}}(t) & \quad \text{coupled CVT mechanics,} \\
\dot{\mathbf{x}}(t) \longrightarrow \mathbf{x}(t') & \quad \text{time integration.}
\end{aligned} \tag{2.5.1}$$

Chapter 3 supplies the mechanics contained in the first arrow. Geometry is developed first because the current shift state fixes the instantaneous pulley configuration. The axial, rotational, belt-loop, and contact relations are then exposed in sequence before they are assembled into the evolution law. This ordering supports the derivation; the completed relations are evaluated together as one coupled instantaneous problem.

# Chapter 3

## Derivation

### 3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley, and the resulting pair of pulley radii determines the transmission ratio.

Both pulleys contain a movable sheave. Let  $x_p$  denote the axial position of the movable primary sheave and  $x_s$  the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

#### 3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in [Section 2.3.2](#) and illustrated in [Figure 2.3](#). The local coordinates  $x_p$  and  $x_s$  follow the same convention on their respective pulleys.

For either pulley  $j \in \{p, s\}$ ,  $x_j = 0$  denotes the fully open configuration. Increasing  $x_j$  moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing  $x_j$  opens it. The corresponding velocity follows the same sign convention:  $\dot{x}_j > 0$  denotes closing motion and  $\dot{x}_j < 0$  denotes opening motion.

The primary therefore begins at  $x_p = 0$  and moves in the positive direction as it closes. The secondary begins in a closed position, with  $x_s > 0$ , and moves toward smaller values of  $x_s$  as it opens.

Consider a belt seated between two straight sheave faces with half-angle  $\beta$ , as shown in [Figure 3.1](#). Let  $b_{\text{in}}$  denote the width of the belt at its inner surface. As assumed in [Assumption 10 \(Fixed Belt](#)

Cross-Section), this width remains constant.

Suppose the movable sheave closes the pulley by a distance  $\Delta x_j$ . At the original belt radius, the available space between the sheaves is therefore reduced from  $b_{in}$  to

$$b_{in} - \Delta x_j.$$

The belt responds by moving outward through a radial distance  $\Delta r_j$ . At its new position, the angled sheave faces provide an additional axial distance  $d$  on each side.

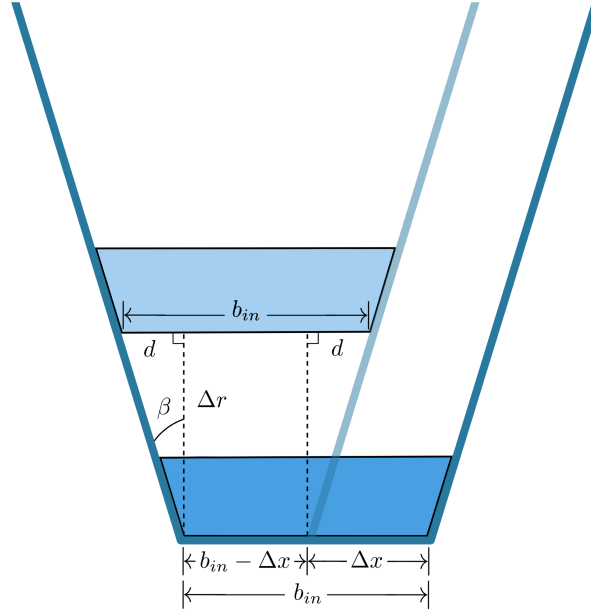


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley  $j$ . Closing the pulley by  $\Delta x$  reduces the available width at the original belt position to  $b_{in} - \Delta x$ . Moving outward by  $\Delta r$  provides an additional axial distance  $d$  at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from **Figure 3.1**,

$$b_{in} = 2d + (b_{in} - \Delta x_j). \quad (3.1.1)$$

The right triangle beside either sheave face relates  $d$  to the radial movement. For the sheave half-angle  $\beta$ ,

$$\tan \beta = \frac{d}{\Delta r_j}, \quad (3.1.2)$$

and therefore

$$d = \Delta r_j \tan \beta. \quad (3.1.3)$$

Substituting Equation (3.1.3) into Equation (3.1.1) gives

$$b_{\text{in}} = 2\Delta r_j \tan \beta + (b_{\text{in}} - \Delta x_j). \quad (3.1.4)$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2 \tan \beta}. \quad (3.1.5)$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2 \tan \beta}. \quad (3.1.6)$$

Because the sheave faces are straight, this slope is constant.

**Radius convention** The derivation above follows the inner surface of the belt, where  $b_{\text{in}}$  is defined. The belt also has the finite radial height  $h_b$ , so its inner and outer surfaces occupy different radii. Let  $r_{j,\text{inner}}$  and  $r_{j,\text{outer}}$  denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \quad (3.1.7)$$

The fixed-cross-section assumption in Assumption 10 (Fixed Belt Cross-Section) also makes  $h_b$  constant. Consequently, the inner and outer surfaces undergo the same radial displacement, and the axial-to-radial relationship derived above applies to either one.

An absolute pulley radius must nevertheless identify which belt surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \quad (3.1.8)$$

The derivative in Equation (3.1.6) describes how this radius changes, but one known configuration is needed to establish its absolute value. Let  $x_{j,\text{ref}}$  denote a known axial position at which the belt is seated, and let  $r_{j,\text{ref}}$  denote the outer pulley radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2 \tan \beta}. \quad (3.1.9)$$



The pair  $(x_{j,\text{ref}}, r_{j,\text{ref}})$  simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing  $x_s$ , [Equation \(3.1.9\)](#) produces the corresponding decrease in  $r_s$ .

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At  $x_p = 0$ , the distance between the primary sheaves can therefore exceed  $b_{\text{in}}$ . Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let  $x_{p,\text{dz}}$  denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \leq x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At  $x_p = x_{p,\text{dz}}$ , the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\text{ref}} = x_{p,\text{dz}}, \quad r_{p,\text{ref}} = r_{p,\text{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\text{min}}, & 0 \leq x_p < x_{p,\text{dz}}, \\ r_{p,\text{min}} + \frac{x_p - x_{p,\text{dz}}}{2 \tan \beta}, & x_{p,\text{dz}} \leq x_p \leq x_{p,\text{max}}. \end{cases} \quad (3.1.10)$$

The linear axial-to-radial relationship in [Equation \(3.1.9\)](#) is standard in reduced CVT geometry models ???. It applies to both pulleys whenever the belt is seated, while [Equation \(3.1.10\)](#) extends the primary description through the deadzone before contact.

The local geometry now tells us the pulley radius associated with either movable-sheave position. To determine which primary and secondary positions can occur together, the two local descriptions must next be joined through the geometry of the complete belt.

### 3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the radius on each pulley. The belt, however, passes around both pulleys. To connect the two radii, the belt must now be followed around the complete loop.

The belt path consists of four parts: a wrapped arc on the primary pulley, a wrapped arc on the secondary pulley, and two equal straight spans. Under [Assumption 13 \(Uniform Contact Radius Around Each Wrap\)](#), each wrapped portion is represented by a single radius over its full contact angle. If each straight span has length  $l$ , and the primary and secondary wrap angles are  $\phi_p$  and  $\phi_s$ , the total geometric length of the loop is

$$L_{\text{geo}} = r_p \phi_p + r_s \phi_s + 2l. \quad (3.1.11)$$

The quantities  $l$ ,  $\phi_p$ , and  $\phi_s$  follow directly from the geometry shown in [Figure 3.2](#).

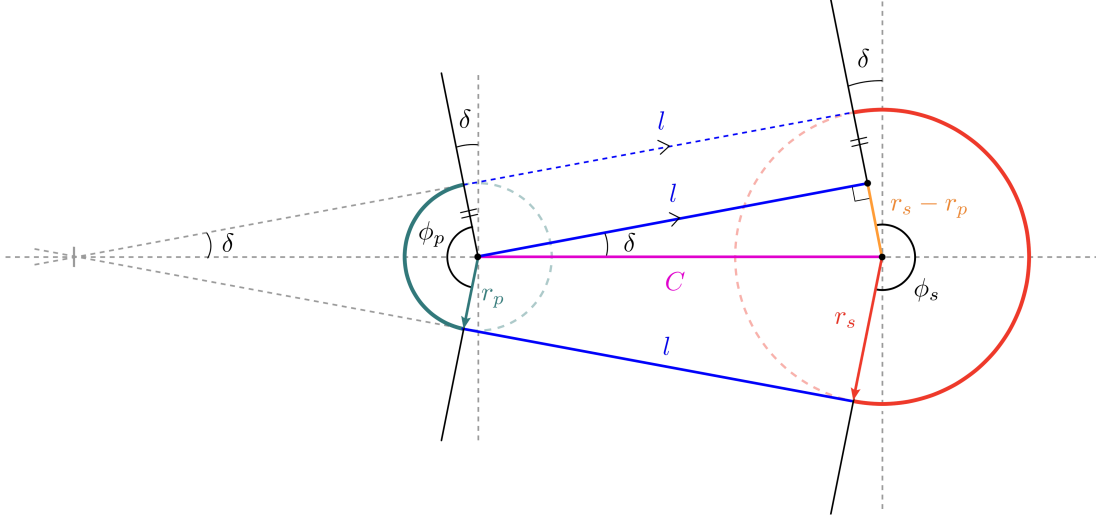


Figure 3.2: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length  $l$ . Unequal pulley radii incline the straight spans by the angle  $\delta$ , producing the primary and secondary wrap angles  $\phi_p$  and  $\phi_s$ .

Consider the right triangle in the upper span in [Figure 3.2](#). The line joining the pulley centres forms its hypotenuse and has length  $C$ . One side lies along the straight belt span and therefore has length  $l$ .

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown by the orange construction in [Figure 3.2](#), the difference between their lengths gives the remaining side of the triangle,  $r_s - r_p$ . The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \quad (3.1.12)$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \quad (3.1.13)$$

This expression requires

$$|r_s - r_p| < C. \quad (3.1.14)$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring  $C > r_s + r_p$ , and  $r_s + r_p$  is necessarily greater than  $|r_s - r_p|$ .

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to  $\pi$ .

When the radii differ, the straight spans rotate through the angle  $\delta$  shown in [Figure 3.2](#). The same right triangle gives

$$\sin \delta = \frac{r_s - r_p}{C}, \quad (3.1.15)$$

and therefore

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{C} \right). \quad (3.1.16)$$

The angle  $\delta$  is signed. For the configuration shown,  $r_s > r_p$ , so  $\delta > 0$ . The primary wrap loses an angle  $\delta$  at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.17)$$

If  $r_p > r_s$ , then  $\delta < 0$ , and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting [Equations \(3.1.13\)](#), [\(3.1.16\)](#) and [\(3.1.17\)](#) into [Equation \(3.1.11\)](#) gives

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between; align-items: center; border-bottom: 1px solid black; margin-bottom: 10px;"> <div><b>Exact geometric belt length</b></div> <div><b>(3.1.18)</b></div> </div> <div style="text-align: center; padding: 10px;"> <math display="block"> \begin{aligned} L_{\text{geo}}(r_p, r_s, C) = &amp; \pi (r_p + r_s) \\ &amp; + 2 (r_s - r_p) \sin^{-1} \left( \frac{r_s - r_p}{C} \right) \\ &amp; + 2 \sqrt{C^2 - (r_s - r_p)^2} \end{aligned} </math> </div> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of  $r_p$ ,  $r_s$ , and  $C$ , [Equation \(3.1.18\)](#) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under [Assumption 11 \(Constant Belt Length\)](#), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \quad (3.1.19)$$

where  $L_b$  is the fixed belt length expressed using the same outer-surface reference adopted for  $r_p$  and  $r_s$  in Equation (3.1.8). This is the geometric compatibility condition for the belt loop. Once any two of  $r_p$ ,  $r_s$ , and  $C$  are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance  $C$ . The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled,  $C$  remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by  $r_{p,\text{ref}}$  and  $r_{s,\text{ref}}$ , the only unknown in Equation (3.1.19) is  $C$ .

In the general unequal-radius case,  $C$  appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for  $C$ . Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \quad (3.1.20)$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L(r_{p,\text{ref}}, r_{s,\text{ref}}, C) = 0. \quad (3.1.21)$$

The root solver varies  $C$  until this condition is satisfied.

### Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{(r_s - r_p)^2 + \frac{1}{4}[L_b - \pi(r_p + r_s)]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in [Appendix A](#).

Once the initial root solve has determined  $C$ , both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius  $r_p$  is known. Substituting  $r_p$ ,  $C$ , and  $L_b$  into [Equation \(3.1.20\)](#) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_s. \quad (3.1.22)$$

The resulting  $r_s$  is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If  $r_s$  is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_p. \quad (3.1.23)$$

Thus, once  $C$  has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. [Budynas and Nisbett \(2015\)](#) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations  $r_p(x_p)$  and  $r_s(x_s)$  now describe how each pulley produces its radius, while [Equation \(3.1.19\)](#) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

### 3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley:  $x_p$  determines  $r_p(x_p)$ , and  $x_s$  determines  $r_s(x_s)$ . A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length  $L_b$ . The remaining problem is therefore to identify the pairs  $(x_p, x_s)$  that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \quad (3.1.24)$$

Every pair that makes [Equation \(3.1.24\)](#) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take  $x_p$  simply as an arbitrary choice. For each value of  $x_p$ , the primary relation gives  $r_p(x_p)$ . The shaft spacing fixes  $C$ , and  $L_b$  remains fixed by [Assumption 11 \(Constant Belt Length\)](#), so the belt-length residual can then be solved for the required secondary radius  $r_s$ . The local secondary relation finally gives the corresponding position  $x_s$ :

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in [Figure 3.3](#).

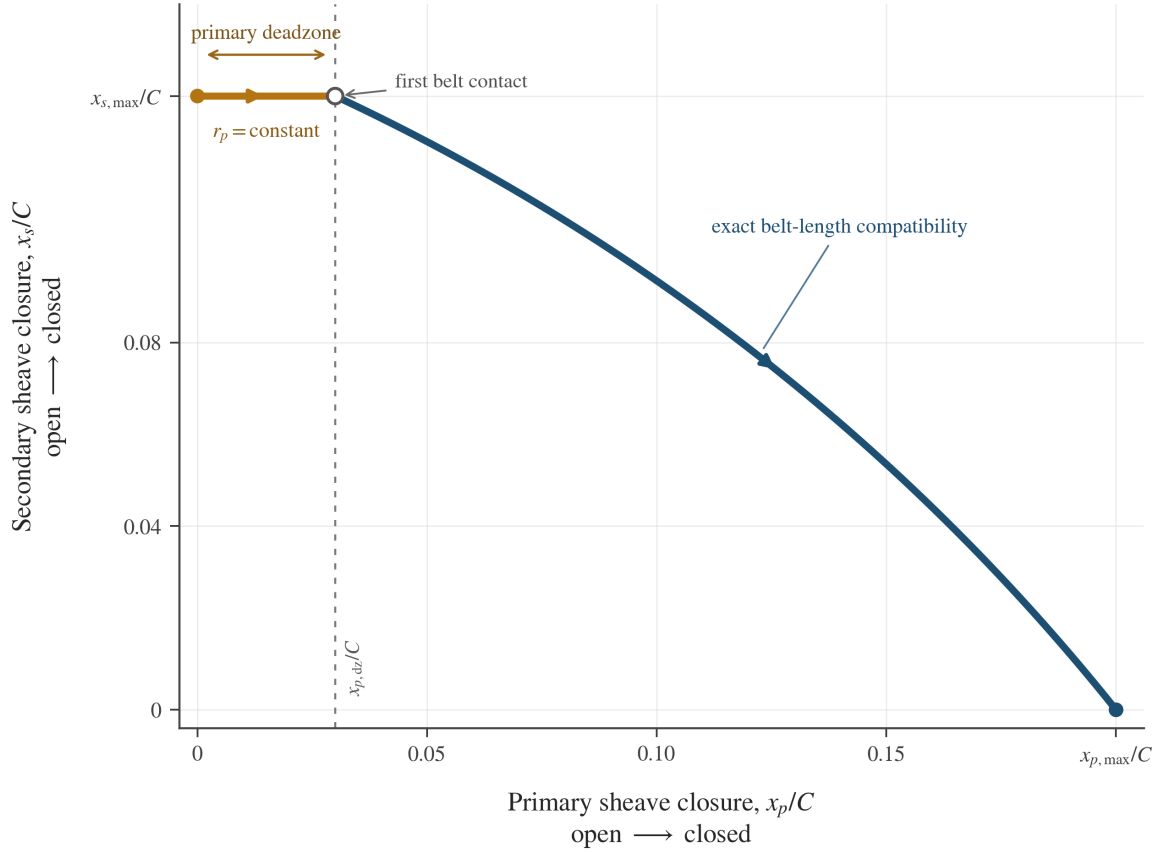


Figure 3.3: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance  $C$ , so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While  $0 \leq x_p < x_{p,dz}$ , the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases  $r_p$ . The fixed belt loop then requires  $r_s$  to decrease, which means that the secondary opens and  $x_s$  decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any  $x_p$  meets the curve once, giving one compatible  $x_s$ . At the deadzone height, however, a horizontal line meets every  $x_p$  between 0 and  $x_{p,dz}$ . Thus  $x_s$  is a unique function of  $x_p$  over the full travel, whereas  $x_p$  is not a unique function of  $x_s$ .

The primary coordinate is therefore adopted as the system shift coordinate. This is the same  $s$  included in the state vector of [Section 2.4.1](#):

**System shift coordinate**

**(3.1.25)**

$$s \equiv x_p, \quad x_p(s) = s, \quad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \leq s < x_{p,\text{dz}}, \\ \text{root}_{0 \leq x_s \leq x_{s,\max}} [\mathcal{R}_x(s, x_s) = 0], & x_{p,\text{dz}} \leq s \leq x_{p,\max}. \end{cases} \quad (3.1.26)$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give  $x_{s,\max}$  at  $s = x_{p,\text{dz}}$ , so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when  $s$  changes with time.

### 3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount  $ds$ , how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit  $s$ . Throughout this subsection, a prime denotes differentiation with respect to  $s$ . For either pulley  $j \in \{p, s\}$ , the corresponding time rates are then

$$\dot{x}_j = x'_j(s)\dot{s}, \quad \dot{r}_j = r'_j(s)\dot{s}. \quad (3.1.27)$$

**Primary radius rate** The primary side follows immediately from the definition  $x_p(s) = s$ :

$$x'_p(s) = 1, \quad \dot{x}_p = \dot{s}. \quad (3.1.28)$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating [Equation \(3.1.10\)](#) therefore gives

$$r'_p(s) = \begin{cases} 0, & 0 \leq s < x_{p,\text{dz}}, \\ \frac{1}{2 \tan \beta}, & x_{p,\text{dz}} < s \leq x_{p,\max}. \end{cases} \quad (3.1.29)$$

Multiplying this slope by  $\dot{s}$  gives



### Primary Radius Rate

(3.1.30)

$$\dot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

**Secondary coordinate and radius rates** The secondary requires one additional step. Its local pulley relation tells us how  $r_s$  changes when  $x_s$  changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\text{geo}}(r_p(s), r_s(s), C) = L_b. \quad (3.1.31)$$

The centre distance  $C$  is fixed by the pulley shafts, while  $L_b$  is constant under [Assumption 11 \(Constant Belt Length\)](#). A small change in  $s$  therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\text{geo}}}{ds} = 0. \quad (3.1.32)$$

Both radii change during this motion. The change caused by  $r_p$  is the sensitivity of the belt length to  $r_p$ , multiplied by the actual rate  $r'_p$ . The secondary contributes in the same way. Applying the chain rule to [Equation \(3.1.31\)](#) gives

$$\frac{\partial L_{\text{geo}}}{\partial r_p} r'_p + \frac{\partial L_{\text{geo}}}{\partial r_s} r'_s = 0. \quad (3.1.33)$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in [Equations \(3.1.13\)](#) and [\(3.1.16\)](#). Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{C} \right), \quad l = \sqrt{C^2 - (r_s - r_p)^2}, \quad (3.1.34)$$

Here,  $\delta$  sets the wrap angles through [Equation \(3.1.17\)](#), while  $l$  is the length of either straight span. Using these quantities, the exact path length from [Equation \(3.1.18\)](#) is

$$L_{\text{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \quad (3.1.35)$$

First, hold  $r_s$  and  $C$  fixed and increase  $r_p$ . The radius difference  $r_s - r_p$  then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \quad (3.1.36)$$

Using this result in the definitions of  $\delta$  and  $l$  gives

$$\begin{aligned} \frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(-\frac{1}{C}\right) \\ &= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} = -\frac{1}{l}, \end{aligned} \quad (3.1.37)$$

$$\begin{aligned} \frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [2(r_s - r_p)] \\ &= \frac{r_s - r_p}{l}. \end{aligned} \quad (3.1.38)$$

We can now differentiate the complete path length in [Equation \(3.1.35\)](#). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned} \frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2 \left[ (-1)\delta + (r_s - r_p) \left(-\frac{1}{l}\right) \right] + 2 \left( \frac{r_s - r_p}{l} \right) \\ &= \pi - 2\delta \\ &= \phi_p. \end{aligned} \quad (3.1.39)$$

The two terms containing  $(r_s - r_p)/l$  cancel. The remaining sensitivity is the primary wrap angle.

The secondary calculation follows the same path, but increasing  $r_s$  increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \quad (3.1.40)$$

The corresponding derivatives of  $\delta$  and  $l$  are

$$\begin{aligned}\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{1}{C}\right) \\ &= \frac{1}{l},\end{aligned}\tag{3.1.41}$$

$$\begin{aligned}\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [-2(r_s - r_p)] \\ &= -\frac{r_s - r_p}{l}.\end{aligned}\tag{3.1.42}$$

Differentiating the full path length with these signs gives

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2 \left[ (1)\delta + (r_s - r_p) \left(\frac{1}{l}\right) \right] + 2 \left( -\frac{r_s - r_p}{l} \right) \\ &= \pi + 2\delta \\ &= \phi_s.\end{aligned}\tag{3.1.43}$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \quad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s.\tag{3.1.44}$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition [Equation \(3.1.33\)](#), the two radial changes must therefore satisfy

$$\phi_p r'_p + \phi_s r'_s = 0.\tag{3.1.45}$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since  $x'_p = 1$ ,

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{1}{2 \tan \beta} x'_s.\tag{3.1.46}$$

Substituting these two slopes into [Equation \(3.1.45\)](#) gives

$$\phi_p \left( \frac{1}{2 \tan \beta} \right) + \phi_s \left( \frac{1}{2 \tan \beta} x'_s \right) = 0.\tag{3.1.47}$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x'_s = 0. \quad (3.1.48)$$

Solving for the slope of the secondary mapping gives

|                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Coordinate Slope</b></div> <div><b>(3.1.49)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $x'_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor  $x_s$  changes. After contact,  $x'_s < 0$ : primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in [Figure 3.3](#) is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by  $\dot{s}$ :

$$\dot{x}_s = x'_s(s)\dot{s} = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\dot{s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases} \quad (3.1.50)$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

|                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Radius Rate</b></div> <div><b>(3.1.51)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $\dot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The two pulley-radius rates are now determined by  $s$  and  $\dot{s}$ . However, the secondary multiplier  $-\phi_p/\phi_s$  changes as the radii and wrap angles change. Even if  $\dot{s}$  is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

### 3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in [Equations \(3.1.30\)](#) and [\(3.1.51\)](#) were obtained by multiplying the slope of each geometric mapping by the shift velocity  $\dot{s}$ . Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in [Equation \(3.1.49\)](#) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

**Primary radius acceleration** The primary coordinate requires no additional geometry. The system-coordinate definition in Equation (3.1.25) gives  $x_p(s) = s$ , so

$$x'_p(s) = 1, \quad x''_p(s) = 0, \quad (3.1.52)$$

and therefore

$$\dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.1.53)$$

The general shift-to-time relation in Equation (3.1.27) gives the primary radius rate in the form

$$\dot{r}_p = r'_p(s)\dot{s}. \quad (3.1.54)$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt} [r'_p(s)\dot{s}] \\ &= \frac{dr'_p}{dt}\dot{s} + r'_p(s)\ddot{s}. \end{aligned} \quad (3.1.55)$$

Because  $r'_p$  is itself a function of  $s$ ,

$$\frac{dr'_p}{dt} = r''_p(s)\dot{s}. \quad (3.1.56)$$

Substituting Equation (3.1.56) into Equation (3.1.55) gives

$$\ddot{r}_p = r''_p(s)\dot{s}^2 + r'_p(s)\ddot{s}. \quad (3.1.57)$$

Within either smooth branch of the primary radius map,  $r'_p(s)$  is constant. Hence

$$r''_p(s) = 0. \quad (3.1.58)$$

Using the primary slopes from Equation (3.1.29), the primary radius acceleration is

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div> <b>Primary Radius Acceleration</b> </div> <div> <b>(3.1.59)</b> </div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> <div style="text-align: center; margin-top: 20px;"> <math display="block">\ddot{r}_p = \begin{cases} 0, &amp; 0 \leq s &lt; x_{p,\text{dz}}, \\ \frac{\ddot{s}}{2 \tan \beta}, &amp; x_{p,\text{dz}} &lt; s \leq x_{p,\text{max}}. \end{cases}</math> </div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.29).

**Secondary coordinate curvature** The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.48):

$$\phi_p + \phi_s x'_s = 0.$$

The balance in Equation (3.1.48) contains the information needed for the second derivative. We will differentiate it once more with respect to  $s$ , which first requires the rates at which the two wrap angles change with  $s$ .

The required geometric quantities were defined in Equations (3.1.16) and (3.1.17). Written together, they are

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \quad \phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.60)$$

As  $s$  changes, both  $r_p$  and  $r_s$  change, so the radius difference inside  $\delta$  changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r'_s - r'_p. \quad (3.1.61)$$

Applying the chain rule to  $\delta$ , and using the straight-span length from Equation (3.1.13) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{r'_s - r'_p}{C}\right) \\ &= \frac{r'_s - r'_p}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \frac{r'_s - r'_p}{l}. \end{aligned} \quad (3.1.62)$$

After primary contact, the local pulley slopes are given by Equation (3.1.46):

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{x'_s}{2 \tan \beta}.$$

Substituting these slopes into Equation (3.1.62) gives

$$\begin{aligned} \delta' &= \frac{\frac{x'_s}{2 \tan \beta} - \frac{1}{2 \tan \beta}}{l} \\ &= \frac{x'_s - 1}{2l \tan \beta}. \end{aligned} \quad (3.1.63)$$

The wrap-angle derivatives now follow directly from Equation (3.1.60):

$$\phi'_p = -2\delta', \quad \phi'_s = 2\delta'. \quad (3.1.64)$$

We can now differentiate the secondary slope balance in Equation (3.1.48). The primary term contributes  $\phi'_p$ , while the secondary term requires the product rule because both  $\phi_s$  and  $x'_s$  change with  $s$ :

$$\begin{aligned} 0 &= \frac{d}{ds} (\phi_p + \phi_s x'_s) \\ &= \phi'_p + \phi'_s x'_s + \phi_s x''_s. \end{aligned} \quad (3.1.65)$$

Substituting the wrap-angle derivatives from Equation (3.1.64) gives

$$-2\delta' + 2\delta' x'_s + \phi_s x''_s = 0. \quad (3.1.66)$$

Solving first for  $x''_s$ ,

$$\begin{aligned} \phi_s x''_s &= 2\delta'(1 - x'_s), \\ x''_s &= \frac{2\delta'(1 - x'_s)}{\phi_s}. \end{aligned} \quad (3.1.67)$$

Using the expression for  $\delta'$  from Equation (3.1.63),

$$\begin{aligned} x''_s &= \frac{2}{\phi_s} \left( \frac{x'_s - 1}{2l \tan \beta} \right) (1 - x'_s) \\ &= -\frac{(1 - x'_s)^2}{\phi_s l \tan \beta}. \end{aligned} \quad (3.1.68)$$

On the engaged branch, the first-derivative result in Equation (3.1.49) gives  $x'_s = -\phi_p/\phi_s$ . The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x'_s &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \quad (3.1.69)$$

where  $\phi_p + \phi_s = 2\pi$  follows from Equation (3.1.17). Substituting this result into Equation (3.1.68) gives the curvature of the secondary coordinate mapping:

### Secondary Coordinate Curvature

(3.1.70)

$$x_s''(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{4\pi^2}{\phi_s^3 l \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The deadzone branch in Equation (3.1.26) is horizontal, so both  $x_s'$  and  $x_s''$  are zero there. On the engaged branch,  $x_s'' < 0$ : as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

**Secondary axial and radial accelerations** The curvature in Equation (3.1.70) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.50):

$$\dot{x}_s = x_s'(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt} [x_s'(s)\dot{s}] \\ &= \frac{dx_s'}{dt}\dot{s} + x_s'(s)\ddot{s}. \end{aligned} \tag{3.1.71}$$

Because  $x_s'$  changes only through  $s$ ,

$$\frac{dx_s'}{dt} = x_s''(s)\dot{s}. \tag{3.1.72}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x_s'(s)\ddot{s} + x_s''(s)\dot{s}^2. \tag{3.1.73}$$

Substituting the slope from Equation (3.1.49) and the curvature from Equation (3.1.70) gives

### Secondary Axial Acceleration

(3.1.74)

$$\ddot{x}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\ddot{s} - \frac{4\pi^2}{\phi_s^3 l \tan \beta}\dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant  $1/(2 \tan \beta)$ . Applying it to the secondary gives



$$\ddot{r}_s = \frac{1}{2 \tan \beta} \ddot{x}_s. \quad (3.1.75)$$

Substituting Equation (3.1.74) into Equation (3.1.75) gives

---

**Secondary Radius Acceleration** **(3.1.76)**

---

$$\ddot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\ddot{s}}{2 \tan \beta} - \frac{2\pi^2}{\phi_s^3 l \tan^2 \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.49) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.70): even when  $\ddot{s} = 0$ , moving through the engaged branch changes  $-\phi_p/\phi_s$ , so the secondary velocity continues to change.

**Remark (Deadzone Boundary)**

The idealized maps in Equations (3.1.10) and (3.1.26) change slope when the primary first reaches the belt at  $s = x_{p,dz}$ . Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.59) and (3.1.76) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

**Remark (Practical Consequences of the Pulley Geometry)**

The radius, rate, and acceleration relations derived in this section describe how the pulley geometry evolves as the CVT shifts. Appendix D explores the practical consequences of these results in more detail. In particular, it examines the resulting geometric CVT ratio and ratio-change rate across the shift travel, together with how the current pulley geometry and sheave angle shape the available ratio range and ratio-change sensitivity.

### 3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in Equation (3.1.25) sets the primary position directly, while Equation (3.1.26) gives the unique compatible secondary position. The local pulley maps in Equations (3.1.9) and (3.1.10) convert those two positions into pulley radii. Finally, the rate and acceleration relations in Equations (3.1.30), (3.1.51), (3.1.59) and (3.1.76) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \longmapsto (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s).$$

These results follow from the geometric representation used throughout the section. The belt cross-section is fixed under Assumption 10 (Fixed Belt Cross-Section), the belt length is fixed under Assumption 11 (Constant Belt Length), and Assumption 13 (Uniform Contact Radius Around Each Wrap) represents the belt path on each pulley by a single radius over its wrap. The centre distance

is established once by [Equation \(3.1.21\)](#) and then remains fixed by the pulley-shaft spacing. Under these conditions, the two wrapped arcs and straight tangent spans in [Figure 3.2](#) can form only one modeled belt path for each admissible value of  $s$ .

**Relation to prior transient models** These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

[Duan et al. \(2016\)](#) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through [Assumption 11 \(Constant Belt Length\)](#) and uses one contact radius over each pulley through [Assumption 13 \(Uniform Contact Radius Around Each Wrap\)](#). Consequently, elastic path-length change and deformation-induced variation of the local contact radius are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of [Sorge \(2008\)](#) and [Cammalleri and Sorge \(2009\)](#), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of [Juli'o and Plante \(2011\)](#) and [Ballew \(2015\)](#), instead obtain the evolving belt geometry from the motion and contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in [Equation \(3.1.19\)](#) to find the representative pulley-radius pair and its derivatives directly from  $s$ ,  $\dot{s}$ , and  $\ddot{s}$ . The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

## 3.2 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate  $s(t)$  changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory ?? (??)) gives

$$m \ddot{s} = F_{p,ax} - F_{s,ax}, \quad (3.2.1)$$

where  $F_{p,ax}$  and  $F_{s,ax}$  denote the axial forces generated by the primary and secondary mechanisms, respectively, and  $m$  is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for  $F_{p,ax}$  and  $F_{s,ax}$  as functions of the current system state. Once these force models are established, the shift acceleration  $\ddot{s}$  follows directly, completing the dynamic description of axial motion.

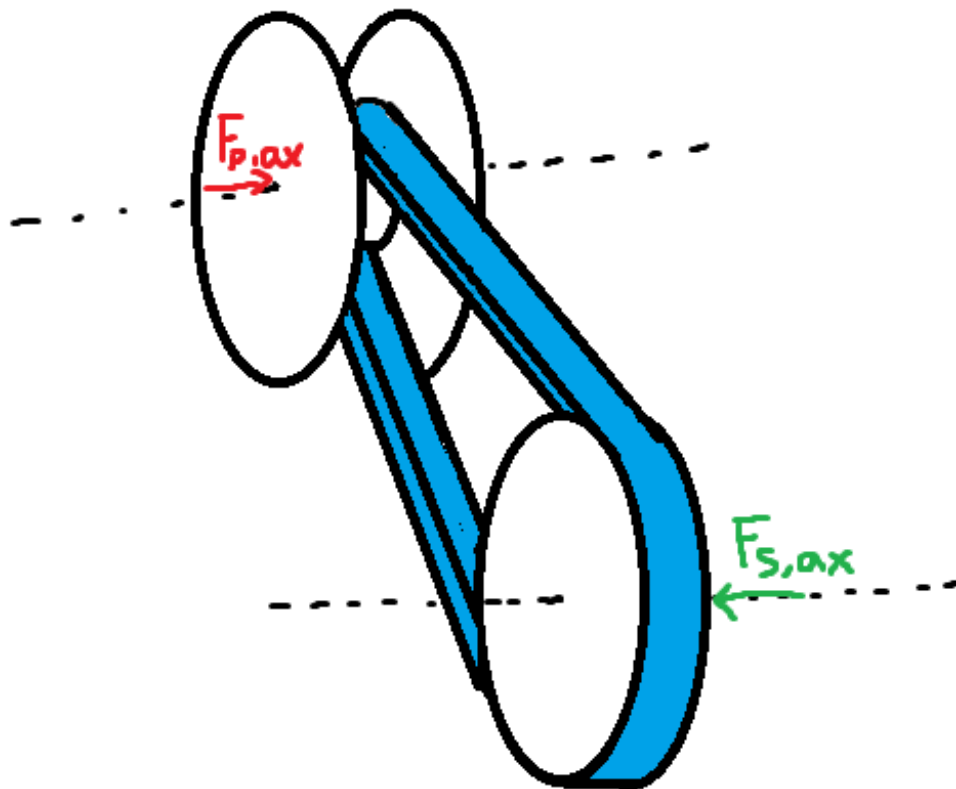


Figure 3.4: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration  $\ddot{s}$ .

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

### 3.2.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate  $s$  evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate  $s$ . Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

**Primary Ramp Geometry (Axial-to-Radial Mapping)** On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see [Figure 3.5](#)). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \quad (3.2.2)$$

where  $r_f(s)$  is the effective flyweight radius.

An incremental axial displacement  $ds$  produces a radial displacement  $dr_f$ . The local slope of the ramp is therefore  $dr_f/ds$ . The instantaneous primary ramp angle is denoted by  $\alpha_p(s)$  and is defined with respect to the axial direction as

$$\tan \alpha_p(s) = \frac{dr_f}{ds}, \quad \alpha_p(s) = \tan^{-1} \left( \frac{dr_f}{ds} \right). \quad (3.2.3)$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger  $dr_f/ds$ ) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

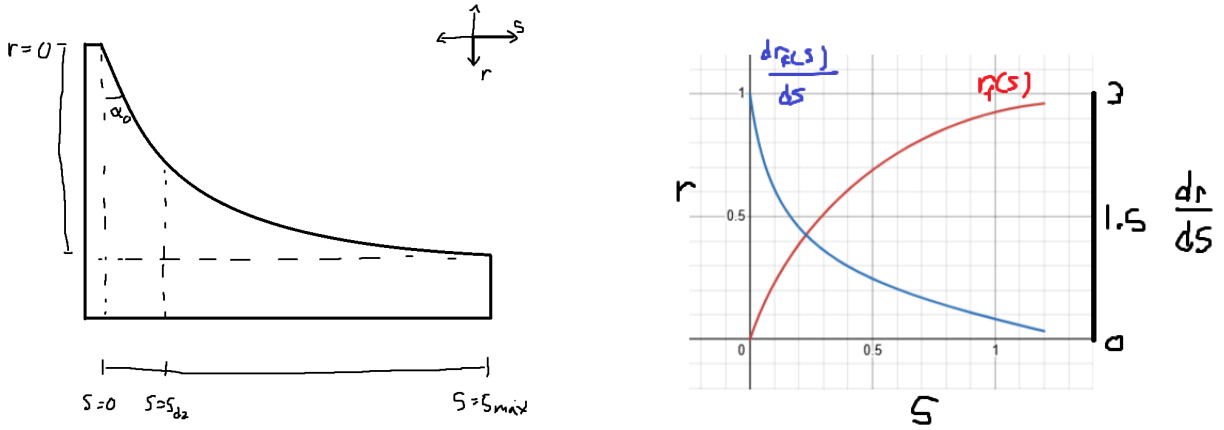


Figure 3.5: Primary ramp geometry. **Left:** As axial displacement  $s$  increases, the flyweight slides along the ramp, producing radial change  $dr_f$ . The instantaneous ramp angle satisfies  $\tan \alpha_p = dr_f/ds$ . **Right:** Example profiles of  $r_f(s)$  and its slope  $dr_f/ds$ .

**Primary Admissibility Conditions** In order to be a valid primary ramp profile  $r_f(s)$ , it must satisfy:

1.  $r_f(s) \in C^1([0, s_{\max}])$  (continuously differentiable),
2.  $\frac{dr_f}{ds} \geq 0$  over  $[0, s_{\max}]$ ,
3.  $\frac{dr_f}{ds}$  is finite over  $[0, s_{\max}]$ .

Condition (1) ensures the ramp angle  $\alpha_p(s)$  is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below  $\pi/2$  and the axial gain  $\tan \alpha_p$  remains finite.

**Secondary Helix Geometry (Axial-to-Rotation Mapping)** On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see [Figure 3.6](#)).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \quad (3.2.4)$$

where  $\theta_s(s)$  is the helix-induced rotation.

An incremental rotation  $d\theta_s$  corresponds to a circumferential displacement  $r_h d\theta_s$  along a cylindrical surface of effective radius  $r_h$ . The instantaneous secondary helix angle is denoted by  $\alpha_s(s)$  and is defined from the local slope of the ramp surface:

$$\tan \alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan \alpha_s(s) = \frac{1}{r_h \frac{d\theta_s}{ds}}. \quad (3.2.5)$$

Thus, a shallower helix (smaller  $\alpha_s$ ) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

#### Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by  $\tan \alpha_p = dr_f/ds$ .

The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition  $ds/(r_h d\theta_s)$  and the inverse relationship in [Equation \(3.2.5\)](#).

In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

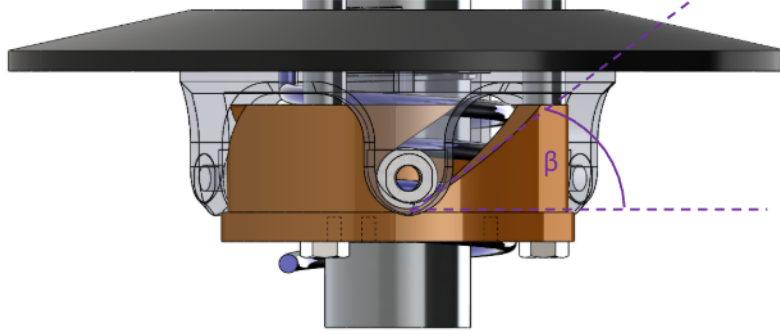


Figure 3.6: Secondary helix geometry. Axial travel  $s$  induces rotation  $\theta_s(s)$  around a cylindrical surface of radius  $r_h$ . The instantaneous helix angle satisfies  $\tan \alpha_s = ds/(r_h d\theta_s)$ .

**Secondary Admissibility Conditions** The helix rotation mapping  $\theta_s(s)$  must satisfy:

1.  $\theta_s(s) \in C^1([0, s_{\max}])$  (continuously differentiable),
2.  $\frac{d\theta_s}{ds} \geq 0$  over  $[0, s_{\max}]$ ,
3.  $\frac{d\theta_s}{ds}$  is finite over  $[0, s_{\max}]$ .

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below  $\pi/2$  and the geometric gain remains finite.

#### Remark (Piecewise Profiles in Practice)

The  $C^1$  requirement above is sufficient for all derivations in this document. In practice, piecewise  $C^1$  profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

### 3.2.2 Primary Axial Force Model

The primary axial clamping force arises from the competition between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 3.7, each flyweight of mass  $m_f$  rotates with angular speed  $\omega_p$  at an instantaneous radius  $r_{f,0} + r_f(s)$ , where  $r_{f,0}$  is the flyweight radius in the reference configuration  $s = 0$  and  $r_f(s)$

is the additional radial displacement induced by the ramp. The resulting outward centrifugal force is redirected by the ramp geometry described in [Section 3.2.1](#) into an axial closing force on the movable primary sheave.

Opposing this motion is the primary compression spring, characterized by stiffness  $k_p$  and preload  $x_{p,0}$ . As the sheaves close ( $s$  increases), the spring compression increases, producing a larger restoring force that resists further closure.

The resulting primary axial force is therefore the ramp-redirectioned centrifugal contribution minus the spring force. This quantity, denoted  $F_{p,ax}(s, \omega_p)$ , provides the primary-side contribution to the generalized axial force used in the shift equation of motion.

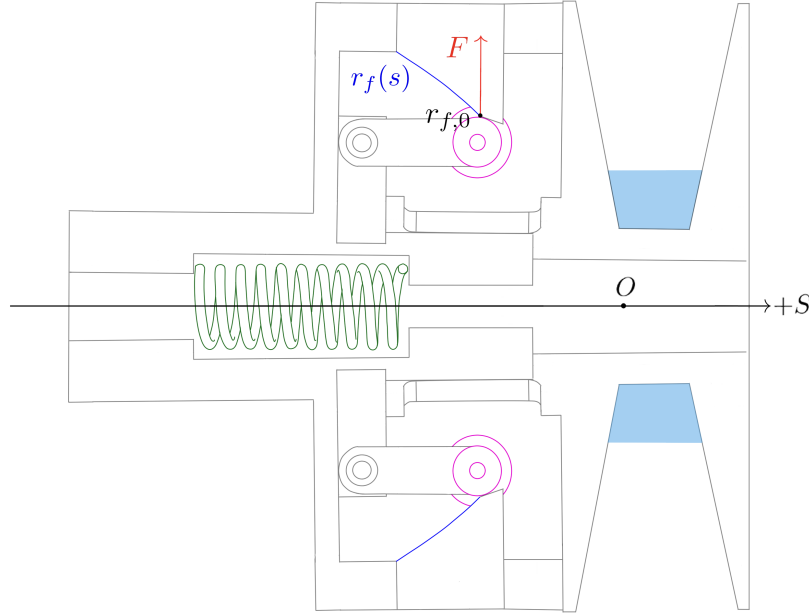


Figure 3.7: Primary CVT axial-force mechanism. Each flyweight of mass  $m_f$  rotates at angular speed  $\omega_p$  and experiences an outward centrifugal force at total radius  $r_{f,0} + r_f(s)$ . The ramp geometry redirects this radial loading into an axial closing force on the movable primary sheave in the  $+s$  direction, while the primary compression spring ( $k_p, x_{p,0}$ ) provides an opposing restoring force. The lower flyweight–ramp pair is symmetric and shown unlabeled for clarity.

**Centrifugal Loading of Flyweights** From Theory ?? (??),

$$F_c = m\omega^2 r.$$

Let

- $m_f$  denote the flyweight mass,
- $r_{f,0}$  denote the initial flyweight radius at  $s = 0$ ,
- $r_f(s)$  denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\text{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)). \quad (3.2.6)$$

**Radial-to-Axial Projection** The ramp redirects radial force into axial force. As shown in [Figure 3.8](#), the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \quad (3.2.7)$$

From [Equation \(3.2.3\)](#), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds}. \quad (3.2.8)$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.



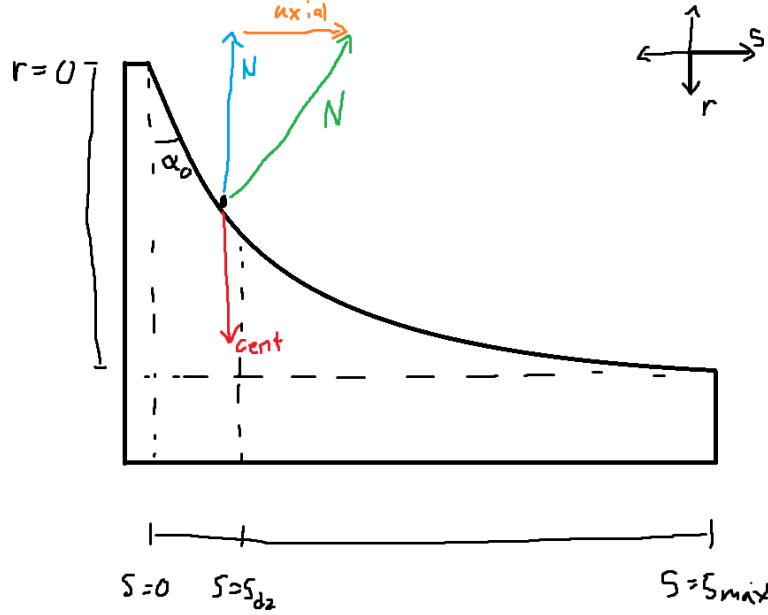


Figure 3.8: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of  $\tan \alpha_p$ , which equals  $dr_f/ds$  by definition of the ramp slope.

**Primary Compression Spring** From Theory ?? (??),

$$F = k\Delta x.$$

Let  $k_p$  denote primary stiffness and  $x_{p,0}$  preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \quad (3.2.9)$$

**Net Primary Axial Force** The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

|                            |                 |
|----------------------------|-----------------|
| <b>Primary Axial Force</b> | <b>(3.2.10)</b> |
|----------------------------|-----------------|

$$F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

### 3.2.3 Secondary Axial Force Model

The secondary pulley generates axial clamping force in two ways. First, the secondary spring directly presses the sheaves together through axial compression. Second, the helix mechanism converts

torque carried by the secondary pulley into additional axial force. The secondary is therefore torque-reactive: larger transmitted torque can increase the clamping force applied to the belt.

The goal of this section is to derive the net secondary axial force  $F_{s,ax}$ . The derivation follows the physical mechanism directly: determine the torque reacted through the helix, convert that reacted torque into axial force through the helix geometry, and then add the direct compression spring force.

The secondary assembly consists of a fixed sheave, a movable sheave, a helical slot between them, and a spring. The fixed sheave is attached to the secondary shaft. The movable sheave can translate axially, but the helix constrains that translation to occur with relative rotation between the movable and fixed sheaves. Under **Assumption 2 (Rigid Structural Components)**, this coupled motion is treated as a rigid geometric constraint.

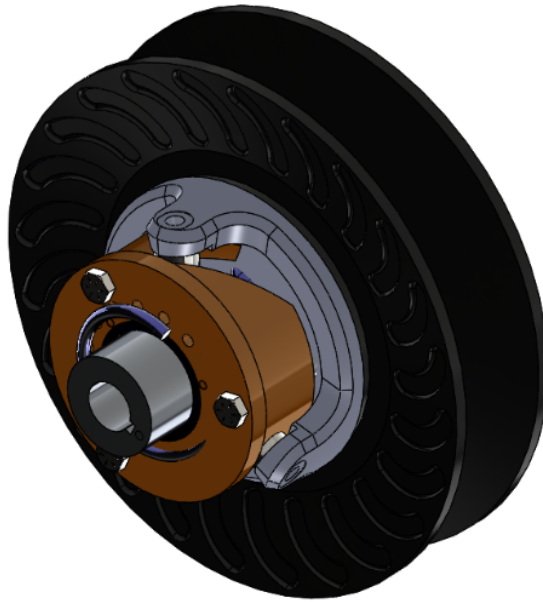


Figure 3.9: Secondary CVT assembly showing the helix mechanism and spring. The helix is modeled as a captured slot, so axial displacement of the movable sheave is coupled to relative rotation between the movable and fixed sheaves.

### Physical Role of the Helix

The helix acts like a screw-like constraint between the fixed and movable secondary sheaves. A screw produces axial force only when relative rotation is resisted. The same idea applies here: torque applied to the movable sheave tends to rotate it relative to the fixed sheave. Because the movable sheave is captured by the helix, that relative rotation is tied to axial motion. The helix therefore reacts torque, and the inclined helix path converts that reacted torque into axial force.

This gives the structure of the force model. The helix contribution is not determined by secondary speed alone. It is determined by the torque that the movable sheave attempts to pass through the helix, after accounting for the torque required to accelerate the movable sheave itself.

Let  $\tau_s$  denote the total signed torque transmitted from the belt to the secondary pulley. Under

**Assumption 8 (Equal Secondary Sheave-Face Torque Partition)**, this torque is treated as equally shared between the fixed and movable secondary sheave faces. The belt-applied torque acting on the movable sheave is therefore

$$\tau_{b,M} = \frac{\tau_s}{2}. \quad (3.2.11)$$

This is the portion of the belt torque that appears on the movable-sheave free body and can therefore load the helix.

### Movable-Sheave Kinematics

The helix geometry was parameterized earlier by the relative rotation mapping

$$\theta_s = \theta_s(s), \quad (3.2.12)$$

as introduced in **Equation (3.2.4)**. Here,  $\theta_s$  is the relative rotation between the movable and fixed secondary sheaves.

The fixed sheave is attached to the secondary shaft, so its angular velocity is

$$\omega_F = \omega_s. \quad (3.2.13)$$

The movable sheave does not generally rotate at this same speed while the CVT is shifting. With the sign convention used here, positive helix rotation means that the movable sheave lags the fixed sheave, so

$$\dot{\theta}_s = \omega_s - \omega_M. \quad (3.2.14)$$

Since the helix angle is a function of the shift coordinate,

$$\dot{\theta}_s = \frac{d\theta_s}{ds} \dot{s}. \quad (3.2.15)$$

Combining these two relations gives the angular velocity of the movable sheave:

$$\omega_M = \omega_s - \frac{d\theta_s}{ds} \dot{s}. \quad (3.2.16)$$

Thus, secondary shift does not only move the sheave axially. It also changes the movable sheave's angular speed relative to the fixed sheave by an amount set by the helix slope.

Differentiating **Equation (3.2.16)** gives

$$\dot{\omega}_M = \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s}. \quad (3.2.17)$$

This acceleration is needed because torque applied to the movable sheave has two possible roles: it can be reacted through the helix, or it can angularly accelerate the movable sheave.

## Torque Reacted Through the Helix

The movable sheave is now treated as a rotating rigid body. Under [Assumption 14 \(Lumped Inertia Representation\)](#), its inertia about the secondary shaft is represented by the scalar  $I_M$ .

The torques acting on the movable sheave are the belt-applied torque  $\tau_s/2$ , the torsional spring torque, and the torque applied by the helix. The secondary spring is twisted by the same relative sheave rotation that is imposed by the helix. With torsional stiffness  $k_{s,\theta}$  and preload deflection  $\theta_{s,0}$ , the spring torque is

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}(\theta_{s,0} + \theta_s(s)). \quad (3.2.18)$$

Let  $\tau_{\text{helix} \rightarrow M}$  denote the torque applied by the helix to the movable sheave. Applying [Theory ??](#) to the movable sheave gives

$$\frac{\tau_s}{2} + \tau_{s,\text{spring}} + \tau_{\text{helix} \rightarrow M} = I_M \dot{\omega}_M. \quad (3.2.19)$$

The quantity needed for axial clamping is the equal and opposite torque applied by the movable sheave into the helix:

$$\tau_{M \rightarrow \text{helix}} = -\tau_{\text{helix} \rightarrow M}. \quad (3.2.20)$$

Solving the movable-sheave torque balance for this reacted torque gives

$$\tau_{M \rightarrow \text{helix}} = \frac{\tau_s}{2} + \tau_{s,\text{spring}} - I_M \dot{\omega}_M. \quad (3.2.21)$$

Substituting [Equation \(3.2.18\)](#) and [Equation \(3.2.17\)](#) gives the reacted helix torque in terms of the state and acceleration variables:

$$\tau_{M \rightarrow \text{helix}} = \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left[ \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right]. \quad (3.2.22)$$

This equation is the core of the secondary mechanism. The helix reacts the movable-face belt torque and the torsional spring torque, except for the portion of torque required to angularly accelerate the movable sheave.

## Conversion from Reacted Torque to Axial Force

The remaining step is geometric. The helix relates axial displacement to relative rotation. For a virtual axial displacement  $\delta s$ , the corresponding relative rotation is

$$\delta\theta_s = \frac{d\theta_s}{ds} \delta s.$$

By virtual work, the axial force associated with the reacted helix torque is

$$F_{s,\text{helix}} = \tau_{M \rightarrow \text{helix}} \frac{d\theta_s}{ds}. \quad (3.2.23)$$

A steeper torque-to-shift mapping therefore produces a larger axial force from the same reacted torque.

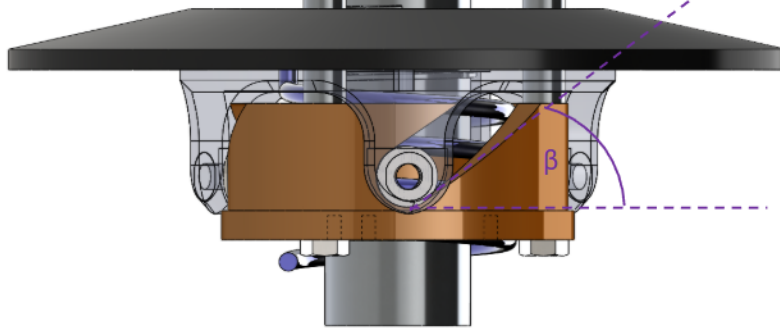


Figure 3.10: Helix free-body concept. Torque applied to the movable sheave is reacted through the helix. Because the helix is inclined, the reacted torque produces an axial force proportional to  $d\theta_s/ds$ .

Substituting Equation (3.2.22) into Equation (3.2.23) gives

$$F_{s,\text{helix}} = \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right) \right]. \quad (3.2.24)$$

The terms inside the brackets are the torque that the helix must react. The factor multiplying the brackets is the helix slope that converts that reacted torque into axial force.

### Direct Compression Spring Force

The secondary spring also contributes directly through axial compression. At  $s = 0$ , the spring is pre-compressed by  $x_{s,0}$ . As the secondary sheaves open with increasing shift, the compression increases by  $s$ . From Theory ?? (??), the direct axial spring force is

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \quad (3.2.25)$$

This force exists even if no torque is being transmitted. It provides the baseline secondary clamping contribution, while the helix contribution adds the torque-reactive part.

### Net Secondary Axial Force

The net secondary axial force is the sum of the direct compression-spring force and the helix-generated force:

**Secondary Axial Force****(3.2.26)**

$$F_{s,\text{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right) \right].$$

This form mirrors the physical mechanism. The first term is the direct axial spring force. The second term is the helix force: the torque reacted through the movable sheave and spring is converted into axial force by the helix slope. The inertial term appears because part of the movable-sheave torque is used to accelerate the movable sheave rather than being reacted through the helix.

The secondary force therefore increases with axial spring compression, torsional spring wind-up, and transmitted secondary torque. It is also affected by the motion of the secondary sheave, because the helix couples axial shift to relative rotation. This is why the secondary is a torque-reactive clamping mechanism rather than a simple position-dependent spring.

**3.2.4 Belt Centrifugal Force on the Wrap**

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this section, let

$$j \in \{p, s\}$$

denote either pulley, where  $j = p$  corresponds to the primary and  $j = s$  corresponds to the secondary.

From Theory ?? (??), a mass  $m$  rotating at angular velocity  $\omega$  at radius  $r$  experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around pulley  $j$ , three quantities must be defined:

1. the radius at which the belt mass rotates,
2. the mass of belt material carried around the wrap,
3. the angular velocity of that belt mass.

Each is developed below.

**1. Radius at Which the Belt Mass Rotates** Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner belt surface. The location of this center of mass therefore determines the radius that must be used in the centrifugal force calculation.

The belt cross-section is approximated as a trapezoid (see [Figure 3.11](#)) with:

- radial thickness  $h_b$ ,
- outer width  $b_{\text{out}}$ ,
- inner width  $b_{\text{in}}$ .

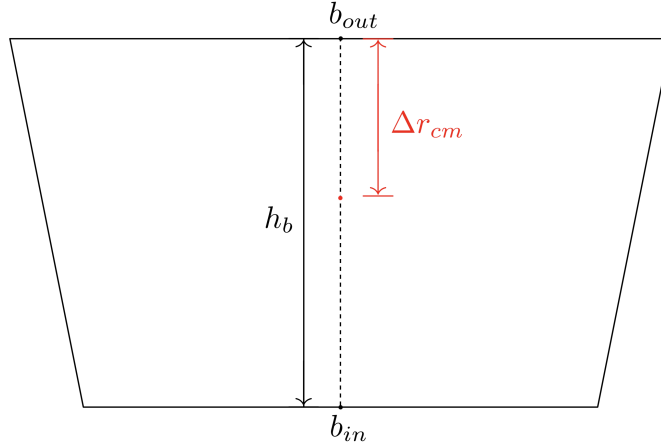


Figure 3.11: Trapezoidal belt cross-section. The outer width  $b_{\text{out}}$  corresponds to the surface away from the pulley axis; the inner width  $b_{\text{in}}$  corresponds to the surface closer to the axis. The radial thickness is  $h_b$  and the centroid is located at  $\Delta r_{\text{cm}}$  from the outer surface.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}). \quad (3.2.27)$$

Let  $y$  denote radial distance measured inward from the outer surface, so that  $y = 0$  at the outer surface and  $y = h_b$  at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with  $y$ :

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \quad (3.2.28)$$

The centroid location  $\Delta r_{\text{cm}}$ , measured inward from the outer surface, follows from the area-centroid definition:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y b(y) \, dy. \quad (3.2.29)$$

Substituting Equation (3.2.28) into Equation (3.2.29) gives

$$\begin{aligned}
\int_0^{h_b} y b(y) \, dy &= \int_0^{h_b} y \left( b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right) \, dy \\
&= \int_0^{h_b} b_{\text{out}} y \, dy + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2 \, dy \\
&= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} (b_{\text{in}} - b_{\text{out}}) \\
&= \frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}}). \tag{3.2.30}
\end{aligned}$$

Dividing by the area from Equation (3.2.27) yields

$$\Delta r_{\text{cm}} = h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \tag{3.2.31}$$

For pulley  $j$ , let  $r_{j,\text{out}}(s)$  denote the outer belt radius. The radius at which the belt mass rotates is therefore

$$r_{j,\text{cm}}(s) = r_{j,\text{out}}(s) - \Delta r_{\text{cm}}. \tag{3.2.32}$$

**2. Mass of Belt Material on the Wrap** To determine the centrifugal loading on pulley  $j$ , we next determine the amount of belt mass associated with an infinitesimal portion of its wrap.

Let  $\theta$  denote the angular coordinate measured from the center of the wrap, so that the wrapped region on pulley  $j$  spans

$$\theta \in \left[ -\frac{\phi_j}{2}, \frac{\phi_j}{2} \right],$$

where  $\phi_j(s)$  is the total wrap angle on pulley  $j$  (see Figure 3.12).



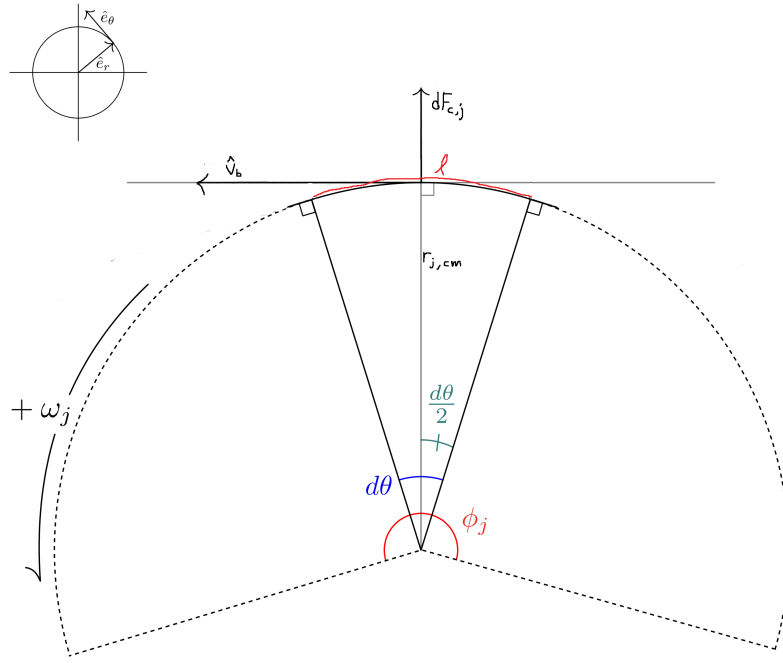


Figure 3.12: Belt wrapped around pulley  $j$ . A differential element at angle  $\theta$  has arc length  $d\ell = r_{j,\text{cm}} d\theta$  and experiences radially outward centrifugal force.

A differential belt element subtending angle  $d\theta$  at the mass-centroid radius  $r_{j,\text{cm}}$  has arc length

$$d\ell = r_{j,\text{cm}} d\theta.$$

Since the belt has cross-sectional area  $A_b$ , the differential belt element has volume

$$dV_j = A_b d\ell. \quad (3.2.33)$$

Substituting  $d\ell = r_{j,\text{cm}} d\theta$  gives

$$dV_j = A_b r_{j,\text{cm}} d\theta. \quad (3.2.34)$$

Let  $\rho_b$  denote the belt material density ( $\text{kg}/\text{m}^3$ ). The corresponding differential mass is then

$$\begin{aligned} dm_j &= \rho_b dV_j \\ &= \rho_b A_b r_{j,\text{cm}} d\theta. \end{aligned} \quad (3.2.35)$$

This gives the amount of belt material associated with an infinitesimal portion of the wrap on pulley  $j$ .

**3. Angular Velocity of the Belt Mass** The rotating belt mass on pulley  $j$  must also be assigned an angular speed. From [Section 2.4](#),  $v_b$  denotes the belt transport speed. The corresponding angular velocity of the belt mass centroid on pulley  $j$  is then

$$\omega_{b,j} = \frac{v_b}{r_{j,\text{cm}}(s)}. \quad (3.2.36)$$

This relation expresses the fact that the same belt transport speed corresponds to different local angular velocities depending on the radius of motion.

**Distributed Centrifugal Load Along the Wrap** With the radius, differential mass, and angular velocity now defined, the centrifugal force on the differential belt element follows from [Theory ?? \(??\)](#):

$$\begin{aligned} dF_{c,j} &= dm_j \omega_{b,j}^2 r_{j,\text{cm}} \\ &= (\rho_b A_b r_{j,\text{cm}} d\theta) \left( \frac{v_b}{r_{j,\text{cm}}} \right)^2 r_{j,\text{cm}} \\ &= \rho_b A_b v_b^2 d\theta. \end{aligned} \quad (3.2.37)$$

This force acts radially outward at each point along the wrap.

**Total Radial Force Over the Wrap** Integrating [Equation \(3.2.37\)](#) over the full wrap angle gives the total radially-directed centrifugal force on pulley  $j$ :

$$\begin{aligned} F_{c,j,\text{rad}} &= \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b v_b^2 d\theta \\ &= \rho_b A_b v_b^2 \int_{-\phi_j/2}^{\phi_j/2} d\theta \\ &= \rho_b A_b v_b^2 \phi_j(s). \end{aligned} \quad (3.2.38)$$

**Radial-to-Axial Conversion Through the Sheave Cone** The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle  $\beta$ , radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see [??](#)), the relationship between radial and axial displacement in the active shift region is

$$\frac{dr}{ds} = \frac{1}{2 \tan \beta}.$$

For an infinitesimal displacement,

$$\delta W = F_{\text{rad}} \, dr = F_{\text{ax}} \, ds.$$

Substituting  $dr = \frac{1}{2 \tan \beta} ds$  gives

$$F_{\text{rad}} \frac{1}{2 \tan \beta} ds = F_{\text{ax}} \, ds.$$

Canceling  $ds$  yields

$$F_{\text{ax}} = \frac{F_{\text{rad}}}{2 \tan \beta}. \quad (3.2.39)$$

Thus, the axial clamping contribution from belt centrifugal force on pulley  $j$  is

$$F_{c,j} = \frac{\rho_b A_b v_b^2 \phi_j(s)}{2 \tan \beta}. \quad (3.2.40)$$

### 3.2.5 Axial Shift Dynamics

The previous subsections derived the axial forces produced by the primary and secondary pulley mechanisms. The remaining task is to use those forces to determine the motion of the shift coordinate  $s(t)$ .

The shift coordinate represents coordinated motion of several physical bodies. As  $s$  increases, the primary movable sheave closes, the secondary movable sheave opens through the belt-length constraint, and belt material migrates through the pulley grooves. The axial equation of motion is therefore written in the generalized coordinate  $s$ , with all force and inertia terms expressed in that coordinate.

From the axial force balance introduced in [Equation \(3.2.1\)](#),

$$m \ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}, \quad (3.2.41)$$

where  $F_{p,\text{ax}}$  acts in the positive  $s$  direction and  $F_{s,\text{ax}}$  opposes it. This section replaces the placeholder mass  $m$  with the equivalent inertia associated with the shift coordinate and then adds the axial belt centrifugal contributions derived in [Section 3.2.4](#).

#### Equivalent Inertia in the Shift Coordinate

Since  $s$  is a generalized coordinate, the inertia associated with shift is constructed from kinetic energy. Following Theory ?? (??), the translational kinetic energy associated with shift is written as

$$T_{\text{shift}} = \frac{1}{2} m_{\text{eq}} \dot{s}^2.$$

The movable primary sheave translates with speed  $\dot{s}$ . The movable secondary sheave translates with the same speed magnitude in the opposite physical direction, so its kinetic energy also contributes to

the same generalized coordinate. Let  $m_{p,m}$  and  $m_{s,m}$  denote the translating masses of the primary and secondary movable sheaves.

The belt also contributes to the shift inertia. Under **Assumption 11 (Constant Belt Length)**, the total belt mass is constant and is

$$m_b = \rho_b A_b L_b, \quad (3.2.42)$$

where  $A_b$  is the trapezoidal belt cross-sectional area from **Equation (3.2.27)**. Under **Assumption 12 (Single Belt Transport Speed)**, the belt is represented by a single lumped transport speed, while the simplified axial contribution associated with radial migration through the groove is represented by the effective axial speed

$$\dot{s}_b = \frac{\dot{s}}{2}. \quad (3.2.43)$$

The belt kinetic energy associated with this axial component is therefore

$$T_b = \frac{1}{2} m_b \left( \frac{\dot{s}}{2} \right)^2 = \frac{1}{2} \left( \frac{m_b}{4} \right) \dot{s}^2.$$

Thus, the base equivalent mass associated with translation in the shift coordinate is

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \quad (3.2.44)$$

This mass accounts for the translating sheaves and the simplified axial belt motion. The rotational inertia coupled through the secondary helix is collected separately below.

### Secondary Force Split

The secondary axial force derived in **Equation (3.2.26)** contains the movable-sheave angular acceleration. This occurs because the helix couples axial shift to relative sheave rotation. Written as a function of the quantities that enter the shift equation,

$$F_{s,\text{ax}} = F_{s,\text{ax}}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}). \quad (3.2.45)$$

Recalling **Equation (3.2.26)**, with the movable-sheave torque contribution written using **Assumption 8 (Equal Secondary Sheave-Face Torque Partition)**,

$$F_{s,\text{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right) \right]. \quad (3.2.46)$$

To place this force into the shift equation, separate the part that does not multiply  $\ddot{s}$  from the part that does:

$$F_{s,\text{ax}} = \hat{F}_{s,\text{ax}} + I_M \left( \frac{d\theta_s}{ds} \right)^2 \ddot{s}. \quad (3.2.47)$$

The acceleration-independent part is

$$\hat{F}_{s,\text{ax}} = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]. \quad (3.2.48)$$

This split is purely algebraic. It keeps the same secondary force model, but makes clear that the movable-sheave rotational inertia appears in the shift coordinate as the reflected inertia term

$$I_M \left( \frac{d\theta_s}{ds} \right)^2.$$

### Generalized Axial Force Balance

The generalized force in the positive  $s$  direction is obtained by collecting the primary-side contributions that drive upshift and subtracting the secondary-side contributions that resist it. The forces entering this balance are:

- the primary mechanism force  $F_{p,\text{ax}}(s, \omega_p)$  from Equation (3.2.10),
- the secondary mechanism force  $F_{s,\text{ax}}$  from Equation (3.2.26),
- the primary belt centrifugal contribution  $F_{c,p}(s, v_b)$  from Equation (3.2.40),
- the secondary belt centrifugal contribution  $F_{c,s}(s, v_b)$  from Equation (3.2.40).

With the sign convention of  $s$ , the generalized force balance is

$$\sum F_s = (F_{p,\text{ax}} + F_{c,p}) - (F_{s,\text{ax}} + F_{c,s}). \quad (3.2.49)$$

Substituting the secondary force split from Equation (3.2.47) gives

$$m_{\text{eq}} \ddot{s} = (F_{p,\text{ax}} + F_{c,p}) - \left( \hat{F}_{s,\text{ax}} + I_M \left( \frac{d\theta_s}{ds} \right)^2 \ddot{s} + F_{c,s} \right). \quad (3.2.50)$$

Collecting the terms proportional to  $\ddot{s}$  gives

$$\left[ m_{\text{eq}} + I_M \left( \frac{d\theta_s}{ds} \right)^2 \right] \ddot{s} = (F_{p,\text{ax}} + F_{c,p}) - (\hat{F}_{s,\text{ax}} + F_{c,s}). \quad (3.2.51)$$

This form has the expected physical structure. The numerator is the net generalized force trying to move the CVT through its shift range. The bracketed factor on the left is the effective inertia associated with that motion: the translating components and the movable secondary sheave inertia reflected through the helix.

### Explicit Shift Acceleration

Substituting the equivalent mass from Equation (3.2.44) into Equation (3.2.51) gives the explicit shift acceleration:

$$\ddot{s} = \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\right) - \left(\widehat{F}_{s,\text{ax}}(s, \dot{s}, \tau_s, \dot{\omega}_s) + F_{c,s}(s, v_b)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4} + I_M \left(\frac{d\theta_s}{ds}\right)^2}.$$

where

$$\widehat{F}_{s,\text{ax}}(s, \dot{s}, \tau_s, \dot{\omega}_s) = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]. \quad (3.2.53)$$

The shift acceleration therefore follows the same structure as any second-order force balance: net generalized force divided by effective inertia. The primary mechanism and primary belt centrifugal contribution drive positive shift, while the secondary mechanism and secondary belt centrifugal contribution oppose it. The secondary term depends on the transmitted secondary torque  $\tau_s$  and secondary angular acceleration  $\dot{\omega}_s$ , so the shift motion remains coupled to the rotational torque-transfer dynamics.

### 3.2.6 Section Summary and Forward Outlook

This section derived the axial force models for the primary and secondary pulley mechanisms and assembled them into an equation of motion for the generalized shift coordinate  $s(t)$ .

The resulting shift equation has the expected force-balance structure: primary mechanism force and primary belt centrifugal loading drive positive shift, while secondary clamping force and secondary belt centrifugal loading oppose it. The effective inertia in the denominator includes the translating sheaves, the simplified axial contribution of belt motion, and the movable secondary sheave inertia reflected through the helix constraint.

Importantly, the axial shift relation is not a standalone subsystem. The quantity  $\ddot{s}$  is written in terms of the current state, but it also depends on two quantities that are not themselves state variables: the transmitted secondary torque  $\tau_s$  and the secondary angular acceleration  $\dot{\omega}_s$ . These appear because the secondary clamping mechanism is torque-reactive and because the helix couples axial shift to relative rotation of the secondary sheaves.

Thus, the result of this section should be interpreted as one equation in the larger coupled CVT model. It gives the axial response that must be satisfied alongside the rotational equations for the primary pulley, belt, and secondary pulley. The following section develops those torque-transfer dynamics, which provide the remaining relations needed to determine  $\tau_s$ ,  $\dot{\omega}_s$ , and the other acceleration and torque unknowns.

With the axial mechanism model established, attention now turns to the torque path through the belt and pulleys. Combining that torque path with the shift relation derived here will provide the coupled dynamic closure used by the full CVT model.

### 3.3 Rotational and Belt Transport Dynamics

The axial force model developed in [Section 3.2](#) gives one equation for the shift coordinate  $s(t)$ . That equation depends on the transmitted secondary torque  $\tau_s$  and on the secondary angular acceleration  $\dot{\omega}_s$ . Those quantities are determined by the torque path through the CVT.

This section derives the common equations of motion for the rotating and translating bodies in that torque path. The primary pulley, belt, and secondary pulley are treated as coupled dynamic bodies. Engine torque enters at the primary pulley, torque is exchanged between the primary pulley and the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain.

The equations derived here are independent of whether the belt is sticking or slipping at either pulley. They describe the bodies being accelerated; the contact state supplies the remaining closure equations later.

### 3.3.1 System Definition and Torque Path

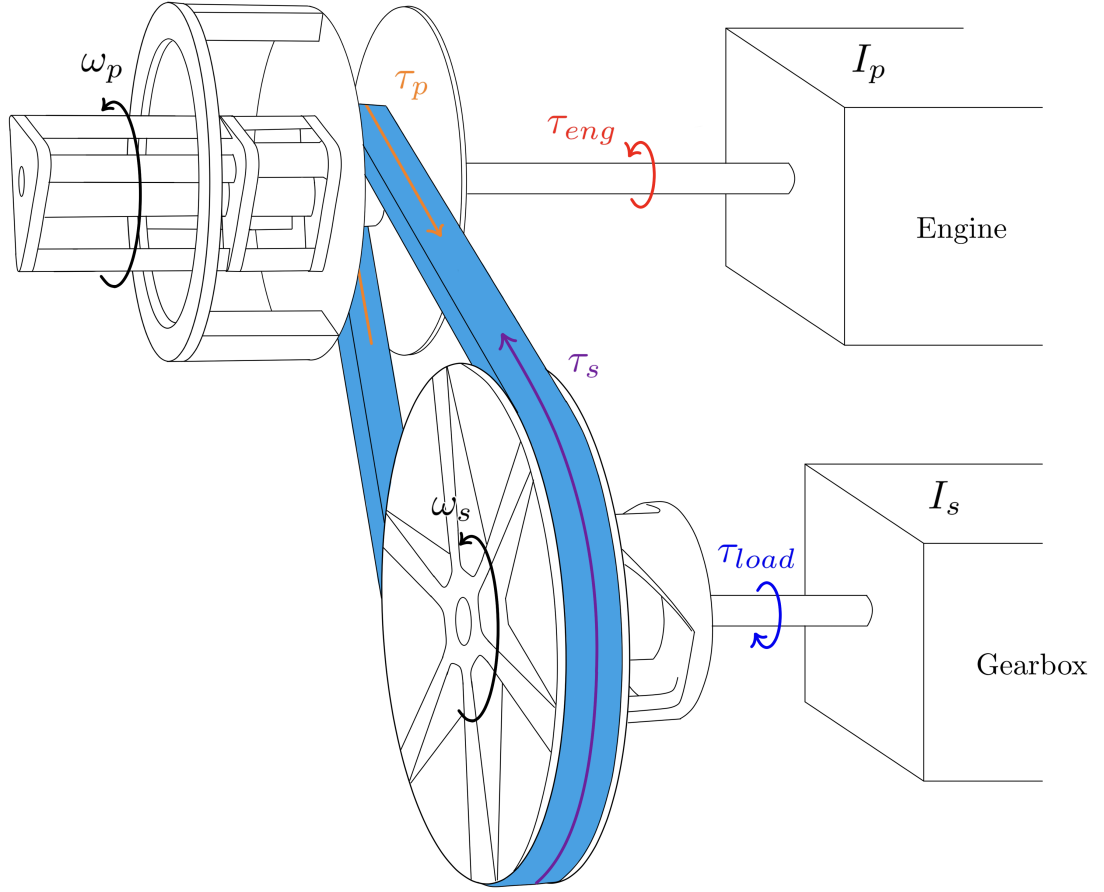


Figure 3.13: Torque path through the CVT. Engine torque  $\tau_{eng}$  drives the primary pulley with angular speed  $\omega_p$ . The primary pulley exchanges torque with the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain while opposing the reflected load torque  $\tau_{load}$ .

The torque path shown in [Figure 3.13](#) is described using three contact torques and three dynamic motions:

- the primary pulley rotates with angular speed  $\omega_p$ ,
- the belt translates with transport speed  $v_b$ ,
- the secondary shaft and fixed sheave rotate with angular speed  $\omega_s$ .

The primary pulley receives engine torque  $\tau_{eng}$ . Let  $\tau_p$  denote the torque exerted by the belt contact on the primary pulley. With the rotational sign convention from [Section 2.3.1](#),  $\tau_p$  opposes positive engine-driven primary rotation.

The belt receives torque from the primary contact and applies torque to the secondary contact. Let  $\tau_s$  denote the total signed torque exerted by the belt on the secondary pulley. This torque acts in



the positive secondary rotational direction, while the reflected drivetrain load  $\tau_{\text{load}}$  opposes it.

### 3.3.2 Engine Torque Model

The engine provides the driving torque  $\tau_{\text{eng}}(t)$  acting on the primary pulley. Under [Assumption 19 \(Reference Full-Throttle Engine Boundary\)](#), engine torque is represented as a prescribed full-throttle function of primary angular speed:

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}(\omega_p(t)). \quad (3.3.1)$$

No specific analytic form is required. The model only requires that the mapping  $\omega_p \mapsto \tau_{\text{eng}}$  be finite over the operating range used in simulation. Outside the supplied range, the torque curve is either saturated or the state is constrained to remain within the admissible engine-speed domain.

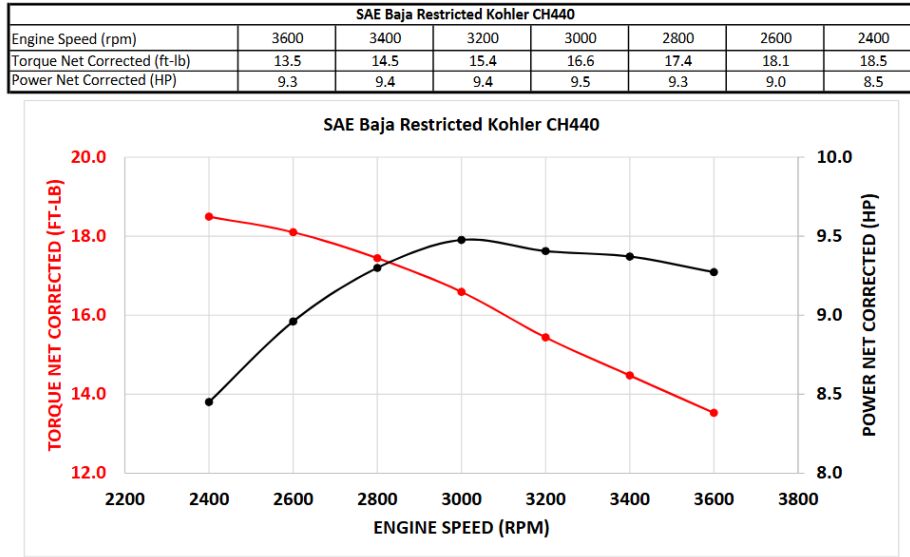


Figure 3.14: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in [Equation \(3.3.1\)](#).

In simulation, the discrete torque curve is interpolated to provide a continuous input torque over the operating range.

### 3.3.3 Load Torque Model

The resisting torque  $\tau_{\text{load}}(t)$  represents the longitudinal vehicle load reflected back to the secondary pulley.

Let  $r_w$  denote the effective wheel rolling radius and let  $G > 0$  denote the fixed gear ratio defined by

$$\omega_s = G \omega_w.$$

The vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G} \omega_s(t). \quad (3.3.2)$$

The total longitudinal road force is modeled as the sum of rolling resistance, grade resistance, and aerodynamic drag:

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \quad (3.3.3)$$

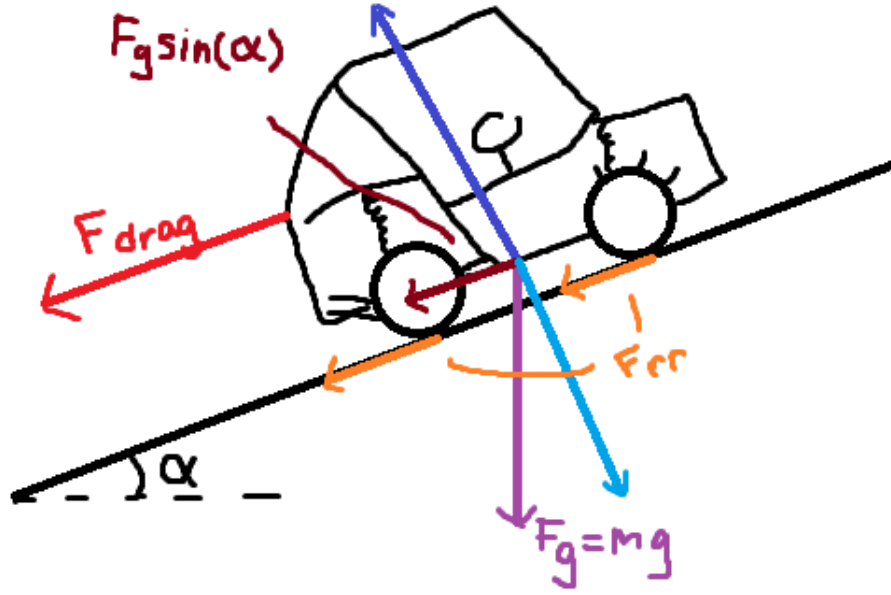


Figure 3.15: Free-body diagram of a vehicle on an incline. The weight  $mg$  resolves into  $mg \sin \alpha$  along the slope and  $mg \cos \alpha$  normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Using Theory ?? (??), Theory ?? (??), and Theory ?? (??), the road-load components are

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos \alpha \operatorname{sgn}(v), \quad (3.3.4)$$

$$F_{\text{grade}}(\alpha) = mg \sin \alpha, \quad (3.3.5)$$

and

$$F_{\text{drag}}(v) = \frac{1}{2} \rho C_d A_f v^2 \operatorname{sgn}(v). \quad (3.3.6)$$

Substituting these terms into Equation (3.3.3) gives

$$F_{\text{road}}(t) = C_{\text{rr}}mg \cos \alpha \operatorname{sgn}(v(t)) + mg \sin \alpha + \frac{1}{2}\rho C_d A_f v(t)^2 \operatorname{sgn}(v(t)). \quad (3.3.7)$$

The wheel torque associated with this road load is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \quad (3.3.8)$$

With the downstream fixed-ratio drive treated as ideal under **Assumption 17 (No Separate Empirical Loss or Thermal Model)**, the resisting torque reflected to the secondary pulley is

$$\tau_{\text{load}}(t) = \frac{1}{G} \tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \quad (3.3.9)$$

Using  $v = (r_w/G)\omega_s$ , this becomes

$$\tau_{\text{load}}(t) = \frac{r_w}{G} \left[ C_{\text{rr}}mg \cos \alpha \operatorname{sgn}\left(\frac{r_w}{G}\omega_s(t)\right) + mg \sin \alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \operatorname{sgn}\left(\frac{r_w}{G}\omega_s(t)\right) \right]. \quad (3.3.10)$$

### 3.3.4 Inertia Definitions

The rotational equations require inertias expressed about the primary and secondary axes. Under **Assumption 14 (Lumped Inertia Representation)**, these inertias are treated as lumped scalar quantities about their corresponding shafts.

#### Primary-Side Inertia

The primary pulley is rigidly connected to the engine side. All components in this lump rotate with angular speed  $\omega_p$ , so the primary-side inertia is

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \quad (3.3.11)$$

where  $I_{\text{eng}}$  is the engine rotational inertia and  $I_{\text{prim}}$  is the primary pulley inertia.

#### Secondary Fixed-Side and Downstream Inertia

The secondary side requires one additional distinction. The fixed secondary sheave and downstream drivetrain rotate with  $\omega_s$ . The movable secondary sheave does not generally rotate with exactly  $\omega_s$  during shift, because the helix couples axial motion to relative rotation. Its rotational motion is therefore handled separately using the helix kinematics from **Equation (3.2.16)** and **Equation (3.2.17)**.

Let  $I_{s,F}$  denote the inertia that rotates directly with the fixed secondary sheave and shaft, excluding the movable secondary sheave inertia  $I_M$ . This fixed-side inertia includes the fixed secondary

components, gearbox-side inertia, wheel inertia reflected through the fixed ratio, and the vehicle translational inertia reflected to the secondary axis.

The vehicle translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2 = \frac{1}{2}m(r_w\omega_w)^2,$$

so the vehicle mass appears as an equivalent wheel-axis inertia  $mr_w^2$ . Since  $\omega_s = G\omega_w$ , wheel-axis inertias reflect to the secondary axis by division by  $G^2$ . Thus,

$$I_{s,F} = I_{\text{sec},F} + I_{\text{gb}} + \frac{I_{\text{wheel}} + mr_w^2}{G^2}. \quad (3.3.12)$$

Here  $I_{\text{sec},F}$  includes the fixed secondary sheave and other secondary components that rotate directly with the secondary shaft. The movable secondary sheave inertia  $I_M$  is not included in  $I_{s,F}$ .

### 3.3.5 Common Equations of Motion

The primary pulley, belt, secondary fixed-side assembly, and shift coordinate provide the four common dynamic equations used in every contact state. The contact state will later determine the two additional closure equations.

#### Primary Pulley Rotational Dynamics

Applying Theory ?? (??) to the primary pulley gives

|                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Primary Pulley Rotational Dynamics</b></div> <div><b>(3.3.13)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> <div style="text-align: center; padding: 10px 0;"> <math display="block">\tau_{\text{eng}}(t) - \tau_p(t) = I_p\dot{\omega}_p(t)</math> </div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The engine torque drives positive primary rotation, while the belt contact torque opposes it.

#### Belt Transport Dynamics

The belt is represented by the transport speed  $v_b(t)$  introduced in [Section 2.4](#). Under [Assumption 11 \(Constant Belt Length\)](#) and [Assumption 12 \(Single Belt Transport Speed\)](#), the belt is modeled as a single moving path with one lumped transport speed.

A contact torque produces an equivalent belt-line force when divided by the effective torque radius from ???. The primary contact contributes  $\tau_p/r_{p,\text{eff}}$  to the belt-line motion, while the secondary contact contributes  $\tau_s/r_{s,\text{eff}}$  in the opposing direction. Applying Theory ?? (??) to the belt gives

|                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Belt Transport Dynamics</b></div> <div><b>(3.3.14)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> <div style="text-align: center; padding: 10px 0;"> <math display="block">m_b\dot{v}_b(t) = \frac{\tau_p(t)}{r_{p,\text{eff}}(s)} - \frac{\tau_s(t)}{r_{s,\text{eff}}(s)}</math> </div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

where  $m_b$  is the total belt mass from [Equation \(3.2.42\)](#).

## Secondary Rotational Dynamics

The secondary side is not treated as a single rigid body during shift. The fixed secondary sheave and downstream drivetrain rotate with the secondary shaft speed  $\omega_s$ , while the movable secondary sheave rotates with  $\omega_M$ . These two angular speeds are equal only when the secondary is not shifting. When the helix is active, axial motion of the movable sheave is accompanied by relative rotation, so the movable sheave must be counted with its own angular acceleration.

The cleanest way to write the secondary rotational balance is to consider the fixed-side secondary assembly and the movable sheave together as one system. The external torques on this combined secondary system are the total belt-applied secondary torque  $\tau_s$  and the reflected load torque  $\tau_{\text{load}}$ . The helix and spring torques between the fixed and movable sheaves are internal to this combined system, so they do not appear in the net external torque balance.

Let  $I_{s,F}$  denote the inertia of the fixed-side secondary assembly, excluding the movable secondary sheave inertia  $I_M$ . The angular momentum of the secondary system about the shaft axis is then

$$L_s = I_{s,F}\omega_s + I_M\omega_M.$$

Applying Theory ?? (??) in angular-momentum form gives

$$\tau_s - \tau_{\text{load}} = \frac{dL_s}{dt} = I_{s,F}\dot{\omega}_s + I_M\dot{\omega}_M. \quad (3.3.15)$$

The movable-sheave acceleration is not independent. From the helix kinematics derived in [Equation \(3.2.17\)](#),

$$\dot{\omega}_M = \dot{\omega}_s - \frac{d^2\theta_s}{ds^2}\dot{s}^2 - \frac{d\theta_s}{ds}\ddot{s}.$$

Substituting this relation into [Equation \(3.3.15\)](#) gives the secondary rotational equation in terms of the common dynamic variables:

|                                                                                                                                          |                 |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Secondary Rotational Dynamics</b>                                                                                                     | <b>(3.3.16)</b> |
| $\tau_s - \tau_{\text{load}} = (I_{s,F} + I_M)\dot{\omega}_s - I_M \frac{d^2\theta_s}{ds^2}\dot{s}^2 - I_M \frac{d\theta_s}{ds}\ddot{s}$ |                 |

This form preserves the distinction between the fixed secondary components and the movable sheave. If the CVT is not shifting, then  $\dot{s} = 0$  and  $\ddot{s} = 0$ , so the movable sheave rotates with the fixed sheave and the secondary inertia reduces to  $I_{s,F} + I_M$ . During shift, the helix introduces the additional coupling terms shown above.

### 3.3.6 Common Dynamic Relations and Remaining Closure

At this point, the model has four dynamic relations that are used regardless of the belt contact state. These are:

- primary rotational dynamics, [Equation \(3.3.13\)](#),

- belt transport dynamics, [Equation \(3.3.14\)](#),
- secondary rotational dynamics, [Equation \(3.3.16\)](#),
- axial shift dynamics, [Equation \(3.2.52\)](#).

The first three equations describe the torque path through the primary pulley, belt, and secondary pulley. The fourth equation carries the axial force balance into the same coupled system.

Together, the common dynamics involve the six unknown quantities

$$\dot{\omega}_p, \quad \dot{\omega}_s, \quad \dot{v}_b, \quad \ddot{s}, \quad \tau_p, \quad \tau_s.$$

The four equations above therefore describe the body dynamics, but they do not yet determine all six unknowns. Two additional relations are still needed to specify how torque is exchanged at the primary and secondary belt–pulley contacts.

This is where the belt contact state enters the model. The equations derived so far describe how the bodies respond to applied forces and torques; the contact relations determine what forces and torques can actually be exchanged between the belt and each pulley. The following section begins with the gross sticking case, where the belt is assumed to remain kinematically compatible with both pulley contacts.

## 3.4 Gross Stick Compatibility and No-Slip Candidate

The previous section established the common equations of motion for the primary pulley, belt, secondary assembly, and shift coordinate. These equations describe how the bodies respond to applied forces and torques, but they do not yet specify the contact state between the belt and the pulleys.

This section considers the gross sticking case. In this contact state, the belt is assumed to remain adhered to both pulleys at their effective torque radii. The resulting solution is the *no-slip candidate*: the accelerations and contact torques that would be required if both belt–pulley contacts remained stuck.

This candidate is not automatically admissible. It is first computed from the sticking constraints, then checked against the available traction limits in the following section.

### 3.4.1 Gross Stick Compatibility

Let  $j \in \{p, s\}$  denote either the primary or secondary pulley. Under [Assumption 12 \(Single Belt Transport Speed\)](#), the belt is represented by a single transport speed  $v_b$ . Under [Assumption 13 \(Uniform Contact Radius Around Each Wrap\)](#), each pulley contact is represented by a single effective torque radius  $r_{j,\text{eff}}$ .

If the belt is grossly stuck to pulley  $j$ , the belt transport speed equals the tangential surface speed of that pulley at the effective torque radius:

$$v_b(t) = r_{j,\text{eff}}(s(t)) \omega_j(t), \quad j \in \{p, s\}. \quad (3.4.1)$$

Differentiating [Equation \(3.4.1\)](#) gives the acceleration-level stick compatibility condition:

**Gross Stick Compatibility****(3.4.2)**

$$\dot{v}_b(t) = \frac{dr_{j,\text{eff}}}{ds} \dot{s}(t) \omega_j(t) + r_{j,\text{eff}}(s(t)) \dot{\omega}_j(t), \quad j \in \{p, s\}.$$

This equation is the contact closure for a stuck belt–pulley interface. It states that the belt-line acceleration must match the acceleration of the pulley surface at the effective radius, including the contribution from changing pulley radius during shift.

Solving Equation (3.4.2) for the pulley angular acceleration gives

$$\dot{\omega}_j(t) = \frac{\dot{v}_b(t) - \frac{dr_{j,\text{eff}}}{ds} \dot{s}(t) \omega_j(t)}{r_{j,\text{eff}}(s(t))}, \quad j \in \{p, s\}. \quad (3.4.3)$$

For the no-slip candidate, Equation (3.4.3) is applied at both the primary and secondary contacts.

**3.4.2 No-Slip Candidate Solve**

For compactness, define

$$r_p = r_{p,\text{eff}}(s), \quad r_s = r_{s,\text{eff}}(s),$$

and

$$\dot{r}_p = \frac{dr_{p,\text{eff}}}{ds} \dot{s}, \quad \dot{r}_s = \frac{dr_{s,\text{eff}}}{ds} \dot{s}.$$

Also define the effective shift inertia from Equation (3.2.52) as

$$M_{\text{shift}} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4} + I_M \left( \frac{d\theta_s}{ds} \right)^2. \quad (3.4.4)$$

This is not a new model assumption; it is the denominator of the axial shift equation already derived.

**Stick-Compatible Angular Accelerations**

Applying the stick condition from Equation (3.4.3) at the primary pulley gives

$$\dot{\omega}_{p,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p}. \quad (3.4.5)$$

Applying the same condition at the secondary pulley gives

$$\dot{\omega}_{s,\text{ns}} = \frac{\dot{v}_{b,\text{ns}} - \dot{r}_s \omega_s}{r_s}. \quad (3.4.6)$$

Thus, once the no-slip belt acceleration  $\dot{v}_{b,\text{ns}}$  is known, both pulley angular accelerations are known.

## Primary No-Slip Torque Demand

The primary rotational equation is

$$\tau_{\text{eng}} - \tau_p = I_p \dot{\omega}_p.$$

Substituting Equation (3.4.5) gives the primary contact torque required to maintain no slip:

|                                                        |                |
|--------------------------------------------------------|----------------|
| <b>Primary No-Slip Torque Demand Before Belt Solve</b> | <b>(3.4.7)</b> |
|--------------------------------------------------------|----------------|

$$\tau_{p,\text{ns}} = \tau_{\text{eng}} - I_p \left( \frac{\dot{v}_{b,\text{ns}} - \dot{r}_p \omega_p}{r_p} \right)$$

This is already expressed in terms of known state quantities and the single remaining unknown  $\dot{v}_{b,\text{ns}}$ .

## Secondary No-Slip Torque Demand

The secondary torque demand is coupled to the axial shift equation because the secondary helix couples  $\tau_s$ ,  $\dot{\omega}_s$ , and  $\ddot{s}$ .

From Equation (3.2.52), the shift acceleration can be written directly as

$$\ddot{s}_{\text{ns}} = \frac{F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} \left[ \frac{\tau_{s,\text{ns}}}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_{s,\text{ns}} - \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{\text{shift}}}. \quad (3.4.8)$$

The secondary rotational equation from Equation (3.3.16) is

$$\tau_{s,\text{ns}} - \tau_{\text{load}} = (I_{s,F} + I_M) \dot{\omega}_{s,\text{ns}} - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - I_M \frac{d\theta_s}{ds} \ddot{s}_{\text{ns}}. \quad (3.4.9)$$

Substituting Equation (3.4.8) into Equation (3.4.9), then collecting terms in  $\tau_{s,\text{ns}}$ , gives

$$\tau_{s,\text{ns}} = \frac{\tau_{\text{load}} + \left[ I_{s,F} + I_M - \frac{I_M^2 \left( \frac{d\theta_s}{ds} \right)^2}{M_{\text{shift}}} \right] \dot{\omega}_{s,\text{ns}} - I_M \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{I_M \frac{d\theta_s}{ds}}{M_{\text{shift}}} \left[ F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2\theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left( \frac{d\theta_s}{ds} \right)^2}{2M_{\text{shift}}}}. \quad (3.4.10)$$



Finally, substituting the secondary stick-compatible acceleration from Equation (3.4.6) gives the secondary torque demand in terms of the belt acceleration:

$$\tau_{\text{load}} + \left[ I_{s,F} + I_M - \frac{I_M^2 \left( \frac{d\theta_s}{ds} \right)^2}{M_{\text{shift}}} \right] \left( \frac{\dot{v}_{b,\text{ns}} - \dot{r}_s \omega_s}{r_s} \right) - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2$$

$$\tau_{s,\text{ns}} = \frac{- \frac{I_M}{M_{\text{shift}}} \frac{d\theta_s}{ds} \left[ F_{p,\text{ax}} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left( \frac{d\theta_s}{ds} \right)^2}{2M_{\text{shift}}}}. \quad (3.4.11)$$

This expression is longer than the primary torque demand because the secondary torque is coupled through the helix to both secondary rotation and axial shift.

### Explicit No-Slip Belt Acceleration

The belt transport equation is

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.4.7) and Equation (3.4.11), then collecting terms in  $\dot{v}_{b,\text{ns}}$ , gives the explicit no-slip belt acceleration:

### No-Slip Belt Acceleration

(3.4.12)

$$\dot{v}_{b,ns} = \frac{\frac{\tau_{eng}}{r_p} + \frac{I_p \dot{r}_p \omega_p}{r_p^2} - \tau_{load} - \left[ I_{s,F} + I_M - \frac{I_M^2 \left( \frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \frac{\dot{r}_s \omega_s}{r_s} - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2 - \frac{1}{r_s} \frac{\frac{I_M \frac{d\theta_s}{ds}}{M_{shift}} \left[ F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left( \frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}}{m_b + \frac{I_p}{r_p^2} + \frac{1}{r_s} \frac{\left[ I_{s,F} + I_M - \frac{I_M^2 \left( \frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \frac{1}{r_s}}{1 - \frac{I_M \left( \frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}}$$

This equation contains only quantities known from the current state, evaluated geometry, force models, and inertial parameters. It therefore determines the belt acceleration required by the no-slip candidate.

### Recovered No-Slip Accelerations and Candidate Torques

Once  $\dot{v}_{b,ns}$  has been computed from Equation (3.4.12), the remaining no-slip quantities follow in direct order.

The pulley angular accelerations are

$$\dot{\omega}_{p,ns} = \frac{\dot{v}_{b,ns} - \dot{r}_p \omega_p}{r_p}, \quad (3.4.13)$$

and

$$\dot{\omega}_{s,ns} = \frac{\dot{v}_{b,ns} - \dot{r}_s \omega_s}{r_s}. \quad (3.4.14)$$

The primary torque candidate is

**Primary No-Slip Torque Demand****(3.4.15)**

$$\tau_{p,ns} = \tau_{eng} - I_p \left( \frac{\dot{v}_{b,ns} - \dot{r}_p \omega_p}{r_p} \right)$$

The secondary torque candidate is

**Secondary No-Slip Torque Demand****(3.4.16)**

$$\tau_{load} + \left[ I_{s,F} + I_M - \frac{I_M^2 \left( \frac{d\theta_s}{ds} \right)^2}{M_{shift}} \right] \left( \frac{\dot{v}_{b,ns} - \dot{r}_s \omega_s}{r_s} \right) - I_M \frac{d^2 \theta_s}{ds^2} \dot{s}^2$$

$$\tau_{s,ns} = \frac{- \frac{I_M}{M_{shift}} \frac{d\theta_s}{ds} \left[ F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \frac{d\theta_s}{ds} \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right]}{1 - \frac{I_M \left( \frac{d\theta_s}{ds} \right)^2}{2M_{shift}}}$$

Finally, the no-slip shift acceleration is obtained from the axial shift equation:

**No-Slip Shift Acceleration****(3.4.17)**

$$\ddot{s}_{ns} = \frac{F_{p,ax} + F_{c,p} - F_{c,s} - k_{s,x}(x_{s,0} + s) - \frac{d\theta_s}{ds} \left[ \frac{\tau_{s,ns}}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_{s,ns} - \frac{d^2 \theta_s}{ds^2} \dot{s}^2 \right) \right]}{M_{shift}}$$

Together, these equations explicitly determine the no-slip candidate:

$$\dot{v}_{b,ns}, \quad \dot{\omega}_{p,ns}, \quad \dot{\omega}_{s,ns}, \quad \tau_{p,ns}, \quad \tau_{s,ns}, \quad \ddot{s}_{ns}.$$

If both belt–pulley contacts were guaranteed to remain stuck, these equations would close the CVT dynamics. In that idealized case, the belt would act as a kinematic constraint between the two pulleys while the shift coordinate changes the effective ratio.

For a frictional belt drive, however, the required no-slip torques must still be checked against the available traction capacity.

**3.4.3 Transition to Traction Limits**

The no-slip candidate answers one question: what accelerations and contact torques are required if the belt remains stuck to both pulleys? It does not yet answer whether the contacts can physically support those torques.

The next section derives the tangential traction limits for the primary and secondary belt–pulley contacts. The candidate torques  $\tau_{p,ns}$  and  $\tau_{s,ns}$  can then be compared with those limits. If both

contact torques are admissible, the no-slip candidate is used. If either required torque exceeds the available traction capacity, the corresponding sticking constraint is replaced by a slip relation.

### 3.5 Traction Limits and Slip Conditions

The previous section derived the no-slip candidate, including the contact torque demands  $\tau_{p,ns}$  and  $\tau_{s,ns}$ . These are the torques that the belt–pulley interfaces would have to exchange if the belt remained grossly stuck to both pulleys.

Those torque demands are not automatically attainable. Torque is transmitted through frictional traction between the belt and the sheave faces, and that traction is limited by the normal load generated by axial clamping. Therefore, each belt–pulley contact can support only a finite contact torque before gross slip must occur.

The purpose of this section is to derive the traction checks used to decide whether the no-slip candidate is admissible. The important distinction is that the dynamic variables  $\tau_p$  and  $\tau_s$  are treated as contact torques, not as tight–slack span-tension differences. The corresponding span tension change can still be reconstructed after the contact torque is known, but the stick/slip admissibility check is posed directly in terms of frictional contact torque.

#### 3.5.1 Contact-Torque Capacity

The same traction argument applies to both pulleys. Let

$$j \in \{p, s\},$$

where  $j = p$  denotes the primary pulley and  $j = s$  denotes the secondary pulley. At pulley  $j$ , the belt–pulley contact is represented by an effective torque radius  $r_{j,\text{eff}}$ . The signed contact torque at that pulley is denoted  $\tau_j$ .

Under **Assumption 4 (Idealized Belt–Pulley Friction Law)** and **Assumption 5 (Constant Effective Friction Coefficients)**, the belt–pulley interface obeys a Coulomb friction law with constant coefficient  $\mu$ . Under **Assumption 7 (Circumferential Traction Approximation)**, the available friction is resolved in the gross tangential torque-transfer direction. Thus, at the differential level,

$$|dF_{t,j}| \leq \mu dN_j, \tag{3.5.1}$$

where  $dF_{t,j}$  is the tangential contact force and  $dN_j$  is the local normal load.

Integrating over the wrap gives the bound on total tangential contact force,

$$|F_{t,j}| \leq \mu N_{\phi,j}, \tag{3.5.2}$$

where

$$N_{\phi,j} \equiv \int_{\text{wrap}} dN_j \tag{3.5.3}$$

is the total normal load carried by the belt–pulley contact on pulley  $j$ .

The corresponding contact torque magnitude is

$$|\tau_j| = r_{j,\text{eff}} |F_{t,j}|. \quad (3.5.4)$$

Therefore, for a prescribed normal load, the maximum gross contact torque that can be supported without slip is

|                                                 |                |
|-------------------------------------------------|----------------|
| <b>Gross Contact-Torque Capacity</b>            | <b>(3.5.5)</b> |
| $ \tau_j  \leq r_{j,\text{eff}} \mu N_{\phi,j}$ |                |

This is the contact-torque capacity used for stick/slip admissibility. It bounds the torque exchanged directly at the belt–pulley interface.

#### Remark (Contact Torque Versus Span Tension)

The contact torque  $\tau_j$  should not be confused with the torque-equivalent of the tight–slack tension difference across the wrap. During transient motion, some contact traction may be used to accelerate belt material on the wrap, so the contact force and the span tension rise need not be equal. The present model uses contact torque as the dynamic coupling variable. A span-tension reconstruction is given in ??, but it is not used as the primary stick/slip admissibility condition.

#### Remark (Other Friction Models)

Although the present simulator adopts a Coulomb friction law for simplicity and analytic transparency, richer friction descriptions are available if greater fidelity near the stick–slip transition is desired. A common next step is a Stribeck-type law, in which the effective friction level varies with relative slip speed so that breakaway friction near zero speed can exceed the kinetic friction observed at larger slip speeds. Such a model can better represent engagement and low-slip behaviour than a purely constant Coulomb coefficient, while still remaining relatively simple to implement [Armstrong-H'elouvry et al. \(1994\)](#); [Canudas de Wit et al. \(1995\)](#).

For still higher fidelity, dynamic friction models introduce internal state variables that represent the evolving contact condition. The LuGre model is a standard example and can capture presliding displacement, hysteresis, friction lag, and smoother stick–slip transitions, while related multistate models such as the generalized Maxwell–slip model can represent more detailed microslip and memory effects [Canudas de Wit et al. \(1995\)](#); [Al-Bender et al. \(2005\)](#). These models are often more physically expressive, but they also require additional parameter identification and increase numerical and modelling complexity. For the present work, Coulomb friction is therefore retained as a deliberate low-complexity approximation, while Stribeck- or state-based friction models remain natural extensions for future refinement [Armstrong-H'elouvry et al. \(1994\)](#).

### 3.5.2 Normal Force from Axial Clamping

The torque capacity in Equation (3.5.5) depends on the total normal load  $N_{\phi,j}$ . We now express that normal load in terms of the axial clamping force generated by pulley  $j$ .

The belt is pressed between two opposing conical sheaves. The two inclined faces apply a net radial normal load on the belt determined by the axial clamp. Using the groove geometry from Equation (3.2.39),

$$F_{\text{rad},j} = 2F_{\text{ax},j} \tan \beta. \quad (3.5.6)$$

The radial load in Equation (3.5.6) represents the total inward force applied by the sheave faces onto the belt. This load is carried by the distributed normal contact forces along the wrap, so

$$N_{\phi,j} = F_{\text{rad},j}. \quad (3.5.7)$$

Only compressive axial clamp can create normal load. To prevent an unphysical negative traction capacity, define the compressive part of any axial force as

$$\langle F \rangle_+ = \max(F, 0). \quad (3.5.8)$$

Therefore,

|                                                               |                |
|---------------------------------------------------------------|----------------|
| <b>Wrap-Normal Load from Axial Clamp</b>                      | <b>(3.5.9)</b> |
| $N_{\phi,j} = 2 \langle F_{\text{ax},j} \rangle_+ \tan \beta$ |                |

Substituting Equation (3.5.9) into Equation (3.5.5) gives the gross contact-torque capacity

|                                                                                    |                 |
|------------------------------------------------------------------------------------|-----------------|
| <b>Pulley Contact-Torque Capacity</b>                                              | <b>(3.5.10)</b> |
| $ \tau_j  \leq 2r_{j,\text{eff}} \mu \langle F_{\text{ax},j} \rangle_+ \tan \beta$ |                 |

For a prescribed axial clamp, this bound is symmetric in positive and negative contact torque. However, the admissible torque range of a complete pulley mechanism may still be asymmetric if the axial clamp itself depends on the signed contact torque. This distinction is important for the torque-reactive secondary pulley.

#### Remark (Wrap Angle in the Reduced Capacity Model)

The wrap angle does not appear explicitly in Equation (3.5.10) because  $N_{\phi,j}$  is the total integrated normal load imposed by the clamp. Under the averaged-contact approximation, increasing wrap angle spreads this same integrated normal load over more contact length rather than increasing the total available gross friction force. Wrap angle still matters for local contact pressure, heat, wear, pressure distribution, and span-tension reconstruction, but it is not an independent multiplier in the reduced gross contact-torque capacity.

### 3.5.3 Primary Pulley No-Slip Admissibility

The generic contact-torque capacity law can now be specialized to the primary pulley. At this stage, the goal is only to test whether the primary contact torque demanded by the no-slip candidate can be supported by the available primary traction.

For compactness, write

$$r_p = r_{p,\text{eff}}(s).$$

The primary axial force is known explicitly from [Equation \(3.2.10\)](#):

$$F_{\text{ax},p}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds} - k_p(x_{p,0} + s). \quad (3.5.11)$$

Using the static friction coefficient  $\mu_s$ , the primary no-slip contact is admissible if

|                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Primary No-Slip Admissibility</b></div> <div><b>(3.5.12)</b></div> </div> <hr style="border: 0.5px solid black; margin: 5px 0;"/> $ \tau_{p,\text{ns}}  \leq 2r_p \mu_s \langle F_{\text{ax},p}(s, \omega_p) \rangle_+ \tan \beta$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

This condition is symmetric in positive and negative primary contact torque because the primary axial force does not depend on the signed primary torque. If [Equation \(3.5.12\)](#) is satisfied, the primary contact can support the torque required by the no-slip candidate. If it is not satisfied, the primary contact cannot remain grossly stuck under the demanded contact torque.

### 3.5.4 Secondary Pulley No-Slip Admissibility

The secondary pulley follows the same contact-torque capacity law as the primary, but with one important difference: the secondary axial clamp is torque-reactive. Through the helix, the same signed secondary contact torque being checked for admissibility also contributes to the axial force that generates the available traction.

To keep this dependence explicit, write the secondary axial force as the function

$$F_{\text{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}) = k_{s,x}(x_{s,0} + s) + \frac{d\theta_s}{ds} \left[ \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,0} + \theta_s(s)) - I_M \left( \dot{\omega}_s - \frac{d^2\theta_s}{ds^2} \dot{s}^2 - \frac{d\theta_s}{ds} \ddot{s} \right) \right]. \quad (3.5.13)$$

For compactness, write

$$r_s = r_{s,\text{eff}}(s).$$

On the no-slip candidate branch, the quantities  $\tau_{s,\text{ns}}$ ,  $\dot{\omega}_{s,\text{ns}}$ , and  $\ddot{s}_{\text{ns}}$  have already been determined by the no-slip solve. Therefore, the secondary clamp used in the traction check is evaluated at the candidate values:

$$F_{\text{ax},s}^{\text{ns}} = F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}).$$

Using the static friction coefficient  $\mu_s$ , the secondary no-slip contact is admissible if

$$\text{Secondary No-Slip Admissibility} \quad (3.5.14)$$

$$|\tau_{s,\text{ns}}| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}) \rangle_+ \tan \beta$$

This condition is written as a magnitude check, but it is not assuming a symmetric secondary torque range. The axial force inside the right-hand side is evaluated using the signed no-slip torque  $\tau_{s,\text{ns}}$ . Therefore, positive and negative secondary torques of equal magnitude generally produce different clamp forces and different available traction. The condition above is a pointwise admissibility test: it asks whether the specific signed no-slip candidate torque creates enough clamp to support itself.

If Equation (3.5.14) is satisfied, the secondary contact can support the torque required by the no-slip candidate. If it is not satisfied, the secondary contact cannot remain grossly stuck under the demanded contact torque.

#### Remark (Optional Secondary Torque Envelope)

For branch selection, the pointwise check in Equation (3.5.14) is sufficient because the no-slip solve has already produced a specific value of  $\tau_{s,\text{ns}}$ . However, it can also be useful to reconstruct the full secondary torque envelope at a fixed instantaneous state, for example when plotting how the torque-reactive helix changes the available torque range.

At a fixed state, the secondary stick envelope is the set of torques satisfying

$$|\tau_s| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_s, \dot{\omega}_s, \ddot{s}) \rangle_+ \tan \beta. \quad (3.5.15)$$

Because  $F_{\text{ax},s}$  contains the signed torque  $\tau_s$ , the set defined by Equation (3.5.15) is generally not symmetric about zero. One torque direction may increase clamp through the helix, while the opposite direction may reduce clamp. This envelope is useful for diagnostic plots or design interpretation, but the branch-selection logic only needs the pointwise candidate check in Equation (3.5.14).

### 3.5.5 Admissibility of the No-Slip Candidate

The full no-slip candidate is admissible only if both pulley contacts satisfy their static traction checks:

$$|\tau_{p,\text{ns}}| \leq 2r_p \mu_s \langle F_{\text{ax},p}(s, \omega_p) \rangle_+ \tan \beta, \quad (3.5.16)$$

and

$$|\tau_{s,\text{ns}}| \leq 2r_s \mu_s \langle F_{\text{ax},s}(s, \dot{s}, \tau_{s,\text{ns}}, \dot{\omega}_{s,\text{ns}}, \ddot{s}_{\text{ns}}) \rangle_+ \tan \beta. \quad (3.5.17)$$



When both conditions hold, the no-slip candidate can be accepted as the contact state for that instant. If either condition fails, at least one contact cannot maintain gross sticking and the corresponding contact relation must be replaced by a slip relation.

### 3.5.6 Kinetic Traction Bound for Slipping Contacts

The same contact-torque capacity structure is used when a contact is sliding, but the kinetic friction coefficient  $\mu_k$  replaces the static coefficient  $\mu_s$ . Thus, the magnitude of the slipping contact torque at pulley  $j$  is

|                                                                                                |                 |
|------------------------------------------------------------------------------------------------|-----------------|
| <b>Kinetic Contact-Torque Magnitude</b>                                                        | <b>(3.5.18)</b> |
| $ \tau_{j,\text{slip}}  = 2r_{j,\text{eff}}\mu_k \langle F_{\text{ax},j} \rangle_+ \tan \beta$ |                 |

The sign of the kinetic torque is chosen so that the friction force opposes the relative sliding motion between the belt and pulley surface. The exact sign convention for  $\tau_p$  and  $\tau_s$  is applied in the contact-state selection section, where the primary and secondary slip branches are written explicitly.

### 3.5.7 Section Conclusion and Transition

This section derived the traction checks needed to test the no-slip candidate from [Section 3.4.2](#). The stick/slip admissibility check was posed directly in terms of contact torque because the rotational and belt transport equations use  $\tau_p$  and  $\tau_s$  as the torques exchanged at the belt–pulley contacts.

The primary check is symmetric because the primary axial force is independent of the signed primary contact torque. The secondary check is not generally symmetric in torque because the helix makes the secondary axial clamp depend on the signed secondary contact torque. For branch selection, this is handled by evaluating the secondary clamp directly on the signed no-slip candidate.

If both pulley contacts satisfy their admissibility conditions, the no-slip candidate is accepted. If one or both contacts cannot support their demanded contact torques, the corresponding contact must be described by a slipping traction law. The next section develops the contact-state selection rules and the resulting slip-branch dynamics.

## 3.6 Slip Branch Dynamics and Contact-State Selection

The previous section determined when the no-slip torque demands can be supported by the available belt–pulley traction. If the primary and secondary contacts both satisfy their admissibility conditions, the no-slip candidate is physically consistent and the belt remains adhered to both pulleys.

When one of these admissibility conditions fails, the no-slip candidate can no longer be used as the final torque-transfer law. The contact that exceeds its available traction cannot supply the demanded torque while remaining stuck, so relative motion begins at that belt–pulley interface. Once this occurs, the contact torque is no longer determined by the no-slip demand. Instead, it is set by the kinetic traction available at the slipping contact.

This section develops the rules used to select the active contact state and the corresponding equations of motion. The same three bodies introduced in [Section 3.3.1](#) are retained: the primary pulley, the belt, and the secondary pulley. What changes from one branch to another is the contact condition

imposed at each pulley. A stuck contact enforces belt-speed compatibility, while a slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion.

The goal is therefore to complete the torque-transfer closure. We first define the relative belt–pulley speeds used to identify sliding, then state the rules for entering, maintaining, and exiting slip. The remaining subsections derive the branch dynamics for primary slip, secondary slip, and simultaneous slip at both contacts.

### 3.6.1 Relative Contact Speeds and Slip-State Logic

To determine whether a belt–pulley contact is sticking or slipping, the model compares the belt transport speed to the local surface speed imposed by each pulley.

For the primary pulley, define the relative contact speed as

$$v_{\text{rel},p} = r_p \omega_p - v_b, \quad (3.6.1)$$

where  $r_p = r_{p,\text{eff}}(s)$ . Thus,  $v_{\text{rel},p} > 0$  means the primary pulley surface is moving faster than the belt at the contact radius and tends to drive the belt forward.

For the secondary pulley, define

$$v_{\text{rel},s} = v_b - r_s \omega_s, \quad (3.6.2)$$

where  $r_s = r_{s,\text{eff}}(s)$ . Thus,  $v_{\text{rel},s} > 0$  means the belt is moving faster than the secondary pulley surface and tends to drive the secondary forward.

These definitions are chosen so that positive relative speed corresponds to the usual forward torque-transfer direction at each contact: primary-to-belt on the primary side, and belt-to-secondary on the secondary side.

A contact can remain stuck only if its relative contact speed is zero, or within a small numerical tolerance:

$$|v_{\text{rel},j}| \leq \varepsilon_v, \quad j \in \{p, s\}. \quad (3.6.3)$$

If both contacts satisfy [Equation \(3.6.3\)](#), the model first tests the no-slip candidate derived in [??](#). This candidate is accepted only if both pulley contacts also satisfy the traction admissibility conditions derived in [Section 3.5](#):

$$|\tau_{p,\text{ns}}| \leq \tau_{p,\text{max}}^{\text{stick}} \quad (3.6.4)$$

and

$$\tau_{s,-}^{\text{stick}} \leq \tau_{s,\text{ns}} \leq \tau_{s,+}^{\text{stick}}. \quad (3.6.5)$$

If these conditions hold, the no-slip candidate becomes the active torque-transfer branch:

$$\tau_p = \tau_{p,\text{ns}}, \quad \tau_s = \tau_{s,\text{ns}}, \quad \dot{v}_b = \dot{v}_{b,\text{ns}}.$$

If either contact fails its admissibility check, that contact cannot support the required no-slip torque and must enter slip. The direction of slip is determined by the direction in which the demanded torque exceeds the admissible interval. For the primary pulley,

$$\tau_{p,\text{ns}} > \tau_{p,\text{max}}^{\text{stick}} \quad \Rightarrow \quad \text{positive primary slip,}$$

while

$$\tau_{p,ns} < -\tau_{p,max}^{stick} \Rightarrow \text{negative primary slip.}$$

For the secondary pulley,

$$\tau_{s,ns} > \tau_{s,+}^{stick} \Rightarrow \text{positive secondary slip,}$$

while

$$\tau_{s,ns} < \tau_{s,-}^{stick} \Rightarrow \text{negative secondary slip.}$$

Once a contact is already sliding, its state is determined primarily by its relative contact speed. If

$$|v_{rel,j}| > \varepsilon_v,$$

then contact  $j$  is treated as slipping, and its torque is set by kinetic traction rather than by the no-slip demand. The slip direction is recorded by

$$\sigma_j = \text{sgn}(v_{rel,j}), \quad j \in \{p, s\}. \quad (3.6.6)$$

Slip persists while the relative contact speed remains nonzero. During this time, the slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion. The no-slip candidate is not reaccepted until the relative speed returns to the sticking tolerance and the static admissibility conditions can again be satisfied.

This gives three possible slipping branches:

- primary slip, where the primary contact slides and the secondary contact remains stuck;
- secondary slip, where the secondary contact slides and the primary contact remains stuck;
- two-contact slip, where both pulley contacts slide simultaneously.

The following subsections derive the equations of motion for each branch.

#### Remark (Power Flow During Slip)

In the no-slip branch, belt-speed compatibility constrains the primary pulley, belt, and secondary pulley to move together kinematically. Once slip occurs, that compatibility is lost at the slipping contact. The contact torque is then limited by kinetic traction, while the pulley and belt speeds evolve according to their own equations of motion.

As a result, the instantaneous mechanical power at the primary contact,  $\tau_p \omega_p$ , does not need to match the instantaneous mechanical power at the secondary contact,  $\tau_s \omega_s$ . The difference is associated with changes in stored kinetic energy of the rotating components and belt, together with slip dissipation. Thus, an apparent short-term excess of output-side power does not imply energy creation; it reflects energy being drawn from inertia while the slipping contact breaks the ideal no-slip power constraint.

### 3.6.2 Kinetic Traction Law for a Slipping Contact

Once a belt–pulley contact is slipping, the torque at that contact is no longer determined by the no-slip torque demand. Instead, the contact applies the largest kinetic traction torque available in the direction opposing relative motion.

The generic torque-capacity expression from Equation (3.5.10) still applies, but the static coefficient  $\mu_s$  is replaced by the kinetic coefficient  $\mu_k$ . For a slipping contact on pulley  $j \in \{p, s\}$ , define

$$\tau_{j,\max}^{\text{slip}}(\dot{v}_b) = r_j [2\mu_k F_{\text{ax},j} \tan \beta - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_{j,\text{cm}} v_b)], \quad (3.6.7)$$

where  $r_j = r_{j,\text{eff}}(s)$ . The notation  $\tau_{j,\max}^{\text{slip}}(\dot{v}_b)$  emphasizes that the available slipping torque is not generally a fixed number; it depends on the belt acceleration through the inertial correction term.

The direction of the slipping torque is determined by the relative contact speed defined in Section 3.6.1. Let

$$\sigma_j = \text{sgn}(v_{\text{rel},j}), \quad j \in \{p, s\}. \quad (3.6.8)$$

With the relative-speed definitions adopted in Equation (3.6.1) and Equation (3.6.2),  $\sigma_j > 0$  corresponds to the usual forward torque-transfer direction at contact  $j$ . The slipping contact torque is then modeled as the saturated kinetic traction torque

$$\tau_j = \sigma_j \tau_{j,\max}^{\text{slip}}(\dot{v}_b). \quad (3.6.9)$$

Equation (3.6.9) should not be interpreted as a completed solution for  $\tau_j$ , because the right-hand side still contains  $\dot{v}_b$ . In a slip branch, the belt acceleration must be solved simultaneously with the saturated contact torque using the belt equation of motion Equation (3.3.14). The following subsections carry out this solve for primary slip, secondary slip, and simultaneous slip at both contacts.

### 3.6.3 Primary-Slip Branch

We first consider the branch where the primary contact is slipping while the secondary contact remains stuck. In this case, the primary pulley no longer enforces belt-speed compatibility. Instead, the primary contact torque is set by the saturated kinetic traction law from Section 3.6.2.

The secondary contact, however, is still adhered. Therefore, the secondary pulley remains kinematically compatible with the belt:

$$\dot{\omega}_s = \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s}, \quad (3.6.10)$$

Using the secondary rotational equation of motion Equation (3.3.16), the secondary contact torque is then

$$\tau_s = \tau_{\text{load}} + I_s \left( \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \quad (3.6.11)$$

At the slipping primary contact, let

$$\sigma_p = \text{sgn}(v_{\text{rel},p}).$$

Applying Equation (3.6.9) with  $j = p$  gives

$$\tau_p = \sigma_p r_p [2\mu_k F_{\text{ax},p} \tan \beta - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b)], \quad (3.6.12)$$

where  $r_p = r_{p,\text{eff}}(s)$ .

The remaining unknown is the belt acceleration  $\dot{v}_b$ . This is found by substituting Equation (3.6.11) and Equation (3.6.12) into the belt equation of motion Equation (3.3.14):

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}. \quad (3.6.13)$$

Substituting gives

$$m_b \dot{v}_b = \sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p (r_{p,cm} \dot{v}_b + \dot{r}_{p,cm} v_b)] - \frac{\tau_{load}}{r_s} - \frac{I_s}{r_s} \left( \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \quad (3.6.14)$$

Collecting the  $\dot{v}_b$  terms yields the primary-slip belt acceleration:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Primary-Slip Belt Acceleration</b></div> <div><b>(3.6.15)</b></div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> $\dot{v}_{b,p\text{-slip}} = \frac{\sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p \dot{r}_{p,cm} v_b] - \frac{\tau_{load}}{r_s} + \frac{I_s \dot{r}_s \omega_s}{r_s^2}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,cm} + \frac{I_s}{r_s^2}}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Once  $\dot{v}_{b,p\text{-slip}}$  is known, the branch torques follow by substitution into Equation (3.6.12) and Equation (3.6.11). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\tau_{eng} - \tau_p}{I_p}, \quad (3.6.16)$$

and

$$\dot{\omega}_s = \frac{\dot{v}_{b,p\text{-slip}} - \dot{r}_s \omega_s}{r_s}. \quad (3.6.17)$$

Thus, in the primary-slip branch, the primary contact torque is set by kinetic traction, the secondary contact still enforces belt-speed compatibility, and the belt equation of motion determines the resulting belt acceleration. This branch remains valid only while the secondary contact can remain adhered. If the secondary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

### 3.6.4 Secondary-Slip Branch

We next consider the branch where the secondary contact is slipping while the primary contact remains stuck. In this case, the secondary contact torque is set by saturated kinetic traction, while the primary contact still enforces belt-speed compatibility.

The primary contact remains adhered, so the primary pulley must satisfy

$$\dot{\omega}_p = \frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p}, \quad (3.6.18)$$

where  $r_p = r_{p,\text{eff}}(s)$ . Using the primary rotational equation of motion [Equation \(3.3.13\)](#), the primary contact torque is therefore

$$\tau_p = \tau_{\text{eng}} - I_p \left( \frac{\dot{v}_b - \dot{r}_p \omega_p}{r_p} \right). \quad (3.6.19)$$

At the slipping secondary contact, let

$$\sigma_s = \text{sgn}(v_{\text{rel},s}).$$

Applying [Equation \(3.6.9\)](#) with  $j = s$  gives

$$\tau_s = \sigma_s r_s [2\mu_k F_{\text{ax},s}(s, \tau_s) \tan \beta - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b)], \quad (3.6.20)$$

where  $r_s = r_{s,\text{eff}}(s)$ .

Unlike the primary slip branch, this expression is implicit in  $\tau_s$  because the secondary axial force is torque-reactive. Substituting the secondary axial force model from [Equation \(3.2.26\)](#) gives

$$\begin{aligned} \tau_s = \sigma_s r_s \left[ \mu_k \tan \beta (\tau_s + k_{s,\theta}(\theta_{s,0} + \theta_s(s))) \frac{d\theta_s}{ds} \right. \\ \left. + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]. \end{aligned} \quad (3.6.21)$$

Collecting the  $\tau_s$  terms gives

$$\tau_s = \frac{\sigma_s r_s \left[ \mu_k \tan \beta k_{s,\theta}(\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}. \quad (3.6.22)$$

The remaining unknown is the belt acceleration  $\dot{v}_b$ . Substituting [Equation \(3.6.19\)](#) and [Equation \(3.6.22\)](#) into the belt equation of motion [Equation \(3.3.14\)](#),

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s},$$

and collecting the  $\dot{v}_b$  terms gives the secondary-slip belt acceleration:

### Secondary-Slip Belt Acceleration

(3.6.23)

$$\dot{v}_{b,s-\text{slip}} = \frac{\frac{\tau_{\text{eng}}}{r_p} + \frac{I_p \dot{r}_p \omega_p}{r_p^2} - \frac{\sigma_s \left[ \mu_k \tan \beta k_{s,\theta}(\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\text{cm}} v_b \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}{m_b + \frac{I_p}{r_p^2} - \frac{\sigma_s \rho_b A_b \phi_s r_{s,\text{cm}}}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}$$

Once  $\dot{v}_{b,s-\text{slip}}$  is known, the branch torques follow by substitution into [Equation \(3.6.19\)](#) and [Equation \(3.6.22\)](#). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\dot{v}_{b,s-\text{slip}} - \dot{r}_p \omega_p}{r_p}, \quad (3.6.24)$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \quad (3.6.25)$$

Thus, in the secondary-slip branch, the secondary contact torque is set by kinetic traction, including the torque-reactive helix feedback, while the primary contact still enforces belt-speed compatibility. This branch remains valid only while the primary contact can remain adhered. If the primary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

### 3.6.5 Two-Contact Slip Branch

Finally, consider the branch where both belt-pulley contacts are slipping. In this case, neither pulley enforces belt-speed compatibility. Both contact torques are instead determined by saturated kinetic traction, and the belt acceleration is obtained directly from the belt equation of motion.

Let

$$\sigma_p = \text{sgn}(v_{\text{rel},p}), \quad \sigma_s = \text{sgn}(v_{\text{rel},s}).$$

At the primary contact, applying [Equation \(3.6.9\)](#) with  $j = p$  gives

$$\tau_p = \sigma_p r_p [2\mu_k F_{\text{ax},p} \tan \beta - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b)], \quad (3.6.26)$$

where  $r_p = r_{p,\text{eff}}(s)$ .

At the secondary contact, applying [Equation \(3.6.9\)](#) with  $j = s$  gives

$$\tau_s = \sigma_s r_s [2\mu_k F_{\text{ax},s}(s, \tau_s) \tan \beta - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b)], \quad (3.6.27)$$

where  $r_s = r_{s,\text{eff}}(s)$ . As in [Section 3.6.4](#), the secondary expression is implicit because the secondary axial force is torque-reactive. Substituting [Equation \(3.2.26\)](#) and collecting the  $\tau_s$  terms gives

$$\tau_s = \frac{\sigma_s r_s \left[ \mu_k \tan \beta k_{s,\theta} (\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x} (x_{s,0} + s) - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_{s,\text{cm}} v_b) \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}. \quad (3.6.28)$$

The belt equation of motion is still

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.6.26) and Equation (3.6.28), then collecting the  $\dot{v}_b$  terms, gives the two-contact-slip belt acceleration:

---

**Two-Contact-Slip Belt Acceleration** (3.6.29)

---

$$\dot{v}_{b,ps\text{--slip}} = \frac{\sigma_p [2\mu_k F_{ax,p} \tan \beta - \rho_b A_b \phi_p \dot{r}_{p,cm} v_b] - \frac{\sigma_s \left[ \mu_k \tan \beta k_{s,\theta} (\theta_{s,0} + \theta_s(s)) \frac{d\theta_s}{ds} + 2\mu_k \tan \beta k_{s,x} (x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,cm} v_b \right]}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,cm} - \frac{\sigma_s \rho_b A_b \phi_s r_{s,cm}}{1 - \sigma_s r_s \mu_k \tan \beta \frac{d\theta_s}{ds}}}$$


---

Once  $\dot{v}_{b,ps\text{--slip}}$  is known, the branch torques follow by substitution into Equation (3.6.26) and Equation (3.6.28). Since neither contact is adhered, the pulley accelerations are obtained directly from the rotational equations of motion:

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_p}{I_p}, \quad (3.6.30)$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \quad (3.6.31)$$

Thus, in the two-contact slip branch, both pulley contacts are governed by kinetic traction, and the belt is not kinematically constrained to either pulley. The belt acceleration is determined only by the balance of the two saturated contact torques through the belt equation of motion.

### 3.6.6 Contact-State Selection

The preceding subsections define the possible torque-transfer closures. The active branch is selected by checking which contacts can remain adhered.

Let the primary sticking condition be

$$\mathcal{S}_p \equiv (|v_{\text{rel},p}| \leq \varepsilon_v) \wedge \left( |\tau_{p,ns}| \leq \tau_{p,\text{max}}^{\text{stick}} \right),$$

and let the secondary sticking condition be

$$\mathcal{S}_s \equiv (|v_{\text{rel},s}| \leq \varepsilon_v) \wedge \left( \tau_{s,-}^{\text{stick}} \leq \tau_{s,ns} \leq \tau_{s,+}^{\text{stick}} \right).$$

The branch closure for the belt acceleration and contact torques is then



$$(\dot{v}_b, \tau_p, \tau_s) = \begin{cases} (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}), & \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,p\text{--slip}}, \tau_p \text{ from Equation (3.6.12)}, \tau_s \text{ from Equation (3.6.11)}), & \neg \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,s\text{--slip}}, \tau_p \text{ from Equation (3.6.19)}, \tau_s \text{ from Equation (3.6.22)}), & \mathcal{S}_p \wedge \neg \mathcal{S}_s, \\ (\dot{v}_{b,ps\text{--slip}}, \tau_p \text{ from Equation (3.6.26)}, \tau_s \text{ from Equation (3.6.28)}), & \neg \mathcal{S}_p \wedge \neg \mathcal{S}_s. \end{cases} \quad (3.6.32)$$

The corresponding belt accelerations are given by Equation (3.4.12), Equation (3.6.15), Equation (3.6.23), and Equation (3.6.29), respectively.

Thus, when both contacts can remain adhered, the model uses the no-slip torque demands. When one or both contacts cannot remain adhered, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves the belt acceleration from the belt equation of motion.

### 3.6.7 Section Summary

This section completed the torque-transfer closure for cases where the no-slip candidate cannot be maintained. Relative contact speeds were introduced to identify whether each belt–pulley interface is adhered or sliding, and kinetic traction was used to define the torque applied by a slipping contact.

The resulting branch structure separates the possible contact states into no-slip, primary slip, secondary slip, and two-contact slip. In each slipping case, the saturated contact torque replaces the no-slip demand, and the belt equation of motion determines the corresponding belt acceleration.

Together with the no-slip demand from ?? and the traction admissibility conditions from Section 3.5, the branch selection law in Equation (3.6.32) provides the final torque-transfer rule needed by the complete CVT dynamic model.

## 3.7 Complete CVT Dynamic Model

This section collects the results of the preceding derivations into one closed nonlinear dynamical model. No new physics is introduced here. The purpose is to summarize, in one place,

1. the state variables,
2. the geometric and contact quantities computed from those states,
3. the torque-transfer closure,
4. and the final governing equations of motion.

### 3.7.1 State Definition

The state vector is

$$\mathbf{x}(t) = (\omega_p, \omega_s, s, \dot{s}, v_b), \quad (3.7.1)$$

where

- $\omega_p$  is the primary pulley angular velocity,

- $\omega_s$  is the secondary pulley angular velocity,
- $s$  is the axial shift coordinate,
- $\dot{s}$  is the axial shift velocity,
- $v_b$  is the belt transport speed.

### 3.7.2 Geometry and Contact Kinematics

At each instant, the shift coordinate  $s$  determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived in [Section 3.1](#). In particular,

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s$$

are functions of  $s$ . For compactness in the complete model, write

$$r_p = r_{p,\text{eff}}(s), \quad r_s = r_{s,\text{eff}}(s).$$

The geometric CVT ratio remains

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \quad (3.7.2)$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \quad (3.7.3)$$

as given by ??.

The relative contact speeds at the two pulley interfaces are

$$v_{\text{rel},p} = r_p \omega_p - v_b, \quad v_{\text{rel},s} = v_b - r_s \omega_s, \quad (3.7.4)$$

as defined in [Equation \(3.6.1\)](#) and [Equation \(3.6.2\)](#). These quantities determine whether each contact is eligible to remain adhered or must be treated as sliding.

### 3.7.3 Torque-Transfer Closure

The torque-transfer subsystem is closed by determining the primary contact torque  $\tau_p$ , the secondary contact torque  $\tau_s$ , and the belt acceleration  $\dot{v}_b$ . These quantities are computed algebraically from the current state and contact branch.

The no-slip candidate derived in [Section 3.4.2](#) provides

$$\dot{v}_{b,\text{ns}}, \quad \tau_{p,\text{ns}}, \quad \tau_{s,\text{ns}},$$

from [Equation \(3.4.12\)](#), [Equation \(3.4.15\)](#), and [Equation \(3.4.16\)](#). These are the belt acceleration and contact torque demands required for adhered motion at both pulleys.

The admissibility conditions derived in [Section 3.5.3](#) and [Section 3.5.4](#) determine whether those no-slip demands can be supported. The primary contact is admissible when

$$|\tau_{p,\text{ns}}| \leq \tau_{p,\text{max}}^{\text{stick}}, \quad (3.7.5)$$

as in [Equation \(3.5.12\)](#). The secondary contact is admissible when

$$\tau_{s,-}^{\text{stick}} \leq \tau_{s,\text{ns}} \leq \tau_{s,+}^{\text{stick}}, \quad (3.7.6)$$

as in Equation (3.5.14).

The active contact branch is selected using the closure law from Section 3.6.6. In compact form,

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the branch output selected by Equation (3.6.32)}. \quad (3.7.7)$$

If both contacts remain adhered, the branch output is

$$(\dot{v}_b, \tau_p, \tau_s) = (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}).$$

If one or both contacts slide, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves  $\dot{v}_b$  from the belt equation of motion. The primary-slip, secondary-slip, and two-contact-slip belt accelerations are given by Equation (3.6.15), Equation (3.6.23), and Equation (3.6.29), respectively.

### 3.7.4 Governing ODEs

Once the branch closure has supplied  $\dot{v}_b$ ,  $\tau_p$ , and  $\tau_s$ , the primary pulley equation of motion is

$$\dot{\omega}_p = \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}. \quad (3.7.8)$$

The secondary pulley equation of motion is

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}. \quad (3.7.9)$$

The shift-position kinematics are

$$\frac{d}{dt}s = \dot{s}. \quad (3.7.10)$$

The axial shift acceleration is obtained from Equation (3.2.52):

$$\ddot{s} = \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\right) - \left(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}. \quad (3.7.11)$$

Finally, the belt transport speed evolves according to the branch-selected belt acceleration:

$$\frac{d}{dt}v_b = \dot{v}_b, \quad \dot{v}_b \text{ from Equation (3.7.7)}. \quad (3.7.12)$$

### 3.7.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = (\omega_p, \omega_s, s, \dot{s}, v_b),$$

the model is evaluated in the following order:

1. Compute the pulley geometry from  $s$ , including  $r_p$ ,  $r_s$ ,  $r_{p,\text{cm}}$ ,  $r_{s,\text{cm}}$ ,  $\phi_p$ , and  $\phi_s$ .
2. Compute the engine torque  $\tau_{\text{eng}}(\omega_p)$  and the reflected load torque  $\tau_{\text{load}}(\omega_s)$ .

3. Compute the no-slip candidate

$$\dot{v}_{b,\text{ns}}, \quad \tau_{p,\text{ns}}, \quad \tau_{s,\text{ns}},$$

using Equation (3.4.12), Equation (3.4.15), and Equation (3.4.16).

4. Compute the primary and secondary no-slip admissibility quantities from Section 3.5.3 and Section 3.5.4.
5. Compute the relative contact speeds  $v_{\text{rel},p}$  and  $v_{\text{rel},s}$  using Equation (3.7.4).
6. Select the active torque-transfer branch using Equation (3.6.32). This determines the applied values of

$$\dot{v}_b, \quad \tau_p, \quad \tau_s.$$

7. Evaluate the ODE right-hand sides Equation (3.7.8), Equation (3.7.9), Equation (3.7.10), Equation (3.7.11), and Equation (3.7.12).

### 3.7.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$\begin{aligned} \dot{\omega}_p &= \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}, \\ \dot{\omega}_s &= \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}, \\ \frac{d}{dt}s &= \dot{s}, \\ \frac{d}{dt}\dot{s} &= \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\right) - \left(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}, \\ \frac{d}{dt}v_b &= \dot{v}_b. \end{aligned} \tag{3.7.13}$$

with the algebraic closure

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the active branch output from Equation (3.6.32)}.$$

In summary, the state determines the pulley geometry and relative contact speeds. The no-slip candidate determines the torque demand required for adhered motion. The traction admissibility conditions determine whether that candidate can be maintained. If not, the appropriate slip branch supplies saturated contact torques and the resulting belt acceleration. These branch-selected quantities then feed back into the rotational, belt, and axial shift dynamics, closing the complete CVT state-space model.

## Chapter 4

# Results and Discussion

### 4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows ?? : low ratio means larger  $R$  (greater speed reduction), and high ratio means smaller  $R$ . following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in [Section 4.2](#).

#### 4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

| Case | Description                                    | Purpose in results section                               |
|------|------------------------------------------------|----------------------------------------------------------|
| D    | Default tuning                                 | Baseline reference case                                  |
| F    | Lighter flyweights                             | Shows the effect of reduced primary centrifugal clamping |
| H    | Steeper helix                                  | Shows the effect of reduced secondary torque reactivity  |
| C    | Shallower initial primary ramp + steeper helix | Produces a strongly slip-dominated launch                |

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

#### 4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in [Figure 4.1](#). Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800 \text{ rpm}, \quad \omega_s(0) = 0,$$

where  $\omega_p$  is the primary angular speed and  $\omega_s$  is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

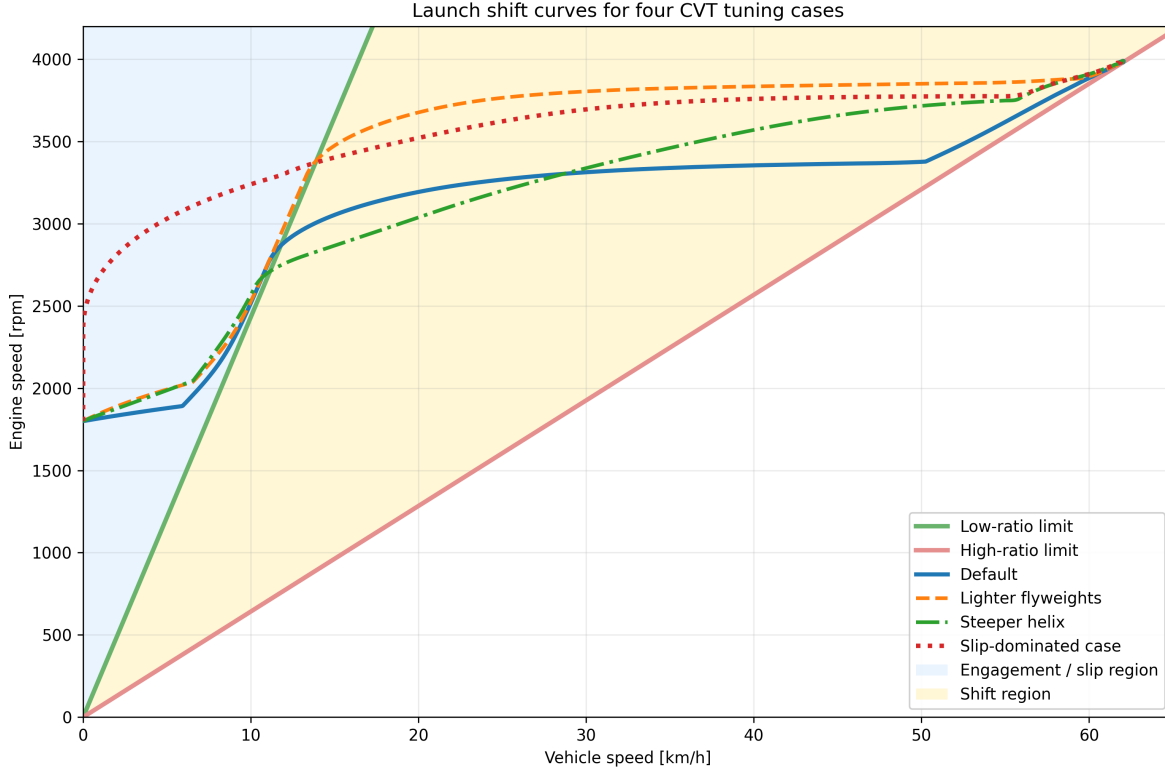


Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio

boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, [Figure 4.1](#) is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

### 4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of [Figure 4.1](#) indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, [Figure 4.2](#) compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.



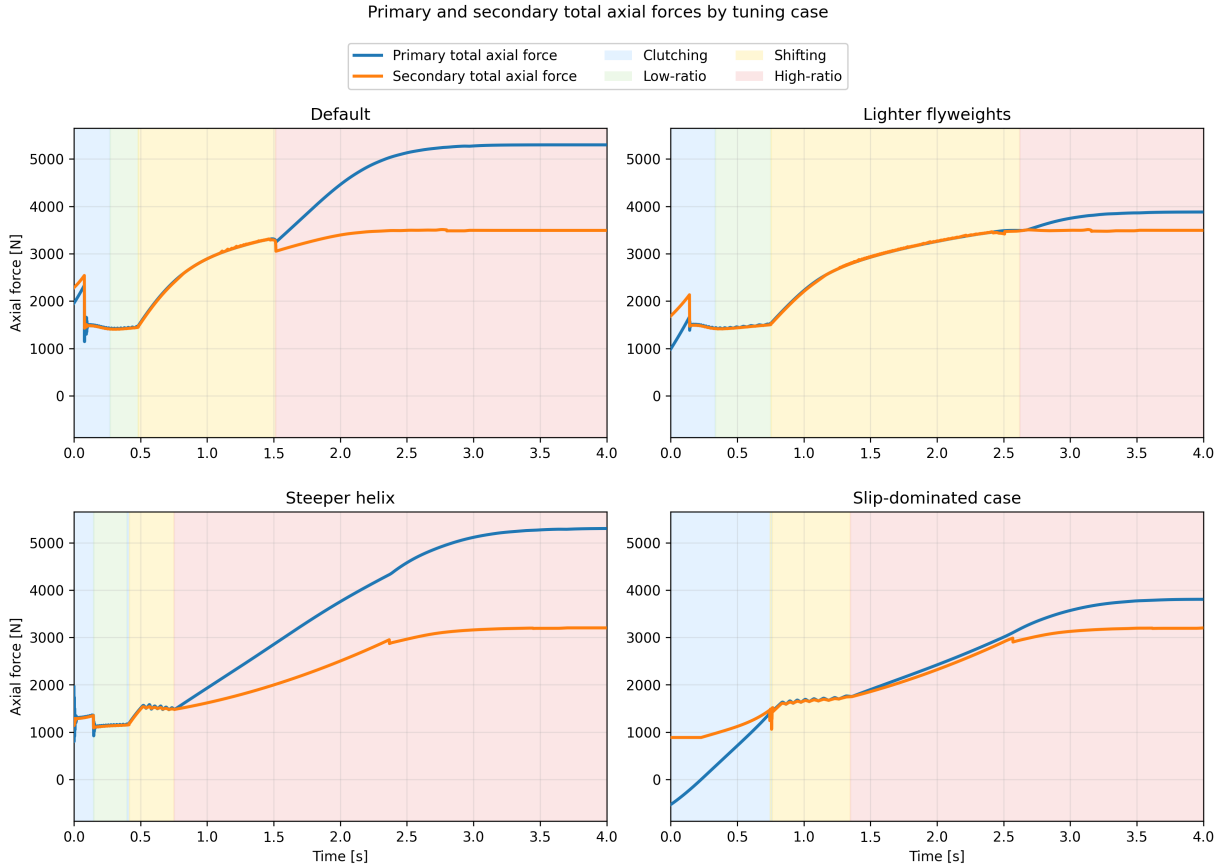


Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in [Figure 4.1](#). As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, [Figure 4.2](#) provides the mechanical explanation for the differences observed in [Figure 4.1](#). A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

#### 4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque  $\tau_{\text{ns}}$ , and the actual transmitted coupling torque  $\tau_c$ .

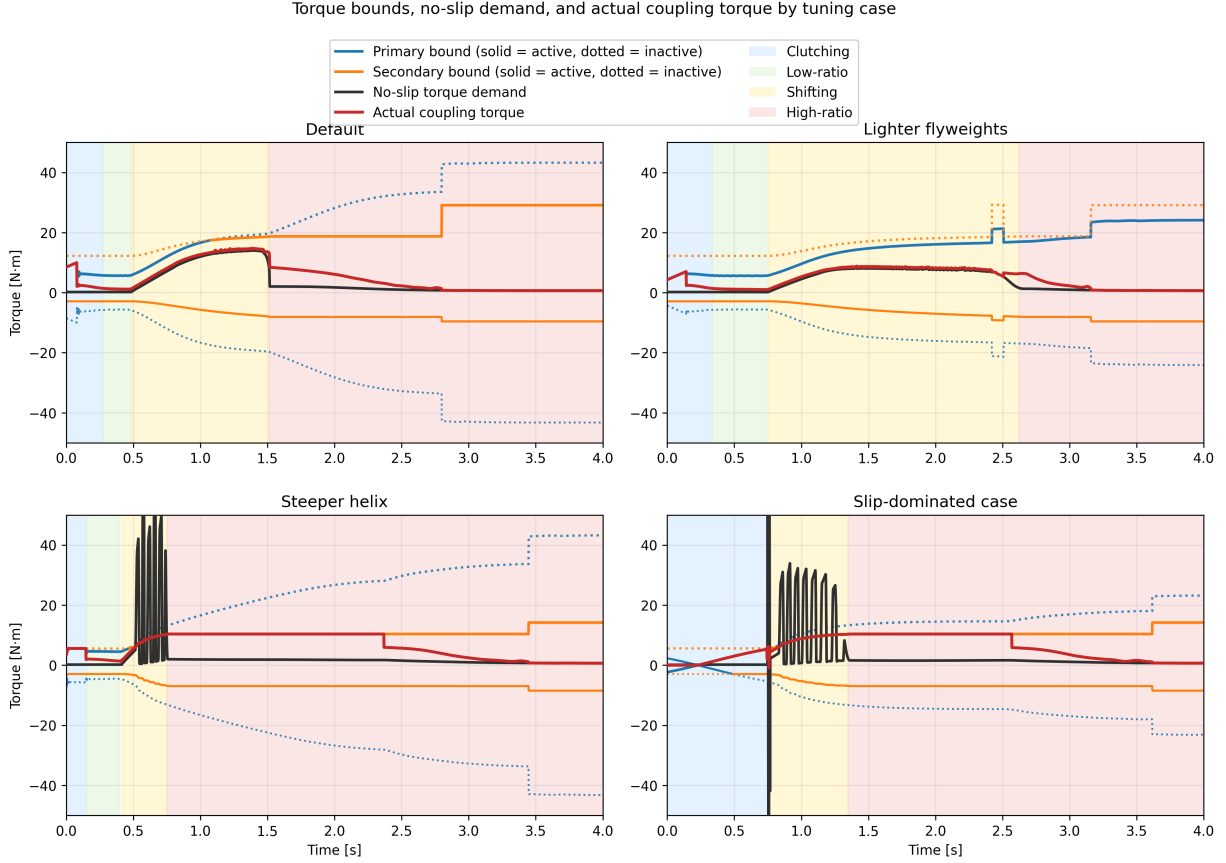


Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand  $\tau_{\text{ns}}$ , and the red curve shows the actual transmitted coupling torque  $\tau_c$ . Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When  $\tau_{\text{ns}}$  lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When  $\tau_{\text{ns}}$  exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a

relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of  $\tau_c$ . In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of  $\tau_{ns}$ , in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however,  $\tau_{ns}$  rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to  $\tau_{ns}$ : if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick-slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix C.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick-slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 4.3 closes the interpretation begun in the previous two subsections. The

shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model’s clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

#### 4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

| Case                | Time to 50 m [s] | Time to 60 km/h [s] |
|---------------------|------------------|---------------------|
| Default             | 3.93             | 2.27                |
| Lighter flyweights  | 4.26             | 2.78                |
| Steeper helix       | 4.19             | 2.82                |
| Slip-dominated case | 4.41             | 2.98                |

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps “kick-start” vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed

primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

## 4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at <https://github.com/gr812b/CVT-Simulator>. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in [Appendix C.3](#).

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): <https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z>,
- archival DOI landing page (placeholder): <https://doi.org/XX.XXXX/zenodo.XXXXXXX>.

Unless otherwise stated, simulation cases in [Section 4.1](#) correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: <https://cvt.kaiarseneau.dev>. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

## 4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

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# Appendix A

## Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

### A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies  $\lambda \approx 0$  and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint ?? reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (\text{A.1.1})$$

### A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute  $r_p = r_{p,\text{outer,min}}$  and  $r_s = r_{s,\text{outer,max}}$  into Equation (A.1.1) and solve for  $C$ .

**Step 1: Isolate the square root.** Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \quad (\text{A.2.1})$$

**Step 2: Square both sides.**

$$[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})]^2 = 4[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2]. \quad (\text{A.2.2})$$

**Step 3: Solve for  $C^2$ .**

$$C^2 = \frac{[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2. \quad (\text{A.2.3})$$

**Step 4: Take the square root.** Expanding the squared term in the numerator:

$$[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2,$$

the approximate center distance becomes

$$C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2}. \quad (\text{A.2.4})$$

This closed-form expression avoids the need for numerical root finding but neglects the  $\sin^{-1}$  term in the exact constraint.

### A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

#### System Parameters

| Parameter                                              | Value     | SI Value      |
|--------------------------------------------------------|-----------|---------------|
| Belt length $L_b$                                      | 37.538 in | 0.9535 m      |
| Belt height $h_b$                                      | 0.613 in  | 0.015 57 m    |
| Groove half-angle $\beta$                              | 11.5°     | 0.2007 rad    |
| Inner primary radius                                   | 0.8125 in | 0.020 637 5 m |
| Primary outer radius $r_{p,\text{outer},\text{min}}$   | 1.4255 in | 0.036 21 m    |
| Secondary outer radius $r_{s,\text{outer},\text{max}}$ | 4.0 in    | 0.1016 m      |

The primary outer radius is computed as the inner radius plus belt height:  $r_{p,\text{outer},\text{min}} = 0.8125 + 0.613 = 1.4255$  in.

**Computed Center Distances** Solving the exact belt-length constraint ?? numerically yields

$$C_{\text{exact}} = 0.251\,74\text{ m} \approx 9.911\text{ in.}$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\text{ m} \approx 10.566\text{ in.}$$

**Error Metrics** The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\text{ m} = 0.655\text{ in,} \quad (\text{A.3.1})$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \quad (\text{A.3.2})$$

**Assessment** The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption  $\lambda \approx 0$  is therefore not justified for this CVT geometry.

Since the center distance  $C$  propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in  $C$  can compound into larger errors in predicted system behavior.

## A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

### A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius  $r_{p,\text{outer}}$  and center distance  $C$ , the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation [Equation \(A.1.1\)](#), this yields a closed-form expression.

**Step 1: Isolate the square root.** From [Equation \(A.1.1\)](#):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \quad (\text{A.4.1})$$

**Step 2: Square both sides.**

$$[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}]^2 = 4[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2]. \quad (\text{A.4.2})$$

**Step 3: Expand and collect terms.** Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of  $r_{s,\text{outer}}$  yields a quadratic equation:

$$(\pi^2 + 4)r_{s,\text{outer}}^2 - 2[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}]r_{s,\text{outer}} + [(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2] = 0. \quad (\text{A.4.3})$$

**Step 4: Apply the quadratic formula.** The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$r_{s,\text{approx}} = \frac{\pi L_b + (4 - \pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2 + 4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2 + 4}. \quad (\text{A.4.4})$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

### A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance  $C_{\text{approx}}$ , so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

**Primary Radius at Zero Shift** At  $s = 0$ , the primary is within the deadzone, so from ??:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer},\text{min}} = 0.036\,21\text{ m}.$$

**Secondary Radius Comparison** Solving the exact belt-length constraint with  $C_{\text{exact}}$  yields

$$r_{s,\text{exact}} = 0.101\,60\text{ m}.$$

Evaluating the approximate formula [Equation \(A.4.4\)](#) with  $C_{\text{approx}} = 0.268\,37\text{ m}$  yields

$$r_{s,\text{approx}} = 0.101\,60\text{ m}.$$

**Transmission Ratio Error** Computing effective radii with  $h_b/2 = 0.007\,79\text{ m}$ :

| Quantity                                | Exact    | Approximate |
|-----------------------------------------|----------|-------------|
| $r_{p,\text{eff}}$                      | 0.0284 m | 0.0284 m    |
| $r_{s,\text{eff}}$                      | 0.0938 m | 0.0938 m    |
| $R = r_{s,\text{eff}}/r_{p,\text{eff}}$ | 3.300    | 3.300       |

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although  $C_{\text{approx}}$  is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ( $s = 0$ ). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

### A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

**Primary Radius at Maximum Shift** Using ?? with  $s = s_{\max} = 0.75 \text{ in} = 0.01905 \text{ m}$  and deadzone  $s_{\text{dz}} = 0.0024892 \text{ m}$ :

$$\begin{aligned}
r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer},\min} + \frac{s_{\max} - s_{\text{dz}}}{2 \tan \beta} \\
&= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5^\circ)} \\
&= 0.03621 + \frac{0.01656}{0.4070} \\
&= 0.0769 \text{ m.}
\end{aligned} \tag{A.4.5}$$

**Secondary Radius Comparison** Solving the exact belt-length constraint ?? numerically with  $C_{\text{exact}} = 0.25174 \text{ m}$  yields

$$r_{s,\text{exact}} = 0.06673 \text{ m.}$$

Evaluating the approximate formula Equation (A.4.4) with  $C_{\text{approx}} = 0.26837 \text{ m}$  yields

$$r_{s,\text{approx}} = 0.05626 \text{ m.}$$

**Transmission Ratio Error** The CVT ratio is defined using effective radii, which account for belt seating depth (??):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With  $h_b/2 = 0.00779 \text{ m}$ :

| Quantity                                | Exact    | Approximate |
|-----------------------------------------|----------|-------------|
| $r_{p,\text{eff}}$                      | 0.0691 m | 0.0691 m    |
| $r_{s,\text{eff}}$                      | 0.0589 m | 0.0485 m    |
| $R = r_{s,\text{eff}}/r_{p,\text{eff}}$ | 0.853    | 0.701       |

Both models predict overdrive ( $R < 1$ ), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ( $s = s_{\max}$ ). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

## A.5 Summary of Approximation Errors

| Configuration                          | $R_{\text{exact}}$ | $R_{\text{approx}}$ | Ratio Error |
|----------------------------------------|--------------------|---------------------|-------------|
| Zero shift ( $s = 0$ )                 | 3.300              | 3.300               | 0.000%      |
| Maximum shift ( $s = s_{\text{max}}$ ) | 0.853              | 0.701               | 17.8%       |

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 3.7), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.



## Appendix B

# Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From [Equation \(3.2.8\)](#), the centrifugal axial force is

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f(s)) \frac{dr_f}{ds}, \quad (\text{B.0.1})$$

where  $m_f$  is the flyweight mass,  $r_{f,0}$  is the initial flyweight radius, and  $r_f(s)$  is the radial displacement along the ramp as a function of axial shift  $s$ .

The profile  $r_f(s)$  must satisfy the admissibility conditions from [Section 3.2](#): continuously differentiable, non-negative slope, and finite slope throughout  $[0, s_{\text{max}}]$ .

### B.1 Linear Ramp

A linear ramp with slope  $k_r$  (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \quad \frac{dr_f}{ds} = k_r = \text{const.} \quad (\text{B.1.1})$$

Substituting into [Equation \(B.0.1\)](#):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 k_r (r_{f,0} + k_r s) = m_f \omega_p^2 k_r r_{f,0} + m_f \omega_p^2 k_r^2 s. \quad (\text{B.1.2})$$

At constant  $\omega_p$ , axial force increases linearly with  $s$  because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

### B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles  $\alpha_1$  and  $\alpha_2$  at the start and end of the segment, where  $|dr_f/ds| = \tan \alpha$  specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius  $R_c$  centered at  $(s_c, r_c)$ , the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \quad (\text{B.2.1})$$

Solving for  $r_f(s)$  on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \quad (\text{B.2.2})$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \quad (\text{B.2.3})$$

**Parameterization from slope angles.** At any point on the arc where the local slope angle is  $\alpha$  (i.e.  $dr_f/ds = \tan \alpha$ ), the circle constraint combined with Equation (B.2.3) gives the parametric form:

$$s = s_c - R_c \sin \alpha, \quad r_f = r_c + R_c \cos \alpha. \quad (\text{B.2.4})$$

The slope angle  $\alpha$  decreases monotonically from  $\alpha_1$  (entry) to  $\alpha_2 < \alpha_1$  (exit) as  $s$  increases along the ramp. Applying Equation (B.2.4) at both boundary shifts  $s_1$  and  $s_2$  and subtracting:

$$s_2 - s_1 = R_c(\sin \alpha_1 - \sin \alpha_2). \quad (\text{B.2.5})$$

With  $\Delta s = s_2 - s_1$  as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin \alpha_1 - \sin \alpha_2}, \quad s_c = s_1 + R_c \sin \alpha_1. \quad (\text{B.2.6})$$

The boundary condition  $r_f(s_1) = 0$  (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos \alpha_1. \quad (\text{B.2.7})$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \quad (\text{B.2.8})$$

To see the cancellation explicitly, split  $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$ :

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[ \underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 (s - s_c) \left[ 1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \quad (\text{B.2.9})$$

This form reveals the structure clearly. The dominant factor is  $-(s - s_c)$ , which is linear in shift: the  $(r_f - r_c)$  terms cancel exactly in the first part of the split. The bracket  $\left[ 1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]$  is a nonlinear correction multiplier. Using the parametric form Equation (B.2.4) with Equation (B.2.7), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos \alpha_1}{R_c \cos \alpha}, \quad (\text{B.2.10})$$

where  $\alpha$  is the local slope angle at shift  $s$ , sweeping from  $\alpha_1$  at entry to  $\alpha_2$  at exit. This makes the effective proportionality between force and  $(s - s_c)$  position-dependent.

The numerator  $r_{f,0} - R_c \cos \alpha_1$  is the key residual: it is controlled entirely by the choice of  $\alpha_1$ ,  $\alpha_2$ , and  $\Delta s$  through  $R_c = \Delta s / (\sin \alpha_1 - \sin \alpha_2)$ . Increasing either  $\alpha_1$  or  $\Delta s$  raises  $R_c$ , making  $R_c \cos \alpha_1$  larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing  $\alpha_1$  and  $\alpha_2$  adjusts both simultaneously.

The condition  $r_{f,0} = R_c \cos \alpha_1$  makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed  $\omega_p$ , the centrifugal force is approximately proportional to  $(s - s_c)$ , growing linearly with shift. This directly matches the linear restoring force of the primary spring (force  $\propto k_s \cdot s$ ). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset  $(r_{f,0} + r_c)$  is small relative to the arc radius  $R_c$ , but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

### B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.2.10), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\text{cent}}}{-m_f \omega_p^2 (s - s_c)} = 1 + \frac{r_{f,0} - R_c \cos \alpha_1}{R_c \cos \alpha}. \quad (\text{B.3.1})$$

This varies with position because  $\alpha$  changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call  $K$ , and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s)) \frac{dr_f}{ds} = Ks, \quad (\text{B.3.2})$$

where  $K > 0$ . This produces centrifugal force exactly proportional to shift:

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 K s. \quad (\text{B.3.3})$$

$K$  is not a free parameter any more than  $R_c$  is for the circular arc: just as  $R_c$  is uniquely determined by  $\alpha_1$ ,  $\alpha_2$ , and  $\Delta s$  (Equation (B.2.6)),  $K$  is uniquely determined by  $\alpha_1$ ,  $\alpha_2$ ,  $s_1$ , and  $s_2$  (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes  $K$  exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

**Deriving the profile.** Equation (B.3.2) is separable. Substituting  $u = r_{f,0} + r_f$ , so  $du/ds = dr_f/ds$ :

$$u du = Ks ds. \quad (\text{B.3.4})$$

Integrating both sides and writing the constant of integration as  $\frac{1}{2}Ks_0^2$ :

$$\frac{1}{2}u^2 = \frac{1}{2}K(s^2 + s_0^2). \quad (\text{B.3.5})$$

Substituting back  $u = r_{f,0} + r_f$ :

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2)}, \quad (\text{B.3.6})$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \quad (\text{B.3.7})$$

In the  $(s, r_f)$  plane, the locus  $(r_{f,0} + r_f)^2 = Ks^2$  (taking  $s_0 = 0$ ) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in [Equation \(B.2.9\)](#) could not be removed, this profile forces the product  $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$  exactly for all  $s$ .

**Local slope and its variation.** To see how  $K$  relates to actual ramp angles, differentiate [Equation \(B.3.6\)](#) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K} s}{\sqrt{s^2 + s_0^2}}. \quad (\text{B.3.8})$$

Since  $dr_f/ds = \tan \alpha(s)$  by definition of the ramp slope angle, the local angle satisfies:

$$\tan \alpha(s) = \frac{\sqrt{K} s}{\sqrt{s^2 + s_0^2}}. \quad (\text{B.3.9})$$

Two observations follow. First, the slope is zero at  $s = 0$  and grows monotonically toward the asymptote  $\sqrt{K}$  as  $s \rightarrow \infty$ . This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm  $(r_{f,0} + r_f)$  at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product  $Ks$  linear. Second, the asymptotic slope  $\lim_{s \rightarrow \infty} \tan \alpha(s) = \sqrt{K}$  gives a simple interpretation:  $K = \tan^2 \alpha_\infty$  where  $\alpha_\infty$  is the slope the profile approaches at large shift.

**Parameterization by entry and exit angles.** Following the same convention as [Appendix B.2](#), specify desired slope angles  $\alpha_1$  and  $\alpha_2$  at entry shift  $s_1$  and exit shift  $s_2$ . From [Equation \(B.3.9\)](#):

$$\tan^2 \alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \quad \tan^2 \alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \quad (\text{B.3.10})$$

Dividing the two equations eliminates  $K$ :

$$\frac{\tan^2 \alpha_1}{\tan^2 \alpha_2} = \frac{s_1^2 (s_2^2 + s_0^2)}{s_2^2 (s_1^2 + s_0^2)}. \quad (\text{B.3.11})$$

Solving for  $s_0^2$ :

$$s_0^2 = \frac{s_1^2 s_2^2 (\tan^2 \alpha_2 - \tan^2 \alpha_1)}{\tan^2 \alpha_1 s_2^2 - \tan^2 \alpha_2 s_1^2}. \quad (\text{B.3.12})$$

Then  $K$  follows from either condition in Equation (B.3.10):

$$K = \frac{\tan^2 \alpha_1 (s_1^2 + s_0^2)}{s_1^2}. \quad (\text{B.3.13})$$

For  $s_0^2 > 0$  a consistent solution requires the numerator and denominator of Equation (B.3.12) to have the same sign. Since the hyperbolic slope increases with  $s$  (Eq. (B.3.8)), the physically consistent case is  $\alpha_2 > \alpha_1$  (exit angle larger than entry), with  $\tan^2 \alpha_1 s_2^2 > \tan^2 \alpha_2 s_1^2$  ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ( $\tan^2 \alpha_1 s_2^2 = \tan^2 \alpha_2 s_1^2$ ).

**Practical impact.** In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc’s partial cancellation is not a fundamental limitation but rather a small correction from the constant offset ( $r_{f,0} + r_c$ ).

**Limitations.** Both profiles address only the centrifugal force term. The secondary pulley torque feedback ( $F_{s,\text{ax}}(s, \tau_s)$  in Equation (3.7.11)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

## B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts  $s_1, s_2, \dots, s_{n-1}$ . This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

**Admissibility at junctions.** The admissibility conditions from Section 3.2 require  $r_f(s)$  to be continuously differentiable ( $C^1$ ). A piecewise ramp satisfies  $C^0$  by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at  $s_k$  differs from the right-segment slope, Eq. (B.0.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

**Numerical stiffness considerations.** A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative  $d^2 r_f / ds^2$  near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores  $C^1$  continuity and bounds  $d^2 r_f / ds^2$  throughout the ramp.

# Appendix C

## Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

### C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius  $r_1$  and fixed center-to-center distance  $C$ , the secondary outer radius  $r_2$  is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from ?? into the residual form:

$$g(r_2; r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1) \arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \quad (\text{C.1.1})$$

where  $L_b$  is the fixed belt length. A separate invocation solves for the center-to-center distance  $C$  itself at initialization, using the same equation evaluated at the boundary radii  $(r_{1,\min}, r_{2,\max})$  with  $C$  as the unknown; this is exactly the initialization step implied by ?? before the ratio law [Equation \(3.7.2\)](#) can be evaluated over the shift range.

**Bracketing strategy.** The arcsin term in [Equation \(C.1.1\)](#) is real only when  $|r_2 - r_1| \leq C$ , so the natural domain for  $r_2$  is the open interval  $(r_1 - C, r_1 + C)$ . Within this interval a sign change is guaranteed: at the lower boundary  $r_2 \rightarrow r_1 - C$ , the arcsin saturates to  $-\pi/2$  and the square root vanishes, giving  $g \rightarrow 2\pi r_1 - 2\pi C - L_b < 0$  for all physically valid parameters. At the upper boundary  $r_2 \rightarrow r_1 + C$ , by symmetry  $g \rightarrow 2\pi r_1 + 2\pi C - L_b > 0$ , satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin  $\varepsilon = 10^{-9}$  m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval  $(\Delta r + \varepsilon, L_b/2)$ , where  $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$  is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed  $L_b$  across a gap of  $L_b/2$ ). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

**Brent’s method.** Both roots are found using `scipy.optimize.brentq`, an implementation of Brent’s algorithm [Brent \(1973\)](#). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order- $\approx 1.84$  convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case  $O(\log_2(1/\varepsilon))$  guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of  $10^{-9}$  m, well below any physically meaningful length precision.

## C.2 ODE Integration for Dynamic Simulation

The full CVT state vector  $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$  is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in [Equation \(3.7.13\)](#), with rotational sub-equations [Equation \(3.7.8\)](#) and [Equation \(3.7.9\)](#), axial dynamics [Equation \(3.7.11\)](#), belt-speed dynamics for  $v_b$ , and algebraic geometry closures [Equation \(3.7.2\)](#) and [Equation \(3.7.3\)](#). Although [Equation \(3.7.11\)](#) is second-order in  $s$ , introducing  $(s, \dot{s})$  as separate states converts the complete model to first-order form for time integration. The time integrator is used with the RK45 method — the Dormand–Prince explicit embedded Runge–Kutta pair [Dormand and Prince \(1980\)](#). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

**Adaptive step size control.** At each candidate step the per-component local error estimate  $\hat{e}_i$  is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\text{atol} + \text{rtol} \cdot |y_i|} \leq 1, \quad (\text{C.2.1})$$

and the next step size is scaled proportionally to  $(\text{tol}/\hat{e})^{1/5}$ . The tolerances used in this work are  $\text{atol} = 10^{-6}$  and  $\text{rtol} = 10^{-4}$  ([Table C.1](#)). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g.  $\dot{s} \approx 0$  at shift equilibrium or  $\omega_p \approx 0$  at engine start), where a purely relative criterion would demand unreachable accuracy.

| Parameter | Value      | Interpretation                                                         |
|-----------|------------|------------------------------------------------------------------------|
| atol      | $10^{-6}$  | absolute floor: 1 $\mu\text{m}$ on $s$ , 1 $\text{prad/s}$ on $\omega$ |
| rtol      | $10^{-4}$  | relative tolerance: 0.01% of current state magnitude                   |
| t_eval    | 10 000 pts | uniform output grid over 30 s                                          |

Table C.1: ODE solver tolerance parameters.

**Event detection.** Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`’s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in [Equation \(3.7.11\)](#) and the full state evolution in [Equation \(3.7.13\)](#). After each accepted step the solver evaluates the event functions on

a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

**Phase splitting at hard constraints.** The shift bounds  $s \in [0, s_{\max}]$  with  $\dot{s} = 0$  at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping [Equation \(3.7.11\)](#) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

### C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in [Equation \(3.6.9\)](#) is physically clear but discontinuous at  $v_{\text{rel}} = 0$ , which can induce step-size chatter near stick-slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \quad (\text{C.3.1})$$

where  $v_{\text{smooth}} > 0$  sets the transition width in slip-speed units. For  $|v_{\text{rel}}| \gg v_{\text{smooth}}$ , this approaches the usual friction-limited value  $\pm\tau_{\max}$ , while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\max}, \tau_{\max}). \quad (\text{C.3.2})$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}}, 0, 1\right), \quad (\text{C.3.3})$$

with  $v_{\text{blend}} > 0$  defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha) \tau_{\text{req}} + \alpha \tau_{\text{sat}}. \quad (\text{C.3.4})$$

This has a useful interpretation. For very small mismatch,  $\alpha \approx 0$ , so the model stays close to the clamped no-slip demand. For larger mismatch,  $\alpha \rightarrow 1$ , so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.



**How to choose  $v_{\text{smooth}}$ .** Choose the smoothing scale small enough that  $\tanh(v_{\text{rel}}/v_{\text{smooth}})$  is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with  $v_{\text{smooth}}$  around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near  $v_{\text{rel}} \approx 0$ .

The blend scale  $v_{\text{blend}}$  should generally be chosen greater than or equal to  $v_{\text{smooth}}$ , so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

## Appendix D

# Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from [Section 3.1.4](#). Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from [??](#):

$$R_g = \frac{r_s}{r_p}, \tag{D.0.1}$$

The individual primary and secondary radius rates and accelerations have already been established in [Section 3.1.4](#). This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

### D.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in [Equation \(D.0.1\)](#) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt} \left( \frac{r_s}{r_p} \right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2} \dot{r}_p. \end{aligned} \tag{D.0.2}$$

The secondary-radius result in [Equation \(3.1.51\)](#) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p. \tag{D.0.3}$$

Substituting this into Equation (D.0.2) gives

$$\begin{aligned}
\dot{R}_g &= \frac{1}{r_p} \left( -\frac{\phi_p}{\phi_s} \dot{r}_p \right) - \frac{r_s}{r_p^2} \dot{r}_p \\
&= -\frac{\dot{r}_p}{r_p} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \\
&= -\frac{\dot{r}_p}{r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right).
\end{aligned} \tag{D.0.4}$$

Finally, Equation (3.1.30) gives  $\dot{r}_p = \dot{s}/(2 \tan \beta)$  in the active-shift branch. Therefore,

|                                                                                                                                                                               |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Rate of Change of Geometric CVT Ratio</b>                                                                                                                                  | <b>(D.0.5)</b> |
| $ \dot{R}_g = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right), & s_{dz} < s \leq s_{max}. \end{cases} $ |                |

For an upshift,  $\dot{s} > 0$ , so the available geometric reduction decreases as the belt moves outward on the primary pulley.

## D.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{D.0.6}$$

Thus, every point in the  $(r_p, r_s)$  plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant- $R_g$  contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length  $L_b$  or center-to-center distance  $C$ —is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure D.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both

panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

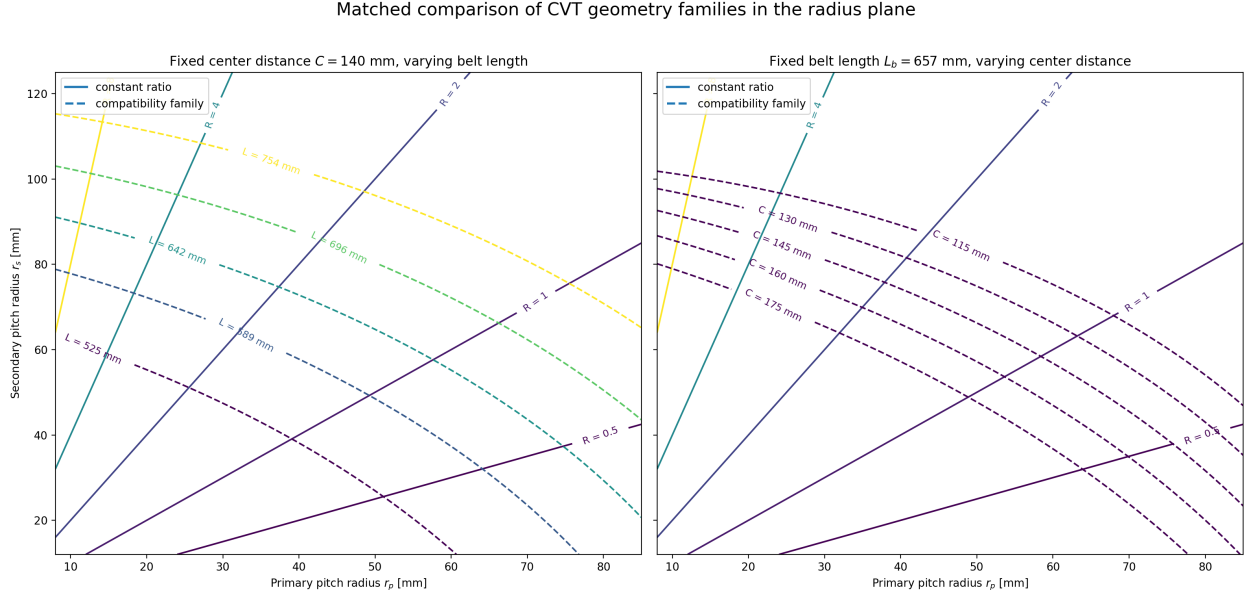


Figure D.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio  $R_g = r_s/r_p$ . In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing  $L_b$  or  $C$  changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, [Figure D.1](#) is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

### D.0.3 Exploring the Local Ratio-Change Map

The expression for  $\dot{R}_g$  gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide

out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right). \quad (\text{D.0.7})$$

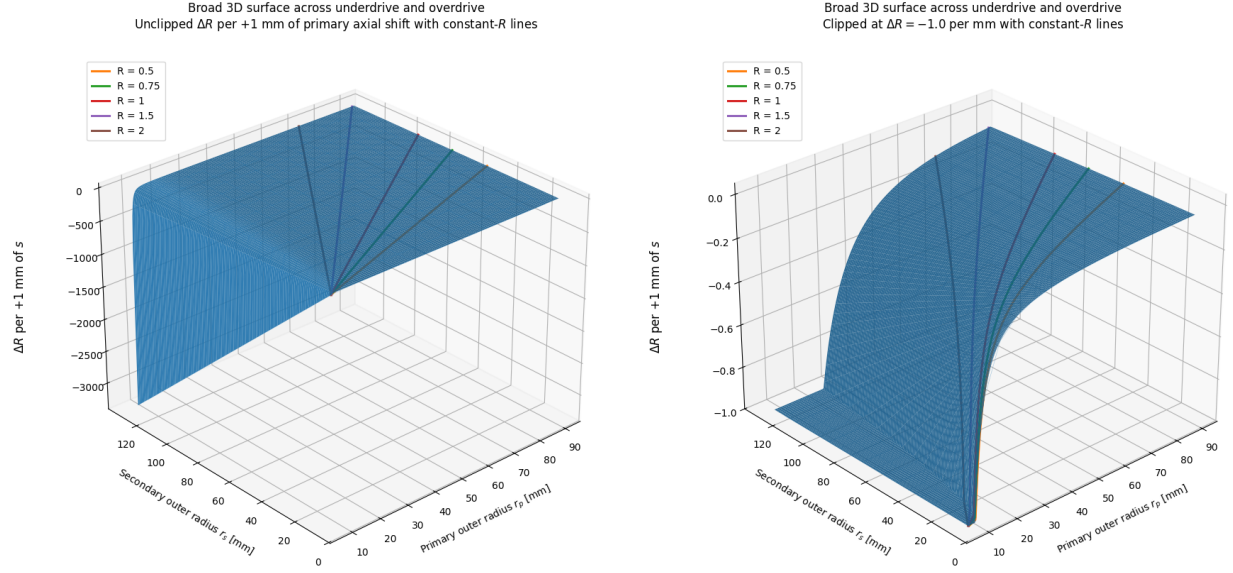
For a small positive increment  $\Delta s$ ,

$$\Delta R_g \approx \frac{dR_g}{ds} \Delta s. \quad (\text{D.0.8})$$

The plots below take  $\Delta s = 1 \text{ mm}$ , so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing  $s$  reduces the geometric reduction ratio.

**A sensitivity field over the radius plane** The first view is a broad surface over the radius plane. Here,  $r_p$  and  $r_s$  are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

**Figure D.2** shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit  $1/r_p$  factor in **Equation (D.0.7)**, and is reinforced by the fact that the geometric ratio  $R_g = r_s/r_p$  is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the  $R_g = 1$  locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small  $r_p$ . (b) The same domain with values below  $-1$  per  $1$  mm clipped, exposing the curvature through the mid-range.

Figure D.2: Local geometric-ratio change for a  $1$  mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the  $R_g = 1$  locus.

**How belt choice and installed geometry select a path** The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the  $(r_p, r_s)$  plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure D.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small- $r_p$  region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

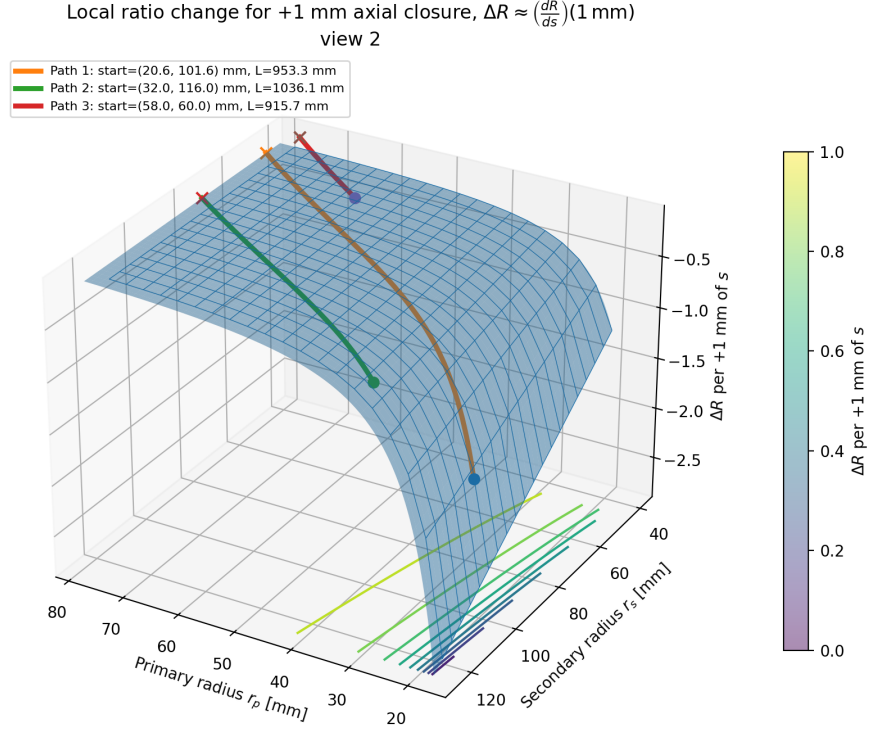
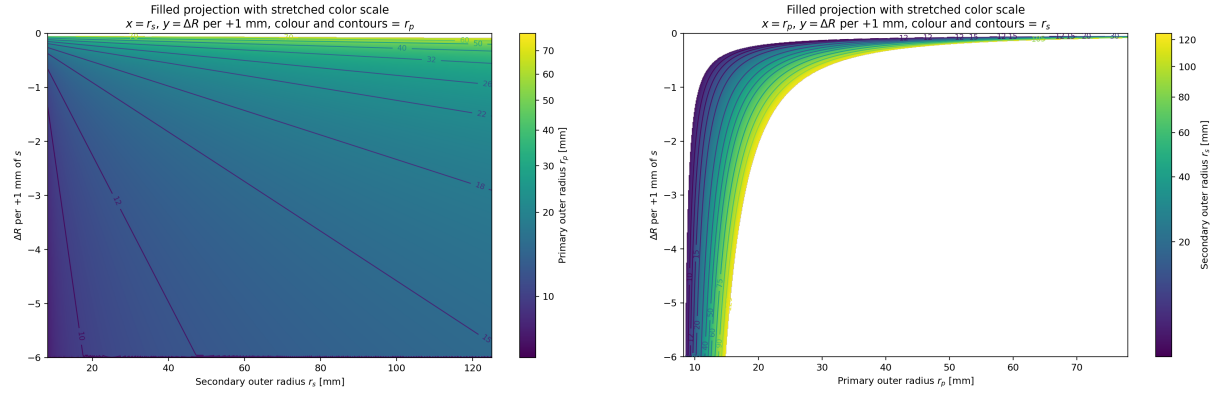


Figure D.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

**Separating the roles of the two pulley radii** The two projections in [Figure D.4](#) help isolate the individual effects of  $r_p$  and  $r_s$ . In [Figure D.4\(a\)](#), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing  $r_s$  increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

[Figure D.4\(b\)](#) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as  $r_p$  approaches its lower limit, then flattens at larger radius. This is the visual form of the  $1/r_p$  dependence in [Equation \(D.0.7\)](#). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. (b) Local ratio change against primary radius. Colour and contours identify the primary radius. Colour and contours identify the secondary radius.

Figure D.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

**Design interpretation** Taken together, [Figure D.1](#) and [Figure D.2](#)–[Figure D.4](#) give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small  $r_p$  can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible  $r_p$  is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by [Chen et al. \(1998\)](#) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, [Childs and Cowburn \(1987\)](#) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small- $r_p$  region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable ([Gerbert, 1996](#); [Bertini et al., 2014](#); [Messick, 2018](#)).



#### D.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (D.0.2) term by term to capture both effects.

For the first term,  $\dot{r}_s/r_p$ , use the quotient rule:

$$\frac{d}{dt} \left( \frac{\dot{r}_s}{r_p} \right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2}. \quad (\text{D.0.9})$$

For the second term,  $r_s \dot{r}_p / r_p^2$ , all three factors may change. Using the product rule,

$$\frac{d}{dt} \left( \frac{r_s \dot{r}_p}{r_p^2} \right) = \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3}. \quad (\text{D.0.10})$$

Subtracting Equation (D.0.10) from Equation (D.0.9) gives

$$\begin{aligned} \ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2} - \left[ \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3} \right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s \dot{r}_p}{r_p^2} - \frac{r_s \ddot{r}_p}{r_p^2} + \frac{2r_s \dot{r}_p^2}{r_p^3}. \end{aligned} \quad (\text{D.0.11})$$

The main derivation gives the required coupled secondary motion in ????:

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p, \quad (\text{D.0.12})$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2. \quad (\text{D.0.13})$$

Substituting these into Equation (D.0.11) gives

$$\begin{aligned} \ddot{R}_g &= \frac{1}{r_p} \left[ -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2 \right] - \frac{2}{r_p^2} \left[ -\frac{\phi_p}{\phi_s} \dot{r}_p \right] \dot{r}_p \\ &\quad - \frac{r_s}{r_p^2} \ddot{r}_p + \frac{2r_s}{r_p^3} \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \ddot{r}_p + \left[ \frac{2}{r_p^2} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right) \ddot{r}_p + \left[ \frac{2}{r_p^2} \left( \frac{\phi_p}{\phi_s} + R_g \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2. \end{aligned} \quad (\text{D.0.14})$$

Finally, use the primary radius rate and acceleration from [Equations \(3.1.30\)](#) and [\(3.1.59\)](#):

#### Acceleration of Geometric CVT Ratio

(D.0.15)

$$\ddot{R}_g = \begin{cases} 0, & 0 \leq s < s_{\text{dz}}, \\ -\frac{\ddot{s}}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right) + \frac{\dot{s}^2}{2 \tan^2 \beta r_p^2} \left( \frac{\phi_p}{\phi_s} + R_g \right) - \frac{2\pi^2 \dot{s}^2}{\phi_s^3 r_p \tan^2 \beta \sqrt{C^2 - (r_s - r_p)^2}}, & s_{\text{dz}} < s \leq s_{\text{max}}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to  $\ddot{s}$  is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity  $dR_g/ds$ , so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to  $\dot{s}^2$  arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation  $R_g(s)$ : as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.